

Equidistribution of primitive quartic torus packets for arbitrary orders

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Abstract

Let K range over totally real quartic fields with no proper intermediate field, and let $\mathcal{O}$ be any order in K . We prove that the packets of periodic diagonal orbits attached to invertible $\mathcal{O}$ -ideal classes, weighted by orbit volume, equidistribute with no escape of mass as $|\text{Disc}(\mathcal{O})|$ tends to infinity. The proof combines measure rigidity with a cubic-resolvent estimate that remains uniform at primes dividing the order index.

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1 Introduction

Ideal classes in a totally real number field give compact orbits of a diagonal group on a space of lattices. A natural arithmetic question is whether the finite collections of orbits obtained in this way become uniformly distributed as their discriminants grow. We prove this for the complete packets attached to invertible ideals of every order in a primitive totally real quartic field. The order may have arbitrarily large index in the ring of integers.

Let

$$X_4 = \mathrm{SL}_4(\mathbb{Z}) \backslash \mathrm{SL}_4(\mathbb{R})$$

be the space of covolume-one row lattices in $\mathbb{R}^4$, and let m_4 be its invariant probability measure. Put

$$H_4 = \{u \in \mathbb{R}^4 : u_1 + u_2 + u_3 + u_4 = 0\}, \quad A_4 = \{\mathrm{diag}(e^{u_1}, e^{u_2}, e^{u_3}, e^{u_4}) : u \in H_4\}.$$

The group A_4 acts on X_4 by right multiplication. Its Haar measure is $du_1 du_2 du_3$, with $u_4 = -u_1 - u_2 - u_3$.

A quartic field K is *primitive* if it has no field strictly between $\mathbb{Q}$ and K . Let K be primitive and totally real, let $\mathcal{O} \subseteq \mathcal{O}_K$ be an order, and fix an ordering $\sigma = (\sigma_1, \dots, \sigma_4)$ of its real embeddings. Here an order is a full-rank subring containing 1. A fractional $\mathcal{O}$ -ideal I is *proper* if its multiplier order $\{a \in K : aI \subseteq I\}$ is $\mathcal{O}$, and it is *invertible* if $IJ = \mathcal{O}$ for some fractional $\mathcal{O}$ -ideal J . The ordinary ideal class group $\mathrm{Pic}(\mathcal{O})$ consists of proper invertible fractional $\mathcal{O}$ -ideals modulo $I \sim aI$ for $a \in K^\times$.

For such an ideal put

$$\Lambda_{I,\sigma} = \mathrm{covol}(\sigma(I))^{-1/4} \sigma(I) \in X_4. \tag{1.1}$$

A covolume-one lattice is represented here by any positively oriented basis; two such bases differ by an element of $\mathrm{SL}_4(\mathbb{Z})$. Let W_{all} be the sixteen diagonal matrices with entries in $\{1, -1\}$. Choose

one ideal in each ordinary class and form the finite set of distinct orbits

$$\mathcal{P}_{\mathcal{O},\sigma} = \{\Lambda_{I,\sigma} w A_4 : I \text{ a class representative, } w \in W_{\text{all}}\}.$$

A matrix in W_{all} acts on the underlying lattice, so a sign matrix of determinant -1 also gives a point of X_4 . Using all signs makes the set independent of the chosen ordinary ideal representatives. Every orbit is compact by the unit theorem; Section 2 proves these assertions and describes the associated measure.

For an orbit $\mathcal{Y} \in \mathcal{P}_{\mathcal{O},\sigma}$, let $\nu_{\mathcal{Y}}$ be its invariant probability and let $\text{vol}(\mathcal{Y})$ be its volume for the specified Haar measure. Define

$$\mu_{\mathcal{O},\sigma} = \frac{\sum_{\mathcal{Y} \in \mathcal{P}_{\mathcal{O},\sigma}} \text{vol}(\mathcal{Y}) \nu_{\mathcal{Y}}}{\sum_{\mathcal{Y} \in \mathcal{P}_{\mathcal{O},\sigma}} \text{vol}(\mathcal{Y})}. \quad (1.2)$$

Each distinct orbit occurs once in these sums.

Theorem 1.1. *Let $(K_i, \mathcal{O}_i, \sigma_i)$ be any sequence in which K_i is a primitive totally real quartic field, $\mathcal{O}_i$ is an order in K_i , and σ_i is an ordering of its real embeddings. If $|\text{Disc}(\mathcal{O}_i)| \rightarrow \infty$, then*

$$\int_{X_4} f d\mu_{\mathcal{O}_i, \sigma_i} \longrightarrow \int_{X_4} f dm_4 \quad \text{for every } f \in C_c(X_4).$$

There is no escape of mass: for every $\varepsilon > 0$ there is a compact $\mathcal{K} \subset X_4$ such that $\mu_{\mathcal{O}_i, \sigma_i}(\mathcal{K}) \geq 1 - \varepsilon$ for all sufficiently large i .

In particular, the field may remain fixed while the order index tends to infinity. The packet in Theorem 1.1 uses every class in $\text{Pic}(\mathcal{O})$, so its ideals are locally free of rank one over the specified order. The proof uses this complete class average in both of its counting arguments.

1.1 History and context

For real quadratic fields, compact diagonal orbits are closed geodesics on the modular surface. Linnik's ergodic method and Skubenko's theorem for positive discriminants established arithmetic distribution under the condition $\left(\frac{d}{p}\right) = 1$ for the discriminant d at a fixed odd prime p [14, 24]. Using Iwaniec's estimates for half-integral weight forms [8], Duke removed that restriction for the geodesic collections associated with positive fundamental discriminants [2, Theorem 1]. Chelluri extended the corresponding statement to the unit tangent bundle, as recorded in [6, Section 1.2, p. 253]. Einsiedler, Lindenstrauss, Michel, and Venkatesh later gave a proof using entropy for all positive nonsquare order discriminants [6, Theorem 2.3].

In higher degree the diagonal group has higher rank, and the arithmetic packets lie in spaces of higher-dimensional lattices. Einsiedler, Lindenstrauss, Michel, and Venkatesh developed the discriminant and volume framework connecting separation of periodic torus orbits to entropy [4]. Their cubic theorem proves equidistribution of packets that include those arising from nonmaximal totally real cubic orders [5, Theorem 1.4]. The local homothety class of a lattice is part of their packet data; a multiplier order alone need not specify it [5, Section 5.5]. Their higher-dimensional discussion also identifies the possibility of proper intermediate homogeneous limits [5, Section 1.6.3]. The theorem of Einsiedler, Katok, and Lindenstrauss makes a diagonal-ergodic probability with positive entropy homogeneous [3, Theorem 1.3]; in prime dimension at least three its homogeneous conclusion is already Haar [3, Corollary 1.4].

For each fixed prime $p \geq 5$, Lemke Oliver, Thorner, and Zaman proved equidistribution for the orbit collections attached to orders in totally real degree- p fields as the order discriminant grows, outside an exceptional set containing $O_{p,\varepsilon}(Q^\varepsilon)$ fields of field discriminant at most Q [12, Theorem 2.3]. The separate preprint [18, Theorem 1.1] treats every packet in each fixed prime degree at least five and permits arbitrary local module types. We adapt its ordinary-vector method in the argument below. For quartic packets, Khayutin obtained entropy bounds for probability weak- $*$ limits of maximal-order packet measures with two-transitive Galois groups, including the primitive quartic types $\mathfrak{A}_4$ and S_4 [9, Theorem 1.1]. Wieser and Yang obtain entropy bounds for certain maximal-order quartic packets whose fields have a quadratic subfield, under further hypotheses [29]. The present argument must establish tightness and positive entropy in almost every diagonal-ergodic component for arbitrary orders, and then exclude the proper homogeneous limits allowed in degree four.

The arithmetic use of a sieve to rule out intermediate algebraic limits has a precedent in Khayutin’s work on joint distributions of CM points [10]. Local Fourier analysis for nonmaximal orders already appears in the cubic work of Einsiedler, Lindenstrauss, Michel, and Venkatesh [5, Proposition 9.9]. Here the object being transformed is a specific quadratic product map on an exterior lattice. The resulting estimate controls arbitrary conductor depth in a cubic-resolvent count, before the remaining multiplicative weights are inserted by a sieve.

1.2 The two counts and the conductor estimate

Write

$$\begin{aligned} R &= \mathcal{O}_K, & D &= |\mathrm{Disc}(K)|, & q &= [R : \mathcal{O}], \\ X &= q\sqrt{D} = \sqrt{|\mathrm{Disc}(\mathcal{O})|}, & \kappa &= \mathrm{Res}_{s=1} \zeta_K(s). \end{aligned}$$

We also write $a_K(n)$ for the number of nonzero integral R -ideals of norm n . Both counting arguments preserve the factor $\sqrt{D}\kappa$ that normalizes the packet after its finite local labels are averaged.

Sections 2 and 3 treat ordinary lattice vectors. An invertible $\mathcal{O}$ -ideal is locally a unit translate of $\mathcal{O}$ inside a maximal-order ideal. Averaging those translates gives an exact unfolding into a weighted sum over integral R -ideals. A local moment bound controls the order index, while Stark’s exceptional-zero result and the absence of a quadratic subfield retain the residue κ in the ideal count. Counting short vectors proves tightness. Counting vectors in boxes away from the coordinate hyperplanes gives an $O(r^4)$ bound for radius- r neighborhoods in X_4 , uniformly on each compact set of centers in every probability limit. For a diagonal flow with distinct weights, a two-sided tube is covered by $O(r^{-3})$ such neighborhoods with centers in a fixed compact enlargement, where r decreases exponentially with the tube length. The resulting exponential decay also holds for normalized averages over sets of A_4 -ergodic components with positive measure, because they are dominated by a multiple of the limit measure. This forces positive entropy in almost every component.

Section 4 applies measure classification and identifies the remaining obstruction. A proper positive-entropy component has two coordinate blocks of dimension two. Each lattice in its supporting orbit then has a vector in its exterior square supported on the two opposite coordinates ij and kl , with both coefficients nonzero. There are three choices of the partition $ij|kl$.

For an exterior vector $v = (v_{ij})_{i < j}$, the positive diagonal action preserves the three signed products

$$(v_{12}v_{34}, -v_{13}v_{24}, v_{14}v_{23}).$$

Their sum is an integer on the exterior square of a unimodular lattice. For the arithmetic lattices in our packets, Section 5 places the products in the three embeddings of a cubic resolvent field F . Fixing their sum at a nonzero integer m puts them on a trace- m plane. If the four exterior coordinates outside ij, kl have size at most r , two product coordinates have size at most r^2 . Thus the geometric obstruction is tested by a rectangle of area $O(r^4)$ on that plane. Theorem 5.2 supplies the corresponding arithmetic rectangle bound, with the multiplicity of every product and the packet normalization included.

The main new work lies in proving that bound uniformly in $\mathcal{O}$. At primes dividing Dq , the difficult local factors in the multiplicity are given by the density of the pushforward of Haar measure on the exterior square of the order lattice, with a unit scalar, under the quadratic product map. Its exact fiber measure accounts for ramification and for the order index. An element of F acts on each opposite-coordinate plane by the value of the corresponding real embedding; the resulting action descends to the rational exterior space $\bigwedge_{\mathbb{Q}}^2 K$. The elements preserving $\bigwedge_{\mathbb{Z}}^2 \mathcal{O}$ form an order C_F . Its discriminant is bounded below by a fixed positive power of X and above by a fixed power of X , even if the field F remains fixed.

Section 6 analyzes these densities through their quadratic Fourier transforms. Denominator ideals organize the frequencies; the stabilizer order controls the decay of the Fourier coefficients and the support and size of the corresponding inverse transforms. The local proof is expressed in terms of the quadratic lattice, its action by a cubic étale algebra, and its stabilizer order, a formulation suitable for analogous cubic-resolvent counts. In Section 7, Poisson summation gives the trace density from scalar frequencies. Complementary estimates control the total nonscalar contribution using discriminant separation, bounds for the Fourier and inverse transforms, and congruences supplied by inverse quadratic forms. The estimate is uniform for small residue moduli and for shrinking rectangles.

At the remaining primes, elementary valuation records give a local factor with mean $1 + (a_K(p) - 1)/p + O(p^{-2})$. Section 8 averages these factors by a weighted Selberg sieve. The subtraction of 1 in the local mean removes the logarithm in the prime-sum bound, leaving κ . Finally, Section 9 unfolds exterior vectors along the diagonal group. The fiber lengths are logarithmic and summable over the rectangles above. The resulting $O(r^4)$ bound excludes every proper homogeneous component and proves Theorem 1.1.

1.3 Conventions

The notation $N_{E/k}$ denotes the algebraic field norm. Ideal norms $N(\mathfrak{a})$ are positive absolute norms, extended multiplicatively to fractional ideals; absolute values are written explicitly for field norms when needed. For a rational prime p , a subscript p denotes tensoring a field with $\mathbb{Q}_p$ or a lattice with $\mathbb{Z}_p$. For a prime $\mathfrak{p}$ or $\mathfrak{q}$ of a local field we use the integer valuation with value 1 on a uniformizer, and write e and f for its ramification index and residue degree. A normalized extension of v_p has $v_p(p) = 1$, and $|p|_p = p^{-1}$. Ideal norms include the residue degrees. We use $\mathfrak{p}$ for primes of K and $\mathfrak{q}$ for primes of F when the two occur together.

All logarithms are natural, and $\log_+ t = \max(0, \log t)$ for $t > 0$. An upper bound $X^{o(1)}$ means $O_\eta(X^\eta)$ for every fixed $\eta > 0$. Unless stated otherwise, constants are uniform in K , $\mathcal{O}$, and the ordering of the embeddings. In the exterior estimates a nonzero trace m and a bounded real region are fixed before X tends to infinity; permitted dependence on them is stated in the relevant results.

2 Packets from arbitrary orders

Fix a primitive totally real quartic field K , an order $\mathcal{O} \subset K$, and an ordering $\sigma = (\sigma_1, \dots, \sigma_4)$ of its real embeddings, as in Theorem 1.1. We describe the packet using finitely many local lattice choices above each ideal class of the maximal order. This description lets us average a containment condition at each finite prime before performing an integral at the real place. The construction is adapted from the packet model in [18, Section 2, Lemma 2.1]; we give the argument with all sixteen sign patterns in degree four.

Throughout the paper put

$$R = \mathcal{O}_K, \quad D = |\text{Disc}(K)|, \quad q = [R : \mathcal{O}], \quad X = q\sqrt{D}, \quad \kappa = \text{Res}_{s=1} \zeta_K(s). \quad (2.1)$$

In particular, $|\text{Disc}(\mathcal{O})| = q^2 D = X^2$. Absolute ideal norms are extended multiplicatively to fractional R -ideals. Thus

$$\text{covol}(\sigma(I)) = \sqrt{D} \, N I \quad (2.2)$$

for every nonzero fractional R -ideal I : this is the defining discriminant identity for $I = R$, follows by index for integral I , and then follows for fractional I after multiplication by an integer. For a full $\mathbb{Z}$ -lattice $J \subset K$, write $J_p = J \otimes_{\mathbb{Z}} \mathbb{Z}_p$, and put

$$K_p = K \otimes_{\mathbb{Q}} \mathbb{Q}_p, \quad R_p = R \otimes_{\mathbb{Z}} \mathbb{Z}_p, \quad \mathcal{O}_p = \mathcal{O} \otimes_{\mathbb{Z}} \mathbb{Z}_p.$$

Here K_p is a product of finite extensions of $\mathbb{Q}_p$, and R_p is the product of their integer rings. We use the usual arithmetic of fractional ideals in the Dedekind domain R ; see [16, Theorems 3.7, 3.20, and 3.29].

2.1 The finite local choices

Choose one fractional R -ideal I in each ordinary ideal class of R . Write h_K for the number of these classes. At every prime choose a generator

$$I_p = \xi_p R_p, \quad \xi_p \in K_p^\times.$$

Such a generator exists because each component of R_p is a discrete valuation ring. Let $\mathcal{L}_{\mathcal{O}}(I)$ be the set of full lattices $L \subset I$ whose completions have the form

$$L_p = \xi_p t_p \mathcal{O}_p, \quad t_p \in R_p^\times \quad \text{for every } p. \quad (2.3)$$

Replacing ξ_p by another generator merely changes t_p by a unit, so this set does not depend on the chosen generators.

Lemma 2.1. *The set $\mathcal{L}_{\mathcal{O}}(I)$ is finite, and its cardinality is*

$$N_{\mathcal{O}} = \prod_{p|q} [R_p^\times : \mathcal{O}_p^\times], \quad (2.4)$$

independently of I . Its members satisfy

$$[I : L] = q, \quad RL = I,$$

and are proper invertible fractional $\mathcal{O}$ -ideals. Every ordinary class of proper invertible fractional $\mathcal{O}$ -ideals is represented, up to multiplication by $K^\times$, by a member of one of the sets $\mathcal{L}_{\mathcal{O}}(I)$. Uniform probability on $\mathcal{L}_{\mathcal{O}}(I)$ is induced by independent normalized Haar probabilities for $t_p \in R_p^\times$.

Proof. The additive group $R/\mathcal{O}$ has order q , so $qR \subset \mathcal{O}$. For every p and every unit t_p , it follows that

$$qI_p = \xi_p t_p qR_p \subset \xi_p t_p \mathcal{O}_p \subset I_p.$$

At $p \nmid q$ the middle lattice is I_p . At the remaining primes its image is a submodule of the finite module I_p/qI_p . The primary decomposition of I/qI identifies its p -primary part with I_p/qI_p . Thus a choice of these local submodules patches to a unique subgroup of I/qI , and its inverse image in I is the unique lattice with the prescribed completions.

The stabilizer of $\mathcal{O}_p$ under multiplication by $R_p^\times$ is exactly $\mathcal{O}_p^\times$. Indeed, $t_p \mathcal{O}_p = \mathcal{O}_p$ implies that both t_p and t_p^{-1} belong to $\mathcal{O}_p$, by applying the equality and its inverse to 1. The converse is immediate. The local choices therefore form the finite homogeneous space $R_p^\times / \mathcal{O}_p^\times$. Its invariant probability is uniform and is the image of normalized unit Haar measure. Patching over the finitely many primes dividing q proves Equation (2.4) and the assertion about probabilities. The index of the lattice in Equation (2.3) is $[R_p : \mathcal{O}_p]$, and its R_p -span is I_p because $1 \in \mathcal{O}_p$. Taking products of the indices and comparing all completions gives $[I : L] = q$ and $RL = I$.

We record why the local condition describes exactly invertible ideals of the order. A full fractional $\mathcal{O}$ -ideal J is invertible if and only if

$$J_p = \eta_p \mathcal{O}_p \quad \text{for some } \eta_p \in K_p^\times \quad \text{at every } p. \quad (2.5)$$

For the forward implication, complete an equality $JJ^{-1} = \mathcal{O}$. The resulting equality over $\mathcal{O}_p$ expresses 1 as a finite sum $\sum_i a_i b_i$, with $a_i \in J_p$ and $b_i \in J_p^{-1}$. At each maximal ideal $\mathfrak{m}$ of the semilocal ring $\mathcal{O}_p$, some $a_i b_i$ is a unit after localization, and that a_i generates $(J_p)_{\mathfrak{m}}$. To obtain one generator for all the maximal ideals, choose such an $a_{\mathfrak{m}}$ for each $\mathfrak{m}$, and use the Chinese remainder theorem [16, Theorem 1.14] to choose $c_{\mathfrak{m}} \in \mathcal{O}_p$ congruent to 1 at $\mathfrak{m}$ and to 0 at every other maximal ideal. Then $\eta_p = \sum_{\mathfrak{m}} c_{\mathfrak{m}} a_{\mathfrak{m}}$ generates $(J_p)_{\mathfrak{m}}$ modulo $\mathfrak{m}$ at each $\mathfrak{m}$; Nakayama's lemma [16, Lemma 1.9] shows that it generates each localization. Hence it generates J_p . Since J_p spans K_p over $\mathbb{Q}_p$, η_p has no zero component and lies in $K_p^\times$.

Conversely, define

$$J^{-1} = \{a \in K : aJ \subset \mathcal{O}\}.$$

This definition commutes with completion: after choosing $\mathbb{Z}$ -bases of J and $\mathcal{O}$, the containment is a finite collection of integrality conditions on the coordinates of multiplication by a , and the same conditions over $\mathbb{Z}_p$ describe the completed lattice. Under Equation (2.5) we therefore have $(J^{-1})_p = \eta_p^{-1} \mathcal{O}_p$. Consequently $(JJ^{-1})_p = \mathcal{O}_p$ for every p , which implies $JJ^{-1} = \mathcal{O}$. This proves the equivalence. It also proves properness: if $aJ \subset J$ for an invertible J , multiplication by J^{-1} gives $a\mathcal{O} \subset \mathcal{O}$, and hence $a \in \mathcal{O}$. Thus its multiplier order is $\mathcal{O}$.

The lattices in Equation (2.3) are stable under $\mathcal{O}_p$ at every prime, so comparison of completions shows that $\mathcal{O}L \subset L$. They are therefore fractional $\mathcal{O}$ -ideals satisfying Equation (2.5). In the other direction, let J be an invertible fractional $\mathcal{O}$ -ideal. Its R -span RJ is a fractional R -ideal. Multiplying J by an element of $K^\times$, we may arrange $RJ = I$ for the chosen representative of this maximal-order class. If $J_p = \eta_p \mathcal{O}_p$, then $\eta_p R_p = I_p = \xi_p R_p$, so $\eta_p = \xi_p t_p$ for a unit $t_p \in R_p^\times$. Thus the resulting J belongs to $\mathcal{L}_{\mathcal{O}}(I)$, as required. $\square$

2.2 Real parameters, identifications, and volume

Let $U = R^\times$, and let U^+ be its totally positive subgroup. Put

$$T_\infty = \{v = (v_1, \dots, v_4) \in (\mathbb{R}^\times)^4 : \prod_i |v_i| = 1\}.$$

We give T_∞ counting measure on its sixteen sign components and measure $du_1 du_2 du_3$ on the logarithms $u_i = \log |v_i|$, with $u_4 = -u_1 - u_2 - u_3$. Denote this measure by dv . The unit theorem [16, Theorems 5.1 and 5.9] says that

$$\ell(U) = \{(\log |\sigma_i(\epsilon)|)_{i=1}^4 : \epsilon \in U\}$$

is a full lattice in the hyperplane $\sum_i u_i = 0$. Its covolume for the deleted-coordinate measure is the ordinary regulator R_K . Its kernel on U consists of 1 and -1 , since these are the roots of unity in a totally real field.

Let $\mathcal{F}_U$ be a measurable fundamental domain for the embedded group $\sigma(U)$ in T_∞ . Reducing logarithms modulo $\ell(U)$ leaves sixteen sign choices, on which the remaining kernel $\{1, -1\}$ identifies pairs of opposite signs. It follows that $\text{vol}(\mathcal{F}_U) = 8R_K$. The analytic class-number formula in signature $(4, 0)$, with two roots of unity, gives

$$h_K \text{vol}(\mathcal{F}_U) = 8h_K R_K = \sqrt{D} \kappa; \quad (2.6)$$

see [17, Chapter V, Theorem 2.4(a)].

For $L \in \mathcal{L}_\mathcal{O}(I)$, set

$$c_I = (X N I)^{-1/4}.$$

Equations (2.2) and $[I : L] = q$ show that $c_I \sigma(L)v$ has covolume one for every $v \in T_\infty$. This notation denotes the corresponding row lattice, and thus a point of X_4 , after choosing a basis of determinant one. This convention also applies when the diagonal v has negative determinant.

Proposition 2.2 (Packet model and volume). *The maps*

$$(L, v) \mapsto c_I \sigma(L)v \quad (L \in \mathcal{L}_\mathcal{O}(I), v \in T_\infty)$$

give a bijection from the disjoint union over the chosen I of the quotients by

$$\epsilon : (L, v) \mapsto (\epsilon L, \sigma(\epsilon)^{-1}v), \quad \epsilon \in U, \quad (2.7)$$

onto $\bigcup_{\mathcal{Y} \in \mathcal{P}_{\mathcal{O}, \sigma}} \mathcal{Y} \subset X_4$. Counting measure on the L 's times dv descends to the sum of the Haar measures of those orbits. In particular,

$$\sum_{\mathcal{Y} \in \mathcal{P}_{\mathcal{O}, \sigma}} \text{vol}(\mathcal{Y}) = N_\mathcal{O} \sqrt{D} \kappa. \quad (2.8)$$

Writing $\mu_ = \mu_{\mathcal{O}, \sigma}$, for every nonnegative Borel function Φ on X_4 one has*

$$\int_{X_4} \Phi d\mu_* = \frac{1}{h_K N_\mathcal{O} \text{vol}(\mathcal{F}_U)} \sum_I \sum_{L \in \mathcal{L}_\mathcal{O}(I)} \int_{\mathcal{F}_U} \Phi(c_I \sigma(L)v) dv. \quad (2.9)$$

The identity permits infinite values on both sides.

Proof. The action in Equation (2.7) preserves the parameter set, since $\epsilon I = I$ for $\epsilon \in U$, and preserves its image lattice. Lemma 2.1 shows that every ordinary invertible $\mathcal{O}$ -ideal class occurs. If two ideals differ by multiplication by $\alpha \in K^\times$, their normalized embedded lattices differ by the diagonal with coordinates $\sigma_i(\alpha)/|N_{K/\mathbb{Q}}(\alpha)|^{1/4}$, an element of T_∞ . Finally T_∞ is the union of all sixteen sign translates of A_4 . These facts prove surjectivity.

Suppose two displayed normalized lattices coincide. After absorbing the ratio of their positive scalars, this equality gives a diagonal map carrying $\sigma(L)$ onto $\sigma(L')$. Choose $0 \neq b \in L$ and let

$b' \in L'$ be its image. None of the four coordinates of $\sigma(b)$ is zero. The diagonal entries must therefore be $\sigma_i(b')/\sigma_i(b) = \sigma_i(b'/b)$. The map is multiplication by $\alpha = b'/b$, and $L' = \alpha L$. Taking R -spans gives $I' = \alpha I$. The chosen representatives then force $I' = I$ and $\alpha I = I$, so $\alpha \in U$. The scalar normalizations are now equal, and the two parameters differ by exactly Equation (2.7). This proves injectivity, including all identifications involving signs. Equality of covolume-one row lattices is the same as equality in X_4 : any two positively oriented bases of the same lattice differ by an element of $\mathrm{SL}_4(\mathbb{Z})$.

It remains to check that this identification gives the stipulated orbit weights. Write $\mathcal{O}^{\times,+} = \mathcal{O}^{\times} \cap U^+$. The U -stabilizer of a label L is $\mathcal{O}^{\times}$, by its multiplier order. The stabilizer of L together with a sign pattern is therefore $\mathcal{O}^{\times,+}$. This subgroup has finite index in U . For $q = 1$ this follows directly from $\mathcal{O} = R$. For $q > 1$, the kernel of $U \rightarrow (R/qR)^{\times}$ lies in $\mathcal{O}^{\times}$, since both a unit congruent to 1 modulo qR and its inverse belong to $\mathcal{O}$, and imposing positive signs costs at most a further factor of sixteen. The same diagonal-identification argument shows that all periods of the corresponding A_4 -orbit are precisely $\ell(\mathcal{O}^{\times,+})$. They form a full lattice, so the orbit is compact.

Combining a label with a sign produces $16N_{\mathcal{O}}$ finite choices above each I . Each U -orbit of these choices has $[U : \mathcal{O}^{\times,+}]$ elements and parametrizes one A_4 -orbit. On that orbit the quotient of logarithmic Lebesgue measure is its Haar measure, with volume

$$\mathrm{covol}(\ell(\mathcal{O}^{\times,+})) = [\ell(U) : \ell(\mathcal{O}^{\times,+})] R_K = \frac{[U : \mathcal{O}^{\times,+}]}{2} R_K.$$

The last equality uses $\ker(\ell|_U) = \{1, -1\}$ and $-1 \notin \mathcal{O}^{\times,+}$. Consequently the number of distinct orbits times their common volume is

$$\frac{16h_K N_{\mathcal{O}}}{[U : \mathcal{O}^{\times,+}]} \cdot \frac{[U : \mathcal{O}^{\times,+}]}{2} R_K = 8h_K N_{\mathcal{O}} R_K.$$

This proves both the assertion about the descended measure and Equation (2.8), by Equation (2.6).

One can also take $\mathcal{L}_{\mathcal{O}}(I) \times \mathcal{F}_U$ as a fundamental domain for the diagonal action in Equation (2.7): reducing v into $\mathcal{F}_U$ transports L by the same unique unit, outside the usual null boundary set. Counting measure times dv is invariant under this action. These domains have total measure $h_K N_{\mathcal{O}} \mathrm{vol}(\mathcal{F}_U)$, equal to the total volume just computed. Dividing their integral by that quantity proves Equation (2.9). $\square$

3 Ordinary vectors and probability limits

We first control vectors in the lattices themselves. A count near the origin will prevent escape into the cusp, and a count near a vector with no zero coordinate will bound the mass of small neighborhoods in X_4 . The latter bound is the input to the entropy argument in Section 4. We follow the method of [18, Sections 2–5], proving the estimates with constants uniform over primitive totally real quartic fields, their orders, and the orderings of their embeddings. The absence of a quadratic subfield supplies the residue estimate needed in this degree.

3.1 Containment probabilities and unfolding

For a rational prime p , write

$$R_p = \prod_{\mathfrak{p}|p} R_{\mathfrak{p}}.$$

Here $R_{\mathfrak{p}}$ is the integer ring of the completion $K_{\mathfrak{p}}$. Let $v_{\mathfrak{p}}$ be its integer valuation, normalized to be 1 on a uniformizer, and put

$$e_{\mathfrak{p}} = v_{\mathfrak{p}}(p), \quad f_{\mathfrak{p}} = [R/\mathfrak{p} : \mathbb{Z}/p\mathbb{Z}].$$

Thus $N_{\mathfrak{p}} = p^{f_{\mathfrak{p}}}$ and $\sum_{\mathfrak{p}|p} e_{\mathfrak{p}} f_{\mathfrak{p}} = 4$; see [16, Theorem 3.34]. For $p \mid q$ and a tuple $\mathbf{l}_p = (l_{\mathfrak{p}})_{\mathfrak{p}|p}$ of nonnegative integers, call the set of $x \in R_p$ with $v_{\mathfrak{p}}(x_{\mathfrak{p}}) = l_{\mathfrak{p}}$ in every component the valuation shell $\mathbf{l}_p$. Choose x in this shell and define

$$P_p(\mathbf{l}_p) = \mathbb{P}_{t_p \in R_p^\times} \{x \in t_p \mathcal{O}_p\}, \quad (3.1)$$

where the probability is normalized unit Haar measure. The definition is independent of x : the unit group acts transitively on each set with specified component valuations.

For a nonzero integral R -ideal $\mathfrak{a}$, set

$$P(\mathfrak{a}) = \prod_{p|q} P_p((v_{\mathfrak{p}}(\mathfrak{a}))_{\mathfrak{p}|p}). \quad (3.2)$$

An empty product, including the case $q = 1$, is 1. If $0 \neq b \in I$, its image has no zero local component and $(b)I^{-1}$ is integral. In the coordinates $I_p = \xi_p R_p$, the valuations of $\xi_p^{-1}b$ are those of $(b)I^{-1}$. Lemma 2.1 and the independence of the local choices therefore give

$$\mathbb{P}_{L \in \mathcal{L}_{\mathcal{O}}(I)} \{b \in L\} = P((b)I^{-1}). \quad (3.3)$$

For a nonnegative Borel function f on $\mathbb{R}^4$, write

$$E_f(\Lambda) = \sum_{0 \neq x \in \Lambda} f(x).$$

This is a Borel function on X_4 : in a local basis chart it is a countable sum of the Borel functions obtained by evaluating f at the integer linear combinations of that basis. For $z > 0$, define the integral over the surface with absolute coordinate product z by

$$\mathcal{V}_f(z) = \sum_{\omega \in \{\pm 1\}^4} \int_{\mathbb{R}^3} f(\omega_1 e^{u_1}, \omega_2 e^{u_2}, \omega_3 e^{u_3}, \omega_4 z e^{-u_1 - u_2 - u_3}) du_1 du_2 du_3. \quad (3.4)$$

The signs and logarithmic measure are exactly those of T_{∞} .

Proposition 3.1 (Ordinary vector unfolding). *For every nonnegative Borel f on $\mathbb{R}^4$,*

$$\int_{X_4} E_f d\mu_* = \frac{1}{\sqrt{D} \kappa} \sum_{\substack{\mathfrak{a} \subset R \\ \mathfrak{a} \neq 0}} P(\mathfrak{a}) \mathcal{V}_f(N \mathfrak{a}/X). \quad (3.5)$$

Both sides are allowed to be infinite.

Proof. This is the unfolding of [18, Proposition 2.2] with the packet normalization of Proposition 2.2. Average the vector sum over L in Equation (2.9), and use Equation (3.3). The factor $N_{\mathcal{O}}$ cancels, and the left side becomes

$$\frac{1}{h_K \text{vol}(\mathcal{F}_U)} \sum_I \int_{\mathcal{F}_U} \sum_{0 \neq b \in I} P((b)I^{-1}) f(c_I \sigma(b)v) dv.$$

For fixed I , the map $b \mapsto (b)I^{-1}$ takes $I \setminus \{0\}$ onto the integral ideals in the inverse class of I . Each fiber is one free U -orbit. Indeed, two generators of the same principal ideal differ by a unique unit. The coefficient $P((b)I^{-1})$ is constant on that orbit. Summing its unit translates and integrating over $\mathcal{F}_U$ gives

$$\int_{\mathcal{F}_U} \sum_{\epsilon \in U} f(c_I \sigma(\epsilon b)v) dv = \int_{T_\infty} f(c_I \sigma(b)v) dv.$$

The translates tile T_∞ , including its sign components. All changes in the order of integration and summation are justified by Tonelli's theorem.

The absolute product of the coordinates of $c_I \sigma(b)$ is

$$c_I^4 |N_{K/\mathbb{Q}}(b)| = \frac{N((b)I^{-1})}{X}.$$

Translation of the first three logarithmic coordinates and relabeling of the signs therefore identify the last integral with $\mathcal{V}_f(N((b)I^{-1})/X)$, with Jacobian one. Each integral ideal belongs to the inverse class of exactly one chosen I . Equation (2.6) now gives the coefficient in Equation (3.5). $\square$

3.2 The mass and tail of the local weights

The ideal sum in Equation (3.5) need not have simple coefficients at primes dividing q . We will use its exact total mass over valuation shells and a positive moment of their norms. Both statements are the order case of [18, Lemma 3.1]; the argument below explains why they are uniform even when the local order has large index.

For $p \mid q$, put

$$q_p = [R_p : \mathcal{O}_p], \quad N_p(\mathbf{l}_p) = p^{\sum_{\mathfrak{p} \mid p} f_{\mathfrak{p}} l_{\mathfrak{p}}}, \quad Z_p = \prod_{\mathfrak{p} \mid p} (1 - p^{-f_{\mathfrak{p}}})^{-1}.$$

For a collection $\mathbf{l} = (\mathbf{l}_p)_{p \mid q}$, write

$$d(\mathbf{l}) = \prod_{p \mid q} N_p(\mathbf{l}_p), \quad P(\mathbf{l}) = \prod_{p \mid q} P_p(\mathbf{l}_p), \quad Z_q = \prod_{p \mid q} Z_p.$$

Thus $d(\mathbf{l})$ is the norm of the part of an integral ideal supported above primes dividing q , and $\prod_{p \mid q} q_p = q$.

Lemma 3.2 (Local mass and moment). *With the preceding notation,*

$$\sum_{\mathbf{l}} \frac{P(\mathbf{l})}{d(\mathbf{l})} = \frac{Z_q}{q}. \quad (3.6)$$

For every $\eta > 0$ there is a constant C_η , independent of $K, \mathcal{O}$, such that

$$\sum_{\mathbf{l}} P(\mathbf{l}) d(\mathbf{l})^{-1+1/8} \leq C_\eta \frac{q^\eta}{q}. \quad (3.7)$$

In particular there is an absolute constant C for which

$$\sum_{d(\mathbf{l}) > X^{1/8}} \frac{P(\mathbf{l})}{d(\mathbf{l})} \leq \frac{C}{q} X^{-1/128}. \quad (3.8)$$

Proof. Normalize additive Haar measure m_p on R_p by $m_p(R_p) = 1$. In a component $R_{\mathfrak{p}}$, the set of elements of valuation $l_{\mathfrak{p}}$ has measure $(1 - p^{-f_{\mathfrak{p}}})p^{-f_{\mathfrak{p}}l_{\mathfrak{p}}}$. Consequently the shell $\mathbf{l}_p$ in R_p has measure

$$m_p(\text{shell } \mathbf{l}_p) = \frac{1}{Z_p N_p(\mathbf{l}_p)}. \quad (3.9)$$

Its conditional additive measure is invariant under the transitive unit action. It is therefore the image of unit Haar probability. By Equation (3.1), $P_p(\mathbf{l}_p)$ is exactly the fraction of this shell lying in $\mathcal{O}_p$. The elements with a zero component have measure zero, and $m_p(\mathcal{O}_p) = q_p^{-1}$. Summing Equation (3.9) over the intersections with $\mathcal{O}_p$ gives

$$\sum_{\mathbf{l}_p} \frac{P_p(\mathbf{l}_p)}{N_p(\mathbf{l}_p)} = \frac{Z_p}{q_p}.$$

Multiplication over $p \mid q$ proves Equation (3.6).

For the moment, choose x with normalized additive Haar probability on $\mathcal{O}_p$, and let $v_{\mathfrak{p}} = v_{\mathfrak{p}}(x_{\mathfrak{p}})$. The projection $H_{\mathfrak{p}}$ of $\mathcal{O}_p$ into $R_{\mathfrak{p}}$ contains the embedded copy of $\mathbb{Z}_p$, because $1 \in \mathcal{O}_p$. Its induced probability is normalized Haar on $H_{\mathfrak{p}}$. For an integer $k \geq 1$, the image of this copy of $\mathbb{Z}_p$ modulo the ideal of valuation k has $p^{\lceil k/e_{\mathfrak{p}} \rceil}$ elements. Hence the image of $H_{\mathfrak{p}}$ has at least that many elements, and

$$\mathbb{P}_{\mathcal{O}_p}\{v_{\mathfrak{p}} \geq k\} \leq p^{-\lceil k/e_{\mathfrak{p}} \rceil}. \quad (3.10)$$

This uses only the size of the additive image of the projection; it does not require the projections to be independent.

Set

$$j = \max_{\mathfrak{p} \mid p} \left\lceil \frac{v_{\mathfrak{p}}}{e_{\mathfrak{p}}} \right\rceil.$$

If $j \geq a \geq 1$, then some $v_{\mathfrak{p}}$ is at least $e_{\mathfrak{p}}(a - 1) + 1$. Equation (3.10) and a union bound over at most four components give $\mathbb{P}_{\mathcal{O}_p}\{j \geq a\} \leq 4p^{-a}$. On the other hand,

$$N_p(x) = p^{\sum_{\mathfrak{p} \mid p} f_{\mathfrak{p}} v_{\mathfrak{p}}} \leq p^j \sum_{\mathfrak{p} \mid p} e_{\mathfrak{p}} f_{\mathfrak{p}} = p^{4j}.$$

These assertions hold outside the null sets where a component vanishes. The tail formula for the expectation of $p^{j/2}$ now yields

$$\begin{aligned} \mathbb{E}_{\mathcal{O}_p} N_p(x)^{1/8} &\leq \mathbb{E}_{\mathcal{O}_p} p^{j/2} \\ &= 1 + \sum_{a \geq 1} (p^{a/2} - p^{(a-1)/2}) \mathbb{P}_{\mathcal{O}_p}\{j \geq a\} \leq 1 + 4p^{-1/2}. \end{aligned} \quad (3.11)$$

Applying Equation (3.9) once more gives the exact factorization

$$\sum_{\mathbf{l}} P(\mathbf{l}) d(\mathbf{l})^{-1+1/8} = \frac{1}{q} \prod_{p \mid q} \left(Z_p \mathbb{E}_{\mathcal{O}_p} N_p(x)^{1/8} \right).$$

Each factor is at most $(1 - p^{-1})^{-4}(1 + 4p^{-1/2})$. For any fixed $\eta > 0$, this is at most p^{η} at all sufficiently large primes. The product of the factors at the finitely many remaining primes is bounded by a constant depending only on η . Since $\prod_{p \mid q} p \leq q$, this proves Equation (3.7). Finally, take $\eta = 1/128$ there and use $q \leq X$:

$$\sum_{d(\mathbf{l}) > X^{1/8}} \frac{P(\mathbf{l})}{d(\mathbf{l})} \leq X^{-1/64} \sum_{\mathbf{l}} P(\mathbf{l}) d(\mathbf{l})^{-1+1/8} \leq \frac{C}{q} X^{-1/128}.$$

This proves Equation (3.8). $\square$

3.3 Ideal counts relative to the residue

The shell identity supplies the factor q^{-1} . To use it in Equation (3.5), the count of the remaining ideal part must retain the factor κ . Recall

$$a_K(n) = \#\{\mathfrak{a} \subset R : \mathfrak{a} \neq 0, N \mathfrak{a} = n\}, \quad n \geq 1,$$

and let d_4 be the fourfold divisor function. Unique factorization of ideals makes a_K multiplicative. At every rational prime, including a ramified one, its local generating series is

$$\sum_{j \geq 0} a_K(p^j) T^j = \prod_{\mathfrak{p}|p} (1 - T^{f_{\mathfrak{p}}})^{-1}.$$

Each factor is coefficientwise at most $(1 - T)^{-1}$, and there are at most four factors. Hence $a_K(n) \leq d_4(n)$ for every n ; in particular $a_K(p) \leq 4$.

Lemma 3.3 (Residue and prime sums). *There are absolute constants $D_0 > 1$, $h_0 > 0$, $Y_0 > 1$, and $C \geq 1$ with the following properties for every primitive totally real quartic field K .*

1. *If $D \geq D_0$, then*

$$C^{-1} \kappa \leq h \zeta_K(1 + h) \leq C \kappa \quad (0 < h \leq 4/\log D). \quad (3.12)$$

If $D < D_0$, the same inequalities hold for $0 < h \leq h_0$.

2. *One has*

$$\kappa^{-1} \leq C \log(3D). \quad (3.13)$$

3. *Whenever $Y \geq \max\{Y_0, D^{1/4}\}$,*

$$\exp \left(\sum_{p \leq Y} \frac{a_K(p)}{p} \right) \leq C \kappa \log Y. \quad (3.14)$$

The sum in Equation (3.14) is over rational primes.

Proof. We first treat large D , and put $L = \log D$. Stark's zero theorem [25, Lemma 3] gives at most one nontrivial zero of ζ_K in

$$\Re s \geq 1 - \frac{1}{4L}, \quad |\Im s| \leq \frac{1}{4L};$$

any such zero is real and simple. His Lemma 8 [25, Lemma 8], valid also for nonnormal fields, states in degree four that a real zero $\beta \geq 1 - (96L)^{-1}$ would force a quadratic subfield of K . Primitivity excludes that subfield. It follows that every nontrivial zero ρ satisfies

$$|1 - \rho| \geq \frac{c_0}{L}, \quad c_0 = \frac{1}{192}. \quad (3.15)$$

Indeed, a zero in the smaller open disk would lie in Stark's rectangle, would be real, and would be in the forbidden real interval.

For real $s > 1$, the logarithmic derivative of the completed zeta function gives

$$\begin{aligned} S_K(s) &:= \sum_{\rho} \Re \frac{1}{s - \rho} = \frac{1}{s} + \frac{1}{s - 1} + \frac{L}{2} + \frac{\zeta'_K}{\zeta_K}(s) + G(s), \\ G(s) &= 2 \left(\frac{\Gamma'(s/2)}{\Gamma(s/2)} - \log \pi \right); \end{aligned} \quad (3.16)$$

see [25, Equation (9)]. The real zero sum converges and has nonnegative terms, since $\Re \rho \leq 1$. At $s_0 = 1 + L^{-1}$, the Euler product gives $\zeta'_K / \zeta_K(s_0) \leq 0$. The function G is bounded on $[1, 2]$, so for sufficiently large D ,

$$0 \leq S_K(s_0) \ll L.$$

This estimate persists for all $1 < s \leq 1 + 4/L$. To see this term by term, write $\rho = \beta + i\gamma$. Since $\beta \leq 1$,

$$s - \beta \leq 4(s_0 - \beta), \quad |s - \rho| \geq |1 - \rho|.$$

Equation (3.15) consequently gives

$$|s_0 - \rho| \leq |s - \rho| + \frac{3}{L} \leq (1 + 3/c_0)|s - \rho|.$$

Using $\Re(s - \rho)^{-1} = (s - \beta)/|s - \rho|^2$, we obtain

$$\Re \frac{1}{s - \rho} \leq 4(1 + 3/c_0)^2 \Re \frac{1}{s_0 - \rho}.$$

Thus $S_K(s) \ll L$ on the entire interval.

Let $\mathcal{Z}_K(s) = (s - 1)\zeta_K(s)$. Rearranging Equation (3.16) shows that

$$\frac{\mathcal{Z}'_K}{\mathcal{Z}_K}(s) = S_K(s) - \frac{1}{s} - \frac{L}{2} - G(s) = O(L) \quad (1 < s \leq 1 + 4/L).$$

The function $\mathcal{Z}_K$ extends to $s = 1$ with value $\kappa > 0$. Integrating its logarithmic derivative across an interval of length at most $4/L$ proves Equation (3.12), with an absolute constant. At s_0 , the inequality $\zeta_K(s_0) \geq 1$ then gives $\kappa \gg L^{-1}$.

There are only finitely many fields with $D < D_0$, by Hermite's finiteness theorem [16, Theorem 8.43]. For each of them, $h\zeta_K(1 + h)$ tends to the positive residue κ as $h \downarrow 0$. Taking a common sufficiently small h_0 over this finite set proves the remaining comparison, and their positive residues have a common positive lower bound. Enlarging C proves Equation (3.13) for every field.

Choose Y_0 sufficiently large that $1/\log Y_0 \leq h_0$, and let $Y \geq \max\{Y_0, D^{1/4}\}$. For $h = 1/\log Y$, the comparison just proved applies: in the large D case $h \leq 4/\log D$, and in the bounded D case $h \leq h_0$. Positivity in the Euler logarithm gives

$$\sum_{p \leq Y} a_K(p) p^{-1-h} \leq \log \zeta_K(1 + h).$$

Moreover, $a_K(p) \leq 4$, $1 - p^{-h} \leq h \log p$, and the elementary Chebyshev bound followed by partial summation imply

$$\sum_{p \leq Y} a_K(p)(p^{-1} - p^{-1-h}) \leq 4h \sum_{p \leq Y} \frac{\log p}{p} \ll h \log Y = 1.$$

It follows that

$$\sum_{p \leq Y} \frac{a_K(p)}{p} \leq \log \zeta_K(1 + h) + O(1) \leq \log(\kappa \log Y) + O(1),$$

which proves Equation (3.14). There is no upper restriction on Y . This observation is what permits D to stay bounded while the order index, and hence X , grows. $\square$

We also need an estimate for ideals in an interval. The following uniform form of Shiu's bound [23] is the one recorded in [18, Equation (4.2)].

Lemma 3.4. *There are absolute constants C, u_0 such that, for every nonnegative multiplicative function b with $b(n) \leq d_4(n)$, every real $u \geq u_0$, and every $u^{1/2} \leq v \leq u$,*

$$\sum_{u-v < n \leq u} b(n) \leq C \frac{v}{\log u} \exp \left(\sum_{p \leq u} \frac{b(p)}{p} \right). \quad (3.17)$$

Proof. Apply [19, Theorem 1.1] with no excluded residue classes and with its parameter $\beta = 1/4$. Its two growth conditions hold with bounds independent of b :

$$b(p^j) \leq d_4(p^j) = \binom{j+3}{3} \leq 4^j, \quad b(n) \leq d_4(n) \ll_\epsilon n^\epsilon \quad (\epsilon > 0).$$

The theorem's constant and threshold depend only on these growth bounds and on the fixed parameter. Its strict length condition $u^\beta < v$ is satisfied even at $v = u^{1/2}$ once $u > 1$. This gives Equation (3.17), including both endpoints. $\square$

Proposition 3.5 (Weighted ideal intervals). *There is an absolute constant C such that, for each fixed $A \geq 1$ and $c > 0$, there is a threshold $X_0(A, c)$ with the following property. For every $K, \mathcal{O}, \sigma$ under consideration with $X \geq X_0(A, c)$, and all real x, y satisfying*

$$X^{3/4} \leq x \leq AX, \quad cx \leq y \leq x,$$

one has

$$\sum_{\substack{\mathfrak{a} \subset R, \mathfrak{a} \neq 0 \\ x-y < N\mathfrak{a} \leq x}} P(\mathfrak{a}) \leq C \frac{\kappa y}{q}. \quad (3.18)$$

The constant C is independent of A, c ; the threshold may depend on them.

Proof. The range is empty for $c > 1$, so suppose $0 < c \leq 1$. We adapt the separation of the local norm from the remaining ideal in [18, Proposition 4.2]. Put

$$b(n) = a_K(n) \mathbf{1}_{(n, q)=1}.$$

This is nonnegative and multiplicative, and it is at most d_4 . Every integral ideal splits uniquely into its part above the primes dividing q , described by one tuple $\mathbf{l}$, and a part coprime to qR . The left side of Equation (3.18) is therefore

$$\sum_{\mathbf{l}} P(\mathbf{l}) \sum_{(x-y)/d(\mathbf{l}) < n \leq x/d(\mathbf{l})} b(n). \quad (3.19)$$

First take $d = d(\mathbf{l}) \leq X^{1/8}$, and put $u = x/d, v = y/d$. Then

$$u \geq X^{5/8} \geq D^{1/4}, \quad v \geq cu.$$

The second inequality implies $v \geq u^{1/2}$ when X is large in terms of c . The first inequality puts u in the range of Equation (3.14), once X is above an absolute threshold. To remove the primes dividing q from that Euler bound, observe that

$$\log Z_q = \sum_{p|q} \frac{a_K(p)}{p} + O(1), \quad \sum_{\substack{p|q \\ p > u}} \frac{a_K(p)}{p} \ll \frac{\log(2q)}{u} \ll 1. \quad (3.20)$$

For the first identity, expand $\log Z_p = \sum_{\mathfrak{p}|p} \sum_{j \geq 1} p^{-jf_{\mathfrak{p}}}/j$. The terms of order p^{-1} count exactly the primes of residue degree one, and the remaining terms are $O(p^{-2})$, uniformly in the splitting type. Their sum over rational primes is bounded. For the second estimate, use $a_K(p) \leq 4$, at most $\log(2q)/\log 2$ primes dividing q , and $u \geq X^{5/8}$, $q \leq X$. Consequently Equations (3.14) and (3.20) give

$$\exp \left(\sum_{p \leq u} \frac{b(p)}{p} \right) \ll \frac{\kappa \log u}{Z_q}.$$

Lemma 3.4 and Equation (3.6) now bound the contribution of these tuples to Equation (3.19) by

$$C \frac{\kappa y}{Z_q} \sum_{d(\mathbf{l}) \leq X^{1/8}} \frac{P(\mathbf{l})}{d(\mathbf{l})} \leq C \frac{\kappa y}{q}.$$

It remains to control tuples with $d > X^{1/8}$. Those with $d > x$ contribute nothing. For $z \geq 1$, summing the first three factors of the divisor convolution gives

$$\sum_{n \leq z} d_4(n) = \sum_{a_1 a_2 a_3 a_4 \leq z} 1 \leq z \left(\sum_{a \leq z} \frac{1}{a} \right)^3 \leq z(1 + \log z)^3.$$

Use this with $z = x/d$, and then Equation (3.8). The remaining contribution is at most

$$Cx(1 + \log x)^3 \sum_{d(\mathbf{l}) > X^{1/8}} \frac{P(\mathbf{l})}{d(\mathbf{l})} \leq C \frac{x}{q} X^{-1/128} (1 + \log x)^3.$$

For fixed A , once $X \geq A$ the condition $x \leq AX$ gives $\log x \leq 2 \log X$. By Equation (3.13) and $D \leq X^2$, the ratio of this last bound to $\kappa x/q$ is at most

$$CX^{-1/128} (\log(3X))^4 = o(1).$$

This bound is independent of the field and the order. Increase the threshold in terms of A, c until the ratio is at most c . The tail is then at most $\kappa cx/q \leq \kappa y/q$. The constant in the resulting bound has not changed with A or c , proving the proposition. $\square$

3.4 Short vectors and small neighborhoods

We now apply the weighted interval bound to the two vector counts announced at the start of the section. If $f_r = \mathbf{1}_{[-r, r]^4}$, direct integration in Equation (3.4) gives

$$\mathcal{V}_{f_r}(z) = \frac{16}{6} (\log_+(r^4/z))^3. \quad (3.21)$$

Indeed, in a fixed sign component put $s_i = \log r - u_i$ for $1 \leq i \leq 3$. The cube conditions become $s_i \geq 0$ and $\sum_{i=1}^3 s_i \leq \log(r^4/z)$, a simplex of the displayed volume divided by sixteen.

Proposition 3.6 (Tightness). *There is an absolute constant C such that, for every fixed $0 < r < 1$,*

$$\limsup_{X \rightarrow \infty} \mu_* \{ \Lambda : \text{some } 0 \neq v \in \Lambda \text{ satisfies } \|v\|_{\infty} < r \} \leq Cr^4. \quad (3.22)$$

The limit superior ranges over all primitive totally real quartic fields, all their orders, and all orderings of their real embeddings. Consequently, for every sequence of these packet measures $\mu_i = \mu_{\mathcal{O}_i, \sigma_i}$ with $X_i \rightarrow \infty$ and every $\epsilon > 0$, there is a compact $\mathcal{K}_{\epsilon} \subset X_4$ with $\mu_i(\mathcal{K}_{\epsilon}) \geq 1 - \epsilon$ for all sufficiently large i . Such a sequence is tight, and every weak subsequential limit is an A_4 -invariant probability on X_4 .

Proof. The count E_{f_r} dominates the indicator in Equation (3.22). Put $T = r^4 X$ and $T_0 = X^{3/4}$. By Equations (3.5) and (3.21), it suffices to bound

$$\frac{1}{\sqrt{D} \kappa} \sum_{N \mathfrak{a} \leq T} P(\mathfrak{a}) (\log(T/N \mathfrak{a}))^3. \quad (3.23)$$

The sum is empty when $T < 1$. Otherwise its logarithmic factor is at most $(\log X)^3$, since $N \mathfrak{a} \geq 1$ and $r < 1$. Apply Proposition 3.5 at $x = y = T_0$, with $A = c = 1$. The part with $N \mathfrak{a} \leq \min\{T, T_0\}$ is at most

$$C \frac{\kappa T_0}{q} (\log X)^3.$$

After division by $\sqrt{D} \kappa = X \kappa / q$, this contributes $O(X^{-1/4} (\log X)^3) = o(1)$.

If $T > T_0$, put

$$A_{T_0}(t) = \sum_{T_0 < N \mathfrak{a} \leq t} P(\mathfrak{a}) \quad (T_0 \leq t \leq T).$$

The same interval estimate at $x = y = t$ gives $A_{T_0}(t) \leq C \kappa t / q$, since $T \leq X$. Writing each logarithmic cube as an integral and using Tonelli yields

$$\begin{aligned} \sum_{T_0 < N \mathfrak{a} \leq T} P(\mathfrak{a}) (\log(T/N \mathfrak{a}))^3 &= 3 \int_{T_0}^T A_{T_0}(t) (\log(T/t))^2 \frac{dt}{t} \\ &\leq C \frac{\kappa}{q} \int_0^T (\log(T/t))^2 dt \leq C \frac{\kappa T}{q}. \end{aligned}$$

The last integral is finite after the substitution $t = Ts$. This part of Equation (3.23) is therefore $O(r^4)$. If $T \leq T_0$, it is absent. The constants used at $A = c = 1$ are absolute, so the resulting estimate proves Equation (3.22). This is the compactness argument of [18, Proposition 5.1], now using the quartic weighted interval bound.

Mahler's compactness criterion [15, Section 3, Theorem 2] says that

$$\mathcal{K}_r = \{\Lambda \in X_4 : \|v\|_\infty \geq r \text{ for all } 0 \neq v \in \Lambda\}$$

is compact. Choose r so that $Cr^4 < \epsilon/2$. Equation (3.22) then gives $\mu_i(\mathcal{K}_r) \geq 1 - \epsilon$ for all sufficiently large i . Each of the finitely many preceding packet measures is supported on a finite union of compact orbits, by Proposition 2.2. Enlarging the compact set to include those supports proves tightness of the entire sequence. The space X_4 is locally compact, second countable, and metrizable; the weak compactness theorem for tight probabilities therefore gives probability subsequential limits. For $a \in A_4$ and $f \in C_c(X_4)$, the equality

$$\int f(xa) d\mu_i(x) = \int f(x) d\mu_i(x)$$

passes to a weak limit, since both test functions are continuous with compact support. This proves A_4 -invariance. $\square$

Lemma 3.7 (Boxes away from the coordinate hyperplanes). *Let $V_0 \subset (\mathbb{R}^\times)^4$ be compact, and let $b > 0$. There are constants $C, r_0 > 0$, depending only on V_0, b , such that for every fixed $0 < r < r_0$ there is a threshold $X_0(V_0, b, r)$ for which*

$$\int_{X_4} E_{\mathbf{1}_{v+[-br, br]^4}} d\mu_* \leq Cr^4 \quad (v \in V_0, X \geq X_0(V_0, b, r)). \quad (3.24)$$

The threshold and constants are uniform over $K, \mathcal{O}, \sigma$ and over the centers $v \in V_0$.

Proof. The assertion is immediate if V_0 is empty. Otherwise choose $0 < \epsilon \leq M$ such that $\epsilon \leq |v_i| \leq M$ for every $v \in V_0$ and i . For $r_0 < \epsilon/(2b)$, each box in the statement lies in one coordinate sign component. Its first three logarithmic widths satisfy

$$\log \frac{|v_i| + br}{|v_i| - br} \ll_{V_0, b} r.$$

The product map $x \mapsto \prod_i |x_i|$ has uniformly bounded first derivatives on these boxes. If $t_v = \prod_i |v_i|$, it follows from Equation (3.4) that, for one $C_1 \geq 1$,

$$\mathcal{V}_{1_{v+[-br, br]^4}}(z) \leq C_1 r^3 \mathbf{1}_{[t_v - C_1 r, t_v + C_1 r]}(z) \quad (v \in V_0, 0 < r < r_0). \quad (3.25)$$

Let $t_- > 0$ and $t_+ < \infty$ be the minimum and maximum of t_v on V_0 , and shrink r_0 so that $6C_1 r_0 < t_-$. For a fixed r in this range set

$$x = (t_v + 2C_1 r)X, \quad y = 4C_1 rX.$$

The possible ideal norms in Equation (3.25) lie strictly inside $(x - y, x)$. The enlargement avoids any issue with integer endpoints. There is a fixed $A \geq 1$, depending only on V_0, b , such that $x \leq AX$ for all centers and radii under consideration. For sufficiently large X , uniformly in the centers, $x \geq X^{3/4}$. Also $y \leq x$ by the choice of r_0 , and $y \geq cx$ with $c = 4C_1 r/A > 0$, after decreasing r_0 if necessary so that $c \leq 1$.

Proposition 3.5 applies for every fixed r , with this c and the fixed A . Its constant is independent of c , so the sum of the relevant ideal weights is at most $C\kappa rX/q$, with a constant independent of r . Combining this with Equations (3.5) and (3.25), and using $\sqrt{D} = X/q$, proves Equation (3.24). Only the threshold for X depends on the fixed radius. $\square$

For $r > 0$, define the open identity neighborhood

$$B(r) = \{h \in \mathrm{SL}_4(\mathbb{R}) : \max_{i,j} |h_{ij} - \delta_{ij}| < r\}. \quad (3.26)$$

Proposition 3.8 (Local bounds in probability limits). *Let μ be a weak probability limit of packet measures along $X_i \rightarrow \infty$. For every compact $\Omega \subset X_4$, there are constants $C_\Omega, r_\Omega > 0$ such that*

$$\mu(xB(r)) \leq C_\Omega r^4 \quad (x \in \Omega, 0 < r < r_\Omega). \quad (3.27)$$

Proof. The assertion is immediate if Ω is empty, so assume otherwise. We use the passage from vector boxes to ordinary neighborhoods in [18, Proposition 5.2]. First, every full lattice has a vector with no zero coordinate. If $\Lambda = \mathbb{Z}^4 g$, each coordinate of $(1, N, N^2, N^3)g$ is a nonzero polynomial in N : its coefficients are a column of the invertible matrix g . An integer N outside their finitely many roots gives the required vector.

For each $x_0 \in \Omega$, choose a continuous local lift $g(x)$ of nearby lattices to $\mathrm{SL}_4(\mathbb{R})$, and an integer row z for which $zg(x_0)$ has no zero coordinate. On a sufficiently small neighborhood with compact closure in this chart, $zg(x)$ stays bounded and its coordinates stay bounded away from zero. A finite collection of these neighborhoods covers Ω . Consequently we may select, for every $x \in \Omega$, a vector $v_x \in \Lambda_x$ so that all selected vectors lie in one compact set $V_0 \subset (\mathbb{R}^\times)^4$. The selection is used pointwise and need not be measurable.

Choose M with $|(v_x)_i| \leq M$ for all selected vectors, and fix $b > 4M$. If $y = xh$ with $h \in B(r)$, then $v_x h \in \Lambda_y$, and

$$|(v_x h - v_x)_j| \leq \sum_i |(v_x)_i| |h_{ij} - \delta_{ij}| < br.$$

Thus every lattice in $xB(r)$ has a nonzero vector in $v_x + [-br, br]^4$. Lemma 3.7 gives, for each fixed small $r > 0$ and all sufficiently large i ,

$$\mu_i(xB(r)) \leq \int E_{\mathbf{1}_{v_x + [-br, br]^4}} d\mu_i \leq C_\Omega r^4 \quad (x \in \Omega).$$

The constants are common to all x and r in the stated ranges; the index after which this estimate holds may depend on r . The set $xB(r)$ is open in X_4 . For each fixed x, r , Portmanteau's inequality therefore yields

$$\mu(xB(r)) \leq \liminf_{i \rightarrow \infty} \mu_i(xB(r)) \leq C_\Omega r^4.$$

This proves the bound for every r with the same constant. It uses no interchange between a shrinking radius and the packet limit. $\square$

4 The homogeneous obstruction in degree four

For a sequence with $X \rightarrow \infty$, Proposition 3.6 shows that every weak subsequential limit of the packet measures is an A_4 -invariant probability. Proposition 3.8 supplies the compact ball estimate in Equation (3.27). We first turn that estimate into positive entropy in almost every A_4 -ergodic component. Measure classification then makes those components homogeneous. In degree four, the proper homogeneous components that remain have two coordinate blocks of dimension two. We will show that each of them carries an exterior-square lattice vector in one of three fixed planes. The arithmetic part of the proof will exclude these three events.

4.1 Entropy from the compact ball estimate

Fix distinct real numbers $\lambda_1, \dots, \lambda_4$ with sum zero, and set

$$a(t) = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_4 t}), \quad \lambda_a = \min_{i \neq j} |\lambda_i - \lambda_j| > 0.$$

For an $a(\mathbb{R})$ -invariant probability ξ , let $h_\xi(a(1))$ denote the measure-theoretic entropy of right multiplication by $a(1)$. The argument below is the degree-four specialization of the ordinary-ball argument in [18, Proposition 6.1 and Lemma 6.2], which in turn follows the strategy of [5, Section 2.7.1]. We include the argument because its statement for every dominated invariant probability is needed when passing to ergodic components.

Here is the entropy input. For an open identity neighborhood B in $\text{SL}_4(\mathbb{R})$, write

$$B^{(s,t)} = a(-s)Ba(s) \cap a(-t)Ba(t).$$

The following is the specialization to X_4 of the Einsiedler–Lindenstrauss–Michel–Venkatesh tube criterion [4, Corollary 3.3 and Equation (3.4)]. The flow $a(t)$ is $\mathbb{R}$ -diagonalizable, as required there.

Theorem 4.1 (Entropy criterion for two-sided tubes). *Let ξ_i and ξ be $a(\mathbb{R})$ -invariant Borel probabilities on X_4 , with ξ_i converging weak- $*$ to ξ . Suppose that $t_i > 0$ tend to infinity and that $\eta > 0$. Assume that for every compact $\Omega \subset X_4$ there are an open identity neighborhood $B \subset \text{SL}_4(\mathbb{R})$ and a constant C_Ω such that*

$$\xi_i(xB^{(-t_i, t_i)}) \leq C_\Omega e^{-2\eta t_i} \quad (x \in \Omega, i \geq 1).$$

Then $h_\xi(a(1)) \geq \eta$.

In this criterion B may depend on Ω , and the probabilities need not be ergodic under the flow. The requirement that the weak-* limit is a probability is part of the theorem.

Proposition 4.2 (Entropy for dominated probabilities). *Let μ be an A_4 -invariant probability on X_4 satisfying the compact ball bound in Equation (3.27) for every compact set of centers. If ξ is an A_4 -invariant probability with $\xi \leq c\mu$ for some finite c , then*

$$h_\xi(a(1)) \geq \lambda_a/3.$$

Proof. Use the entrywise neighborhoods $B(r)$ defined in Equation (3.26). Fix $0 < b < 1/4$ and put

$$T_t = a(t)B(b)a(-t) \cap a(-t)B(b)a(t), \quad r = e^{-\lambda_a t} \quad (t \geq 0).$$

For $h \in T_t$, the two conjugations give

$$|h_{ii} - 1| < b, \quad |h_{ij}| < be^{-|\lambda_i - \lambda_j|t} \leq br \quad (i \neq j). \quad (4.1)$$

In the determinant expansion, every term except the product of the diagonal entries contains at least two off-diagonal entries. Since $\det h = 1$, Equation (4.1) implies $\prod_i h_{ii} = 1 + O(r^2)$. The diagonal matrix

$$d(h) = \text{diag}(h_{11}, h_{22}, h_{33}, (h_{11}h_{22}h_{33})^{-1})$$

therefore belongs to a fixed compact subset $\mathcal{D}_\infty \subset A_4$, and $d(h)^{-1}h \in B(Cr)$. The constant C here and below may depend on b and the fixed flow. Indeed, the first three diagonal entries of $d(h)^{-1}h$ are one, the fourth differs from one by $O(r^2)$, and all its off-diagonal entries are $O(r)$.

The logarithmic coordinates on A_4 identify $\mathcal{D}_\infty$ with a bounded subset of $\mathbb{R}^3$. A mesh of spacing comparable to r , followed by a fixed enlargement to account for matrix multiplication, gives

$$T_t \subset \bigcup_{j=1}^{N_t} d_j B(Cr), \quad d_j \in \mathcal{D}_\infty, \quad N_t \leq Cr^{-3}. \quad (4.2)$$

For a compact $\Omega \subset X_4$, the centers xd_j lie in the compact set $\Omega\mathcal{D}_\infty$. Domination and Equation (3.27) now imply

$$\xi(xT_t) \leq C_{\Omega,c} r^{-3} r^4 = C_{\Omega,c} e^{-\lambda_a t} \quad (x \in \Omega) \quad (4.3)$$

for all sufficiently large t . Apply Theorem 4.1 to the constant sequence $\xi_i = \xi$, with $t_i = i$, $B = B(b)$, and $\eta = \lambda_a/3$. The identity $B(b)^{(-t,t)} = T_t$ and $2\lambda_a/3 < \lambda_a$ verify its estimate for large i . For the finitely many earlier i , enlarge $C_{\Omega,c}$, using that ξ is a probability. The criterion gives the assertion. $\square$

Corollary 4.3 (Positive entropy in the ergodic components). *Let μ be an A_4 -invariant probability on X_4 satisfying Equation (3.27) on every compact set of centers. Almost every component in its A_4 -ergodic decomposition has positive $a(1)$ -entropy. In particular, this holds for every probability limit in Proposition 3.8.*

Proof. The group $A_4 \simeq \mathbb{R}^3$ is locally compact and second countable, and acts continuously on the standard Borel space X_4 . The ergodic decomposition theorem [7, Theorem 5.2], specialized to invariant probabilities, gives a measurable decomposition

$$\mu = \int_Z \nu_z d\tau(z)$$

into A_4 -invariant, A_4 -ergodic probabilities over a factor fixed by A_4 . Put $T(x) = xa(1)$.

We record why a restriction to the zero-entropy components is measurable. Choose increasing finite Borel partitions $\mathcal{P}_k$ generating the Borel σ -algebra of X_4 . For any T -invariant probability ν , the Kolmogorov–Sinai approximation theorem and subadditivity give

$$h_\nu(T) = \sup_k \inf_{N \geq 1} \frac{1}{N} H_\nu \left(\bigvee_{j=0}^{N-1} T^{-j} \mathcal{P}_k \right),$$

where H_ν is the Shannon entropy of a finite partition. Each finite-partition entropy is a Borel function of ν , so $z \mapsto h_{\nu_z}(T)$ is measurable. For each probability under consideration, complete the standard Borel probability space; it is then a normalized Lebesgue space. The entropy integral formula for an invariant component factor [21, Theorem 9.8] gives

$$h_\mu(T) = \int_Z h_{\nu_z}(T) d\tau(z), \quad (4.4)$$

and the same formula after restricting and normalizing τ on any measurable subset of positive measure. This formula applies because the component factor is fixed by T . It does not require the individual ν_z to be T -ergodic.

If $Z_0 = \{z : h_{\nu_z}(T) = 0\}$ had measure $w > 0$, then

$$\xi = \frac{1}{w} \int_{Z_0} \nu_z d\tau(z)$$

would be an A_4 -invariant probability with $\xi \leq w^{-1}\mu$. The restricted form of Equation (4.4) would give $h_\xi(T) = 0$, contrary to Proposition 4.2. $\square$

4.2 Finite-volume subgroups containing the diagonal group

We next use the measure-classification theorem of Einsiedler–Katok–Lindenstrauss [3, Theorem 1.3; definition in Conjecture 1.1]. We state the exact specialization that supplies the homogeneous measures needed here.

Theorem 4.4 (Positive-entropy measure classification). *For $n \geq 3$, let A_n be the group of positive diagonal matrices in $\mathrm{SL}_n(\mathbb{R})$. Let ν be an A_n -invariant and A_n -ergodic probability on $\mathrm{SL}_n(\mathbb{R})/\mathrm{SL}_n(\mathbb{Z})$, with the left action. If $h_\nu(a_0(1)) > 0$ for some one-parameter subgroup $a_0 : \mathbb{R} \rightarrow A_n$, then there are a closed connected subgroup $L \subset \mathrm{SL}_n(\mathbb{R})$ containing A_n and a point x such that Lx is closed and ν is the L -invariant probability on Lx .*

No ergodicity for the chosen one-parameter subgroup is assumed. For $n = 4$, inversion $\mathrm{SL}_4(\mathbb{Z})g \mapsto g^{-1}\mathrm{SL}_4(\mathbb{Z})$ turns right multiplication by $a(t)$ into left multiplication by $a(-t)$, and entropy is unchanged upon inverting an invertible transformation. Thus, on our right quotient, the theorem says that a positive-entropy A_4 -ergodic probability is invariant probability on an orbit ΛJ , where J is closed and connected, contains A_4 , and the stabilizer of Λ is a lattice in J .

For a partition $\mathcal{I} = \{I_1, \dots, I_s\}$ of $\{1, 2, 3, 4\}$ into nonempty subsets, define

$$J_{\mathcal{I}} = \left\{ g \in \mathrm{SL}_4(\mathbb{R}) : \begin{array}{l} g_{ij} = 0 \text{ if } i, j \text{ belong to different } I_v, \\ \det(g|_{\mathbb{R}^{I_v}}) > 0 \text{ for every } v \end{array} \right\}. \quad (4.5)$$

This is the identity component of the corresponding block-diagonal group. Lindenstrauss and Weiss identified the equal-block phenomenon for orbit closures containing a compact diagonal orbit [13, Theorem 1.3]; their result also enters the proof of [3, Corollary 1.4]. The next lemma gives a direct finite-volume argument and records the rational structure of the blocks.

Lemma 4.5 (Equal blocks and a rational commutant). *Let J be a closed connected subgroup of $\mathrm{SL}_4(\mathbb{R})$ containing A_4 , and let $\Lambda \in X_4$. Suppose that the stabilizer $\Gamma_\Lambda = \{j \in J : \Lambda j = \Lambda\}$ is a lattice in J . Then $J = J_{\mathcal{I}}$ for a partition $\mathcal{I}$ whose blocks all have the same size.*

More precisely, put $W_{\mathbb{Q}} = \mathbb{Q}\Lambda$ and $W_{\mathbb{R}} = W_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{R} = \mathbb{R}^4$. There is a totally real number field $E \subset \mathrm{End}_{\mathbb{Q}}(W_{\mathbb{Q}})$, of degree $s = |\mathcal{I}|$, whose scalar extension to $\mathbb{R}$ is exactly the commutant of J in $\mathrm{End}_{\mathbb{R}}(W_{\mathbb{R}})$. The coordinate blocks $\mathbb{R}^{I_v}$ are the real eigenspaces of E on $W_{\mathbb{R}}$.

Proof. Let $\mathfrak{j}$ be the Lie algebra of J . It contains the trace-zero diagonal algebra and is invariant under diagonal conjugation. The other weight spaces in $\mathfrak{sl}_4(\mathbb{R})$ are the distinct one-dimensional spaces $\mathbb{R}E_{ij}$, $i \neq j$, where E_{ij} is a matrix unit. An invariant subspace is a sum of these weight spaces, so $\mathfrak{j}$ consists of the diagonal algebra and a selection of the root spaces.

The group J is unimodular. One way to see this standard fact is to use the quotient criterion for Haar measures: existence of an invariant quotient measure and triviality of the modular character of the discrete group Γ_Λ imply that the modular character of J is trivial on Γ_Λ . It therefore descends to $\Gamma_\Lambda \backslash J$. Its pushforward of invariant probability would be invariant under translations by its image. A nontrivial positive dilation admits no invariant probability on $\mathbb{R}_{>0}$, as its disjoint dilates of a fundamental interval all have the same mass. Hence the modular character is trivial on J .

Draw a directed edge $i \rightarrow j$ when $E_{ij} \in \mathfrak{j}$. The trace of the adjoint action of $\mathrm{diag}(u_1, \dots, u_4)$, with $\sum_i u_i = 0$, on $\mathfrak{j}$ is the sum of the selected roots $u_i - u_j$. Unimodularity makes this sum zero. Its coefficient at u_i is the outgoing degree of i minus its incoming degree. These coefficients sum to zero, and a linear form vanishing on $\sum_i u_i = 0$ has all coefficients equal. Thus every vertex has equal outgoing and incoming degrees.

Every edge now lies within a strongly connected component. Otherwise the finite directed acyclic graph of components would have a source with an outgoing edge; summing degree balance over that source would give a positive outgoing degree and zero incoming degree. For distinct indices, the bracket identity $[E_{ij}, E_{jk}] = E_{ik}$ makes the edge relation transitive. Each strongly connected component is consequently a complete directed graph. These components form a partition $\mathcal{I}$, and $\mathfrak{j}$ is the Lie algebra of $J_{\mathcal{I}}$. Connected Lie subgroups with the same Lie algebra agree, giving $J = J_{\mathcal{I}}$.

We use Γ_Λ to recover a rational structure on the block scalars. Its elements preserve Λ , hence act rationally on $W_{\mathbb{Q}}$. Let

$$\mathcal{C}_{\mathbb{Q}} = \{M \in \mathrm{End}_{\mathbb{Q}}(W_{\mathbb{Q}}) : M\gamma = \gamma M \text{ for every } \gamma \in \Gamma_\Lambda\}.$$

In a lattice basis the commutation equations have integer coefficients. A finite subcollection of these linear equations defines the same solution space, by finite dimensionality. It follows that $\mathcal{C}_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{R}$ is the real commutant of Γ_Λ .

We claim that this is also the real commutant of J . Let M be a real endomorphism commuting with Γ_Λ . The continuous map

$$\Gamma_\Lambda \backslash J \longrightarrow \mathrm{End}_{\mathbb{R}}(W_{\mathbb{R}}), \quad \Gamma_\Lambda j \longmapsto j^{-1} M j$$

is well-defined. Its pushforward of invariant probability is invariant under conjugation by A_4 . Each off-diagonal coordinate is multiplied by a nontrivial positive dilation under some element of A_4 . A probability on $\mathbb{R}$ invariant under such a dilation is supported at zero: the annuli $\{x : r^k \leq |x| < r^{k+1}\}$, $k \in \mathbb{Z}$, for a fixed $r > 1$, are disjoint dilates covering $\mathbb{R} \setminus \{0\}$. Thus every off-diagonal coordinate of $j^{-1} M j$ is zero almost everywhere. Invariant probability on the transitive space $\Gamma_\Lambda \backslash J$ has full support; continuity makes $j^{-1} M j$ diagonal for every $j \in J$. In particular M is

diagonal. Conjugating by $1 + tE_{ij}$ for i, j in the same block forces its i -th and j -th diagonal entries to agree. Therefore M is scalar on each block, which is exactly the condition that it commute with J . The reverse inclusion is immediate, proving the claim.

It follows that $\mathcal{C}_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{R} \simeq \mathbb{R}^s$. In particular, $\mathcal{C}_{\mathbb{Q}}$ is commutative and has no nonzero nilpotents. The structure theorem for finite reduced commutative algebras over $\mathbb{Q}$ expresses it as a product of number fields, all totally real because its real scalar extension has no complex factor.

This product has only one factor. If $e \in \mathcal{C}_{\mathbb{Q}}$ were an idempotent other than 0 and 1, then $W = e(W_{\mathbb{R}})$ would be a nonempty proper union of coordinate blocks, rational relative to Λ . Every $\gamma \in \Gamma_{\Lambda}$ preserves the full lattice $\Lambda \cap W$ in W , so $|\det(\gamma|_W)| = 1$. The continuous character

$$\chi : J \longrightarrow \mathbb{R}_{>0}, \quad \chi(j) = |\det(j|_W)|$$

is therefore trivial on Γ_{Λ} . It is surjective: if $k = \dim W$, a diagonal matrix may scale the coordinates in W by $e^{(4-k)t}$ and those in its complement by e^{-kt} . Its total determinant is one and its determinant on W is $e^{k(4-k)t}$. The character χ would consequently push the invariant probability on $\Gamma_{\Lambda} \backslash J$ to a probability on $\mathbb{R}_{>0}$ invariant under all multiplicative translations. The disjoint-interval argument above rules this out. Thus $\mathcal{C}_{\mathbb{Q}}$ is a field, which we denote by E .

Since $E \otimes_{\mathbb{Q}} \mathbb{R} \simeq \mathbb{R}^s$, its degree is s . The space $W_{\mathbb{Q}}$ is a vector space over E , and

$$W_{\mathbb{Q}} \otimes_{\mathbb{Q}} \mathbb{R} \simeq \bigoplus_{\tau: E \hookrightarrow \mathbb{R}} W_{\mathbb{Q}} \otimes_{E, \tau} \mathbb{R}.$$

Each summand has dimension $\dim_E W_{\mathbb{Q}} = 4/s$. These summands are the eigenspaces of the block-scalar algebra, namely the coordinate blocks. The blocks therefore have equal size. $\square$

For a positive-entropy homogeneous probability the lemma leaves $J = \mathrm{SL}_4(\mathbb{R})$ or two blocks of size two. Indeed, the only other equal-block possibility is four singleton blocks, giving $J = A_4$. A lattice in $A_4 \simeq \mathbb{R}^3$ has compact quotient, and translations on this torus have zero entropy: for a flat metric they are isometries, so the iterated Bowen metrics equal the original metric and the topological entropy is zero; the variational principle bounds measure entropy by topological entropy.

For a possible two-block limit orbit ΛJ , the quadratic field $E \subset \mathrm{End}_{\mathbb{Q}}(\mathbb{Q}\Lambda)$ has real scalar extension equal to the commutant of J in $\mathrm{End}_{\mathbb{R}}(\mathbb{R}^4)$. The construction supplies no embedding of E into the packet fields K_i , so their primitivity does not by itself exclude this orbit. We will exclude it by proving that packet limits give zero mass to the exterior obstruction below.

4.3 The three exterior obstruction events

Let $e_1, \dots, e_4$ be the coordinate basis of $\mathbb{R}^4$. For $w \in \bigwedge^2 \mathbb{R}^4$, write w_{ij} , $i < j$, for its coefficients in the basis $e_i \wedge e_j$, and define

$$Q(w) = w_{12}w_{34} - w_{13}w_{24} + w_{14}w_{23}. \quad (4.6)$$

The identity

$$w \wedge w = 2Q(w) e_1 \wedge e_2 \wedge e_3 \wedge e_4$$

shows that $Q((\bigwedge^2 g)w) = \det(g)Q(w)$ for $g \in \mathrm{GL}_4(\mathbb{R})$. Expressing a unimodular lattice in a determinant-one basis gives

$$Q(\bigwedge^2 \Lambda) \subset \mathbb{Z} \quad (\Lambda \in X_4). \quad (4.7)$$

This also covers the sign translates in the packet definition: a coordinate reflection of determinant -1 changes only the sign of Q .

Let

$$\mathcal{B} = \{12|34, 13|24, 14|23\}.$$

For $b = ij|kl \in \mathcal{B}$, with the indices within each pair increasing, put

$$V_b = \text{span}_{\mathbb{R}}\{e_i \wedge e_j, e_k \wedge e_l\}. \quad (4.8)$$

The restriction of Q to V_b is, up to sign, the product of its two displayed coordinates. Define

$$\mathcal{E}_b = \left\{ \Lambda \in X_4 : \text{there exists } w \in \bigwedge^2 \Lambda \cap V_b \text{ with } Q(w) \neq 0 \right\}. \quad (4.9)$$

Each $\mathcal{E}_b$ is A_4 -invariant and Borel. For Borel measurability, take a countable cover of X_4 by local charts admitting continuous lattice bases. In a chart, the six integer coefficients of an exterior vector label it continuously. For each label, membership in V_b is a set of continuous coordinate equalities, and $Q(w) \neq 0$ is an open condition. Taking the countable union over labels and charts proves the claim. Invariance follows because A_4 preserves V_b and Q .

Proposition 4.6 (The quartic homogeneous obstruction). *Let ν be an A_4 -invariant, A_4 -ergodic probability on X_4 . Suppose that $h_\nu(a_0(1)) > 0$ for some one-parameter subgroup $a_0 : \mathbb{R} \rightarrow A_4$. Then either $\nu = m_4$, or there exists $b \in \mathcal{B}$ such that $\nu(\mathcal{E}_b) = 1$. In the latter case every lattice Λ' in the homogeneous orbit carrying ν has a vector in $\bigwedge^2 \Lambda' \cap V_b$ with both coordinates nonzero and with nonzero integer Q -value.*

Proof. Theorem 4.4, in the right-quotient form explained above, represents ν as invariant probability on ΛJ for a connected closed subgroup J containing A_4 . By Lemma 4.5 and the positive-entropy assumption, either $J = \text{SL}_4(\mathbb{R})$, which gives $\nu = m_4$, or J has two coordinate blocks of size two.

In the second case write the blocks as $\{i, j\}$ and $\{k, l\}$, and put $b = ij|kl$. The commutant field E in Lemma 4.5 is quadratic. Choose a nonzero $\alpha \in E$ with $\text{Tr}_{E/\mathbb{Q}}(\alpha) = 0$. Its two scalar values on the blocks are c and $-c$, with $c \neq 0$. The rational endomorphism α of $W_{\mathbb{Q}} = \mathbb{Q}\Lambda$ induces a rational endomorphism

$$D_\alpha : \bigwedge^2 W_{\mathbb{Q}} \longrightarrow \bigwedge^2 W_{\mathbb{Q}}, \quad D_\alpha(v \wedge w) = \alpha(v) \wedge w + v \wedge \alpha(w).$$

On the exterior square of the first block it acts by $2c$, and on that of the second by $-2c$. It vanishes on the cross-block summand. Its real image is therefore exactly

$$\bigwedge^2 \mathbb{R}^{\{i,j\}} \oplus \bigwedge^2 \mathbb{R}^{\{k,l\}} = V_b.$$

Because D_α is rational, V_b is a rational subspace relative to $\bigwedge^2 W_{\mathbb{Q}}$.

It follows that $V_b \cap \bigwedge^2 \Lambda$ is a rank-two lattice in V_b . It contains a vector off both coordinate axes: take two independent lattice vectors; if neither is already off the axes, their sum is. For this vector $Q(w) \neq 0$, and Equation (4.7) makes its value a nonzero integer. Every element of J preserves each of the two exterior lines, multiplying its coefficient by the nonzero determinant of the corresponding block. The translated vector therefore has both coordinates nonzero at every lattice of ΛJ . This proves $\nu(\mathcal{E}_b) = 1$. $\square$

Corollary 4.3 and Proposition 4.6 reduce identification of a packet probability limit μ to proving $\mu(\mathcal{E}_b) = 0$ for all three $b \in \mathcal{B}$. Equation (4.7) lets us test those events one nonzero integer Q -value at a time. The next sections establish the exterior-vector estimate used for this exclusion.

5 The cubic resolvent and exterior multiplicities

The obstruction in Section 4 is detected by an exterior vector concentrated in one pair of opposite coordinates. We shall count such vectors through their three signed opposite products. After packet normalization these products form a triple in a cubic field, and concentration near one exterior plane confines two of them to a small rectangle. Our goal is a uniform count in that rectangle. We first define the count and reduce its multiplicities to local densities. We then construct the integral quadratic structure that will control those densities in the next two sections.

5.1 The three products and the counting problem

Let $\widetilde{K} \subset \mathbb{R}$ be the normal closure of K , and let $G = \text{Gal}(\widetilde{K}/\mathbb{Q})$ act on the four ordered embeddings of K . This permutation action is primitive: a nontrivial block would give a field strictly between $\mathbb{Q}$ and K . A transitive subgroup of S_4 has order 4, 8, 12, or 24. A subgroup of order 4 or 8 is contained in a Sylow 2-subgroup, which preserves a partition into two pairs. It follows that

$$G = \mathfrak{A}_4 \quad \text{or} \quad G = S_4.$$

Both groups act transitively on the six two-element subsets of $\{1, 2, 3, 4\}$ and on the set of pair partitions

$$\mathcal{B} = \{12|34, 13|24, 14|23\}.$$

Define

$$F = \left\{ (z_b)_{b \in \mathcal{B}} \in \widetilde{K}^{\mathcal{B}} : g(z_b) = z_{gb} \text{ for all } g \in G, b \in \mathcal{B} \right\}. \quad (5.1)$$

Evaluation at any partition identifies this algebra with the fixed field of its stabilizer. Thus F is a totally real cubic field, and its three real embeddings are the coordinates in Equation (5.1). In particular, $\text{Tr}_{F/\mathbb{Q}} z = \sum_b z_b$. This is the classical cubic resolvent from the action on pair partitions; for its relation to cubic resolvent rings, see [1, Section 2.3]. We write $R_F = \mathcal{O}_F$ and $D_F = |\text{Disc}(F)|$.

Put

$$V = \bigwedge_{\mathbb{Q}}^2 K, \quad L_V = \bigwedge_{\mathbb{Z}}^2 \mathcal{O}.$$

For $w = \sum_r x_r \wedge y_r \in V$, its embedding coordinates are

$$w_{ij} = \sum_r (\sigma_i(x_r) \sigma_j(y_r) - \sigma_j(x_r) \sigma_i(y_r)) \quad (i < j), \quad w_{ji} = -w_{ij}.$$

Choose a basis $\beta_1, \dots, \beta_4$ of $\mathcal{O}$ whose real embedding matrix $A_{\mathcal{O}} = (\sigma_i(\beta_j))$ has positive determinant. Write $\delta_{\mathcal{O}} = \det A_{\mathcal{O}} \in \widetilde{K}$. Its selected real value is $X = q\sqrt{D}$, and

$$\delta_{\mathcal{O}}^2 = \text{Disc}(\mathcal{O}) = X^2. \quad (5.2)$$

Define the quadratic expression

$$\Psi(w) = \delta_{\mathcal{O}}^{-1} (w_{12}w_{34}, -w_{13}w_{24}, w_{14}w_{23}). \quad (5.3)$$

Lemma 5.1 (The product map). *The expression in Equation (5.3) defines a rational quadratic map $\Psi : V \rightarrow F$. In the exterior basis of L_V induced by (β_i) , write $w = \sum_{i < j} t_{ij} \beta_i \wedge \beta_j$. Then*

$$Q_V(w) := \text{Tr}_{F/\mathbb{Q}} \Psi(w) = t_{12}t_{34} - t_{13}t_{24} + t_{14}t_{23}. \quad (5.4)$$

If $w \in V$ is nonzero, all six w_{ij} , and hence all three components of $\Psi(w)$, are nonzero. For every extension $k/\mathbb{Q}$, $a \in (K \otimes_{\mathbb{Q}} k)^{\times}$, $c \in k$, and $w \in V \otimes_{\mathbb{Q}} k$,

$$\Psi(c(\bigwedge^2 a)w) = c^2 N_{K \otimes k/k}(a) \Psi(w), \quad (5.5)$$

where $\bigwedge^2 a$ is the exterior square of multiplication by a .

Proof. An element $g \in G$ permutes the rows of $A_{\mathcal{O}}$, so it multiplies $\delta_{\mathcal{O}}$ by the sign of that permutation. Permuting the four indices in the three signed products in Equation (5.3) permutes the partitions and introduces the same common sign. Division by $\delta_{\mathcal{O}}$ gives precisely the equivariance in Equation (5.1). This is an identity of quadratic polynomials in the rational coordinates of w , and therefore defines a rational quadratic map to F .

The Pfaffian identity for the exterior square of a four-dimensional linear map is

$$Q((\bigwedge^2 A_{\mathcal{O}})t) = \det(A_{\mathcal{O}})Q(t), \quad Q(t) = t_{12}t_{34} - t_{13}t_{24} + t_{14}t_{23}.$$

It follows either by expanding the six minors or by evaluating $\frac{1}{2}t \wedge t$ in the top exterior power. This proves Equation (5.4). If a coordinate of a rational w vanishes, Galois transitivity on the unordered pairs makes all six coordinates vanish. The exterior embedding map is injective, so $w = 0$. Finally, multiplication by a multiplies w_{ij} by $a_i a_j$ after splitting, where a_i are its four embeddings. Each opposite product is multiplied by $\prod_i a_i = N(a)$, and the scalar c contributes c^2 . This proves Equation (5.5) after splitting and hence over k . $\square$

We relate these products to the packet model of Section 2. Let I be one of its maximal-order ideal-class representatives, let $L \subset I$ be one of its local labels, and let $v = (v_i) \in T_{\infty}$. If $w \in \bigwedge_{\mathbb{Z}}^2 L$, the corresponding exterior vector in the normalized lattice has coordinates

$$\hat{w}_{ij} = (X N I)^{-1/2} w_{ij} v_i v_j.$$

Since $\prod_i |v_i| = 1$, the number $\epsilon(v) = \prod_i v_i$ is 1 or -1 . Thus

$$(\hat{w}_{12}\hat{w}_{34}, -\hat{w}_{13}\hat{w}_{24}, \hat{w}_{14}\hat{w}_{23}) = \epsilon(v) z(I, w), \quad z(I, w) := \frac{\Psi(w)}{N I}. \quad (5.6)$$

In particular $Q(\hat{w}) = \epsilon(v) \text{Tr } z(I, w)$. This records exactly how the diagonal signs act on the products.

Lemma 5.1 and the normalization above show that every lattice in a packet lies outside each $\mathcal{E}_b$. To prove that weak limits give these events zero mass, we need quantitative control of exterior vectors near V_b ; the count below supplies the arithmetic input for that control.

For a rational prime p , fix a generator $I_p = \xi_p R_p$ as in the packet model, and give $R_p^{\times}$ its Haar probability. For $0 \neq w \in \bigwedge_{\mathbb{Z}}^2 I$, define

$$P_p(I, w) = \int_{R_p^{\times}} \mathbf{1}_{\bigwedge_{\mathbb{Z}_p}^2 (\xi_p t \mathcal{O}_p)}(w) d^{\times} t. \quad (5.7)$$

Changing ξ_p by an R_p -unit leaves this probability unchanged. For $p \nmid q$, it is 1. Hence $\prod_p P_p(I, w)$ is a finite product. Multiplication by U^+ preserves these probabilities by translation of the local unit variables. For $e \in U^+$, the algebraic norm $N_{K/\mathbb{Q}}(e) = 1$, so Equation (5.5) also shows that $z(I, w)$ is unchanged. We use U^+ to unfold the positive real parameters on each choice of signs.

Its action on nonzero exterior vectors is free: if a totally positive unit e fixes w , then $e_i e_j = 1$ for every pair by Lemma 5.1; the four positive e_i must all be 1.

For $z \in F$, define the nonnegative weighted multiplicity

$$M(z) = \sum_{\substack{I, w \in (\bigwedge_{\mathbb{Z}}^2 I \setminus \{0\})/U^+ \\ z(I, w) = z}} \prod_p P_p(I, w). \quad (5.8)$$

The sum is initially allowed to have value $+\infty$; the local bounds below prove its finiteness on the nonzero trace slices used here. It depends on $K, \mathcal{O}$, the ordering of the embeddings, and the chosen system of maximal-order ideal representatives, which we suppress from the notation. In fact replacing I by aI and transporting w by $\bigwedge^2 a$ multiplies that class's products by $\text{sgn}(\text{N}_{K/\mathbb{Q}} a)$, since $\text{N}(aI) = |\text{N}_{K/\mathbb{Q}} a| \text{N} I$. We keep the representative system fixed. A term of positive weight admits a local label at every prime. These labels patch to a global L , and local membership implies $w \in \bigwedge_{\mathbb{Z}}^2 L$. Equation (5.6), with positive signs, then shows that $\text{Tr } z(I, w)$ is the integral value of Q on a unimodular exterior lattice. Thus

$$M(z) > 0 \implies z \in F^\times, \quad \text{Tr}_{F/\mathbb{Q}} z \in \mathbb{Z}. \quad (5.9)$$

For $m \in \mathbb{Z} \setminus \{0\}$, put

$$\mathcal{H}_m = \{(y_b)_{b \in \mathcal{B}} \in \mathbb{R}^{\mathcal{B}} : \sum_b y_b = m\}.$$

A rectangle in $\mathcal{H}_m$ means the inverse image of a product of two intervals under any two distinct partition coordinates. Its area is the product of those interval lengths. The following estimate is the arithmetic input needed to exclude the obstruction.

Theorem 5.2 (Planar exterior count). *There is an absolute $\tau > 0$ with the following property. Fix $m \in \mathbb{Z} \setminus \{0\}$ and a bounded set $\Omega \subset \mathcal{H}_m$. For every primitive totally real quartic field K , every order $\mathcal{O} \subset K$, every ordering of its real embeddings, and every chosen system of fractional $\mathcal{R}$ -ideal representatives for the ordinary classes, let M be defined by Equation (5.8). For all sufficiently large X , every rectangle $\mathcal{R} \subset \Omega$ whose two side lengths are at least $X^{-\tau}$ satisfies*

$$\sum_{\substack{z \in F, \text{Tr } z = m \\ z \in \mathcal{R}}} M(z) \ll_{m, \Omega} \kappa \sqrt{D} \text{area}(\mathcal{R}). \quad (5.10)$$

The implied constant and the threshold for X are uniform in the field, the order, the embedding ordering, the ideal representative system, the position of the rectangle, and its side lengths and aspect ratio.

Here and below we identify $z \in F$ with its real triple. To see the scale of this statement, suppose $Q(\hat{w}) = m$ and the four coordinates outside 12|34 have absolute value less than r . The signed products of $\hat{w}$ then have the form $(m - y_2 - y_3, y_2, y_3)$ with $|y_2|, |y_3| < r^2$. Their region has area $O(r^4)$. Equation (5.6) allows either trace m or $-m$ for $z(I, w)$, so the theorem is used at both fixed traces. Sections 6–8 prove Theorem 5.2; the remainder of this section supplies its local and algebraic inputs.

5.2 The local product fibers

Fix $m \neq 0$ for the rest of the section. Choose once a sufficiently large absolute p_0 , and put

$$S = \{p : p \mid Dqm\} \cup \{p : p \leq p_0\}. \quad (5.11)$$

For fixed m , $\prod_{p \in S} p \ll_m Dq \leq X^2$. At primes in S we shall keep the exact additive density of the products. Outside S , integral valuations give a simpler bound.

The following comparison explains what a fiber remembers about an exterior vector. For $k = \mathbb{Q}$ or $k = \mathbb{Q}_p$, put $K_k = K \otimes_{\mathbb{Q}} k$ and $V_k = V \otimes_{\mathbb{Q}} k$, and take embedding coordinates over a splitting field of the etale algebra K_k .

Lemma 5.3 (Comparison of two product fibers). *Let $w, w' \in V_k$ have all six embedding coordinates nonzero. Suppose that their coordinate ratios $r_{ij} = w'_{ij}/w_{ij}$ satisfy*

$$r_{12}r_{34} = r_{13}r_{24} = r_{14}r_{23} = s \in k^\times.$$

There exist $a \in K_k^\times$ and $c \in k^\times$ such that

$$w' = c(\bigwedge^2 a)w, \quad c^2 \mathrm{N}_{K_k/k}(a) = s.$$

The pair (a, c) is unique up to $(a, c) \mapsto (da, d^{-2}c)$, $d \in k^\times$.

Proof. Over the splitting field set

$$c_0 = \frac{r_{12}r_{13}}{r_{23}}, \quad (a_1, a_2, a_3, a_4) = \left(1, \frac{r_{23}}{r_{13}}, \frac{r_{23}}{r_{12}}, \frac{r_{14}r_{23}}{r_{12}r_{13}}\right).$$

The three equal opposite products give $r_{ij} = c_0 a_i a_j$ for every pair. If two such factorizations are divided, all products of two of the four coordinatewise quotients are equal. Comparing pairs that share one index shows that the four quotients are equal. Hence the projective tuple $[a_1 : a_2 : a_3 : a_4]$ is uniquely determined.

The ratios r_{ij} are Galois-equivariant without alternating signs. Uniqueness makes the projective tuple equivariant for the permutation action on the four embeddings. Express it in a k -basis of K_k . The resulting projective coefficient vector is fixed by the usual Galois action. Dividing by one nonzero coefficient makes every coefficient fixed, hence in k . It gives $a \in K_k^\times$, since all four embedding components are nonzero. The remaining coefficient c in $r_{ij} = c a_i a_j$ is Galois-fixed as well. This proves existence over k , including when K_k is a product of nonsplit fields. The projective uniqueness proves the stated ambiguity. Multiplying two opposite equalities gives $c^2 \mathrm{N}(a) = s$. $\square$

For a prime p , $F_p^\times$ denotes the elements nonzero in every field factor of F_p . In the local formulas below, $\mathrm{N}(a)$ denotes the algebraic norm $\mathrm{N}_{K_p/\mathbb{Q}_p}(a)$. For a global pair in Equation (5.8), choose a representative $0 \neq w \in \bigwedge_{\mathbb{Z}}^2 I$ of its U^+ -orbit and put

$$x = (\bigwedge^2 \xi_p)^{-1} w, \quad u = \frac{\mathrm{N}_{K_p/\mathbb{Q}_p}(\xi_p)}{\mathrm{N} I}.$$

The valuations of the numerator and denominator in u are equal: both are the norm valuation of I_p . Thus $u \in \mathbb{Z}_p^\times$, and Equation (5.5) gives $u\Psi(x) = z(I, w)$. Inverting the unit variable in Equation (5.7) gives

$$P_p(I, w) = \Pr_{a \in R_p^\times} [(\bigwedge^2 a)x \in L_{V,p}]. \quad (5.12)$$

These representatives lead us to consider, for arbitrary $z \in F_p^\times$, the full local fiber

$$\mathcal{F}_p(z) = \{(x, u) \in V_p \times \mathbb{Z}_p^\times : u\Psi(x) = z\}.$$

Define the abelian local group and its unit subgroup by

$$\begin{aligned}\mathcal{G}_p &= \{(a, c) \in K_p^\times \times \mathbb{Q}_p^\times : c^2 N(a) \in \mathbb{Z}_p^\times\} / \{(d, d^{-2}) : d \in \mathbb{Q}_p^\times\}, \\ \mathcal{U}_p &= \text{image}(R_p^\times \times \mathbb{Z}_p^\times) \subset \mathcal{G}_p.\end{aligned}\tag{5.13}$$

The quotient in the first line means that d acts by $(a, c) \mapsto (da, d^{-2}c)$. The action on the fiber is

$$(a, c) \cdot (x, u) = \left(c \left(\bigwedge^2 a \right) x, \frac{u}{c^2 N(a)} \right).\tag{5.14}$$

Lemma 5.3, with $s = u/u'$, shows that this action is free and transitive when the fiber is nonempty. The explicit ratios in its proof also give analytic local inverses, so this is an analytic identification of the fiber with $\mathcal{G}_p$. The group $\mathcal{U}_p$ is compact and open. The kernel of its unit parametrization is exactly $d \in \mathbb{Z}_p^\times$: if both a and da are R_p -units, then $v_p(d) = 0$. In particular this assertion holds at ramified primes.

Choose a reference in a nonempty fiber and use it to identify the fiber with $\mathcal{G}_p$. For a coset $\mathfrak{c} \in \mathcal{G}_p/\mathcal{U}_p$, let $\alpha_p(\mathfrak{c}; z)$ be the fraction of that coset for which $x \in L_{V,p}$, using Haar probability on $\mathcal{U}_p$. We call the coset accepted when $\alpha_p(\mathfrak{c}; z) > 0$. The fraction is independent of the representative of the coset. The sum of these fractions is independent of the reference. For an empty fiber we declare this sum to be zero.

A scalar $c \in \mathbb{Z}_p^\times$ preserves $L_{V,p}$. The pushforward of the product Haar probability on $R_p^\times \times \mathbb{Z}_p^\times$ is Haar probability on $\mathcal{U}_p$. Hence the representative constructed from (I, w) occupies a coset $\mathfrak{c}$ for which $P_p(I, w) = \alpha_p(\mathfrak{c}; z(I, w))$.

We now measure the sum over all local cosets. Fix an additive character ψ_p of $\mathbb{Q}_p$ with conductor $\mathbb{Z}_p$, and let dz be the self-dual Haar measure on F_p for the trace pairing. Thus

$$\text{vol}_{dz}(R_{F,p}) = |D_F|_p^{1/2}.\tag{5.15}$$

For every p , consider the law of the random variable

$$u\Psi(x), \quad x \text{ uniform additive in } L_{V,p}, \quad u \text{ uniform in } \mathbb{Z}_p^\times,\tag{5.16}$$

with the two choices independent. The next lemma proves that this law has a density with respect to dz ; we denote its locally constant version on $F_p^\times$ by $f_p^*(z)$. Only the densities for $p \in S$ will enter the multiplicity bound.

Lemma 5.4 (Local density and disintegration). *The law in Equation (5.16) has a density, with a locally constant version on $F_p^\times$. For every $z \in F_p^\times$, only finitely many cosets are accepted, and their fractions satisfy the exact identity*

$$\sum_{\mathfrak{c} \in \mathcal{G}_p/\mathcal{U}_p} \alpha_p(\mathfrak{c}; z) = |D|_p^{-1/2} \frac{1 - p^{-1}}{\prod_{\mathfrak{p}|p, \mathfrak{p} \subset R} (1 - N(\mathfrak{p})^{-1})} f_p^*(z).\tag{5.17}$$

The sum and the density are both zero if $\mathcal{F}_p(z)$ is empty. The identity holds at every prime, including primes dividing q and the small ramified primes.

Proof. Let $\Phi_p(x, u) = u\Psi(x)$. If $\Phi_p(x, u) \in F_p^\times$, all six split coordinates of x are nonzero. After splitting, the derivative in x is the derivative of three products on three disjoint two-dimensional planes. It has rank three, and therefore has rank three over $\mathbb{Q}_p$. The map is a submersion there.

The remaining input set is a finite union of proper algebraic zero loci and has additive measure zero. Local change of variables gives an absolutely continuous pushforward on the submersion locus, hence a density for the full law. To obtain a locally constant version at a fixed $z \in F_p^\times$, cover its inverse image in the compact set $L_{V,p} \times \mathbb{Z}_p^\times$ by finitely many submersion charts, refined to disjoint compact open pieces. The compact complement has image avoiding a sufficiently small neighborhood of z . On that neighborhood each chart contributes a locally constant density. This also treats an empty inverse image.

Give V_p its additive measure for which $L_{V,p}$ has mass one, before restricting to that lattice. We first compute the fiber form for the product input measure after disintegration against dz , and then the mass of $\mathcal{U}_p$ for that form. In a $\mathbb{Q}_p$ submersion chart, the relative absolute four-form is obtained by evaluating the input top form on four fiber basis vectors and three lifts of a target basis, then dividing by the dz -form on that target basis. We may compute this identity after extension to a splitting field, still evaluating on $\mathbb{Q}_p$ tangent vectors and using the fixed extension of $|p|_p = p^{-1}$. We use the same interpretation for the scalar quotient below.

Over a splitting field, the exterior embedding matrix has determinant $\delta_{\mathcal{O}}^3$. Consequently its measure form in the six embedding coordinates is

$$|\delta_{\mathcal{O}}|_p^{-3} \left| \bigwedge_{i < j} dx_{ij} \right|.$$

Likewise dz is the absolute embedding-coordinate form $|\bigwedge_{b \in \mathcal{B}} dz_b|$. To verify the latter normalization, evaluate this form on an $R_{F,p}$ -basis. The square of its embedding determinant is the discriminant, so its absolute value is $|D_F|_p^{1/2}$, as in Equation (5.15).

After splitting, compute the relative form by retaining $x_{12}, x_{13}, x_{14}, u$ and substituting the three product equations for x_{34}, x_{24}, x_{23} . The three change-of-variable factors are $|\delta_{\mathcal{O}}/(ux_{12})|_p$, $|\delta_{\mathcal{O}}/(ux_{13})|_p$, and $|\delta_{\mathcal{O}}/(ux_{14})|_p$. They cancel $|\delta_{\mathcal{O}}|_p^{-3}$, since u is a unit. Haar probability on $\mathbb{Z}_p^\times$ is $(1 - p^{-1})^{-1} |du/u|$. The resulting fiber form is

$$\frac{1}{1 - p^{-1}} \left| \frac{dx_{12}}{x_{12}} \wedge \frac{dx_{13}}{x_{13}} \wedge \frac{dx_{14}}{x_{14}} \wedge \frac{du}{u} \right|. \quad (5.18)$$

The fiber action preserves this measure. Indeed its determinant on V_p is $c^6 N(a)^3 = (c^2 N(a))^3$, of absolute value one, and it multiplies u by a unit while fixing z .

It remains to compute the mass of $\mathcal{U}_p$ in Equation (5.18). Put $\ell_i = da_i/a_i$ and $s = dc/c$ over the splitting field. The pullbacks of its four logarithmic differentials under Equation (5.14) are

$$\ell_1 + \ell_2 + s, \quad \ell_1 + \ell_3 + s, \quad \ell_1 + \ell_4 + s, \quad -\ell_1 - \ell_2 - \ell_3 - \ell_4 - 2s.$$

The vertical scalar direction d has vector $(1, 1, 1, 1, -2)$. Taking the first ℓ_i as its coordinate and using the slice $\ell_1 = 0$, the matrix on the four remaining differentials is

$$\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ -1 & -1 & -1 & -2 \end{pmatrix}, \quad \det = 1.$$

The vertical coordinate has coefficient 1. Equivalently, the wedge of the four displayed pullbacks is the contraction of $(\bigwedge_i \ell_i) \wedge s$ by the vertical vector, up to sign. Evaluating this identity on four $\mathbb{Q}_p$ tangent lifts of a quotient basis gives the same absolute quotient form for nonsplit K_p . On a patch

where a $\mathbb{Q}_p$ -linear coefficient $\lambda(a)$ is nonzero, normalizing it to its value at a reference point gives a genuine $\mathbb{Q}_p$ section; after shrinking a unit-group patch, the normalizing scalar is a unit. Since $\lambda(da) = d\lambda(a)$, the vertical logarithmic derivative is 1. Thus the contraction is the quotient of the parameter form by $|dd/d|$, with quotient Jacobian 1, including at $p = 2$.

For an R_p -basis, the multiplicative embedding form on $K_p^\times$ is the additive basis form multiplied by its embedding determinant and divided by $N(a)$. On $R_p^\times$ the latter is a unit. The mass of this unit set is consequently

$$|D|_p^{1/2} \prod_{\mathfrak{p}|p, \mathfrak{p} \subset R} (1 - N(\mathfrak{p})^{-1}).$$

The c -unit mass and the scalar-kernel d -unit mass are both $1 - p^{-1}$, and cancel when taking the quotient. Haar integration on the compact unit groups assembles the local sections. Including the factor $(1 - p^{-1})^{-1}$ in Equation (5.18), the mass of each $\mathcal{U}_p$ -coset is

$$\frac{|D|_p^{1/2}}{1 - p^{-1}} \prod_{\mathfrak{p}|p, \mathfrak{p} \subset R} (1 - N(\mathfrak{p})^{-1}). \quad (5.19)$$

This calculation uses the actual image of the unit group, so it requires no surjectivity assertion about the norm on units.

The accepted part of the fiber is the inverse image of z in the compact set $L_{V,p} \times \mathbb{Z}_p^\times$. Since $\mathcal{U}_p$ is open, this compact set meets only finitely many of its cosets. Integrating its indicator in the fiber multiplies the mass in Equation (5.19) by $\sum_{\mathfrak{c}} \alpha_p(\mathfrak{c}; z)$. The result is the density $f_p^*(z)$. Solving for the sum proves Equation (5.17). $\square$

In particular the order index has disappeared from the prefactor in Equation (5.17): its contribution through $\delta_{\mathcal{O}}$ cancels in the three products. Its effect remains inside the density of the lattice $L_{V,p}$. The discriminant in the prefactor is D because the translating parameters are the maximal-order units $R_p^\times$.

5.3 Global multiplicity and the unramified factors

The local identity counts every local coset. We next bound how many global pairs can occupy one assignment of those cosets.

Lemma 5.5 (Global pairs in one assignment of cosets). *Fix $z \in F^\times$. Choose a reference in each nonempty local fiber $\mathcal{F}_p(z)$. For any assignment $\mathfrak{c}_p \in \mathcal{G}_p/\mathcal{U}_p$, at most 32 pairs*

$$I, \quad w \in \left(\bigwedge_{\mathbb{Z}}^2 I \setminus \{0\} \right) / U^+, \quad z(I, w) = z$$

occupy the assigned coset at every prime.

Proof. For two such pairs, Lemma 5.3 over $\mathbb{Q}$ gives $w' = c(\bigwedge^2 a)w$ with $a \in K^\times$, $c \in \mathbb{Q}^\times$, since the common opposite ratio is $N(I')/N(I)$. Relative to the local generators ξ_p, ξ'_p , their local comparison is represented by $(a\xi_p/\xi'_p, c)$. Equality of the local cosets means that some $d_p \in \mathbb{Q}_p^\times$ satisfies

$$d_p a \xi_p / \xi'_p \in R_p^\times, \quad d_p^{-2} c \in \mathbb{Z}_p^\times.$$

In particular $v_p(c) = 2v_p(d_p)$ is even at every finite prime. Therefore $c = \epsilon s^2$ with $s \in \mathbb{Q}^\times$ and $\epsilon \in \{1, -1\}$. Absorbing s into a makes $a\xi_p/\xi'_p$ a unit for every p . Hence $I' = aI$ as fractional

maximal-order ideals. The chosen representatives of the ordinary ideal classes then give $I' = I$ and $a \in R^\times = U$.

There are at most $[U : U^+] \leq 16$ possibilities for the unit action modulo U^+ , distinguished by the four real signs, and two possibilities for ϵ . This proves the bound 32. $\square$

At $p \notin S$, both local algebras are unramified, $\mathcal{O}_p = R_p$, and $\delta_{\mathcal{O}}$ is a unit in a splitting field. Let φ_p be Frobenius acting on the four embeddings and on $\mathcal{B}$. Its orbits on $\mathcal{B}$ correspond to primes $\mathfrak{q} \mid p$ of F , of residue degrees $f_{\mathfrak{q}}$. The map from two-element subsets to partitions sends ij to $ij|kl$. Above a Frobenius orbit of partitions its pair labels form either two orbits or one. Define

$$\epsilon_{\mathfrak{q}} = \begin{cases} 1, & \text{if the pair labels form two orbits,} \\ -1, & \text{if they form one orbit,} \end{cases} \quad b_{\mathfrak{q}}(j) = \begin{cases} j+1, & \epsilon_{\mathfrak{q}} = 1, \\ \mathbf{1}_{\{2|j\}}, & \epsilon_{\mathfrak{q}} = -1 \end{cases} = \sum_{r=0}^j \epsilon_{\mathfrak{q}}^r.$$

For $z \in R_{F,p} \cap F_p^\times$, put

$$j_{\mathfrak{q}} = v_{\mathfrak{q}}(z), \quad b_p(z) = \prod_{\mathfrak{q} \mid p} b_{\mathfrak{q}}(j_{\mathfrak{q}}), \quad n_p(z) = \max_{\mathfrak{q} \mid p} j_{\mathfrak{q}}. \quad (5.20)$$

Here $v_{\mathfrak{q}}$ is the integral uniformizer valuation; it has $v_{\mathfrak{q}}(p) = 1$ at these primes. For probability statements one may set $n_p = \infty$, $b_p = 0$ when a component vanishes, a set of Haar measure zero.

Lemma 5.6 (Unramified local data). *Let $p \notin S$. A local fiber with an accepted coset has $z \in R_{F,p}$, and for $z \in R_{F,p} \cap F_p^\times$,*

$$\sum_{\mathfrak{c} \in \mathfrak{G}_p / \mathcal{U}_p} \alpha_p(\mathfrak{c}; z) \leq b_p(z). \quad (5.21)$$

One has $b_p = 1$ when $n_p = 0$, and $0 \leq b_p \leq (1 + n_p)^3$.

Give the affine lattice

$$\{z \in R_{F,p} : \text{Tr}_{F_p/\mathbb{Q}_p} z = m\}$$

its Haar probability. With $t = p^{-1}$ and $f_{\mathfrak{p}}$ the residue degrees of the primes of K above p ,

$$\mathbb{E}_p b_p = (1-t)(1-t^3) \prod_{\mathfrak{p} \mid p} (1-t^{f_{\mathfrak{p}}})^{-1} = 1 + \frac{a_K(p) - 1}{p} + O(p^{-2}), \quad (5.22)$$

$$\mathbb{P}_p(n_p \geq k) \leq 3p^{-k} \quad (k \geq 1). \quad (5.23)$$

The error constant is absolute. In any integral affine coordinates on the trace lattice, $\{n_p \geq k\}$ is determined modulo p^k , and $b_p \mathbf{1}_{\{n_p=k\}}$ is determined modulo p^{k+1} for every finite $k \geq 0$.

Proof. An accepted coset has a representative $x \in L_{V,p} = \bigwedge^2 R_p$. Use the extended valuation with $v_p(p) = 1$ on a splitting field. Its six valuations $k_{ij} = v_p(x_{ij})$ are nonnegative integers, invariant under Frobenius. Since u and $\delta_{\mathcal{O}}$ are units,

$$k_{ij} + k_{kl} = j_{\mathfrak{q}} \quad \text{for the partition } ij|kl \text{ over } \mathfrak{q}.$$

This proves integrality of z . If the pair labels over $\mathfrak{q}$ have two Frobenius orbits, one of their two valuations can be chosen from $0, \dots, j_{\mathfrak{q}}$. If there is one orbit, the two valuations are equal, which requires even $j_{\mathfrak{q}}$. The number of records is therefore at most $b_p(z)$.

Two representatives with the same six valuations have comparison (a, c) satisfying

$$v_p(c) + v_p(a_i) + v_p(a_j) = 0 \quad (i < j).$$

Subtraction makes all four $v_p(a_i)$ equal to a common r , and $v_p(c) = -2r$. Because K_p is unramified, $r \in \mathbb{Z}$. Replacing a by $p^{-r}a$ and c by $p^{2r}c$ puts both in the unit subgroup. Thus a record specifies at most one coset, also for nonsplit unramified factors. The subgroup $\mathcal{U}_p$ preserves $\bigwedge^2 R_p$, so its coset fraction is either 0 or 1. This proves Equation (5.21). The bounds at depth zero and at general depth follow immediately from the definition and the at most three factors.

The trace from an unramified integer ring to $\mathbb{Z}_p$ is onto, as is seen after reduction from the nonzero finite-field trace. If F_p has more than one factor, the trace- m coset therefore projects uniformly onto each chosen factor: the other factors can supply any required trace. More generally, for integers $r_{\mathfrak{q}} \geq 0$, the probability that $v_{\mathfrak{q}}(z) \geq r_{\mathfrak{q}}$ for all $\mathfrak{q}$ is

$$\begin{cases} t^{\sum_{\mathfrak{q}} f_{\mathfrak{q}} r_{\mathfrak{q}}}, & \min_{\mathfrak{q}} r_{\mathfrak{q}} = 0, \\ 0, & \min_{\mathfrak{q}} r_{\mathfrak{q}} > 0. \end{cases}$$

The first line follows by projecting onto the constrained factors while a factor with $r_{\mathfrak{q}} = 0$ supplies the trace. The second uses $p \nmid m$. It also covers the case that F_p is a field.

Expanding $b_{\mathfrak{q}}(j) = \sum_{r=0}^j \epsilon_{\mathfrak{q}}^r$ and summing the absolutely convergent geometric series gives

$$\mathbb{E}_p b_p = \frac{1 - (\prod_{\mathfrak{q}} \epsilon_{\mathfrak{q}}) t^3}{\prod_{\mathfrak{q}} (1 - \epsilon_{\mathfrak{q}} t^{f_{\mathfrak{q}}})}. \quad (5.24)$$

The five Frobenius types give the following data. A pair (f, ϵ) in the second column is attached to one prime of F .

cycle type on four labels	$(f_{\mathfrak{q}}, \epsilon_{\mathfrak{q}})$	$\prod_{\mathfrak{p} p} (1 - t^{f_{\mathfrak{p}}})$	$a_K(p) - 1$
1^4	$(1, +), (1, +), (1, +)$	$(1 - t)^4$	3
$2\ 1^2$	$(1, +), (2, +)$	$(1 - t)^2 (1 - t^2)$	1
2^2	$(1, +), (1, -), (1, -)$	$(1 - t^2)^2$	-1
$3\ 1$	$(3, +)$	$(1 - t)(1 - t^3)$	0
4	$(1, -), (2, -)$	$1 - t^4$	-1

In every row $\prod_{\mathfrak{q}} \epsilon_{\mathfrak{q}} = 1$ and

$$\prod_{\mathfrak{q}} (1 - \epsilon_{\mathfrak{q}} t^{f_{\mathfrak{q}}}) = \frac{\prod_{\mathfrak{p}|p} (1 - t^{f_{\mathfrak{p}}})}{1 - t}.$$

Substitution in Equation (5.24) proves the exact mean. Its linear term is the number of degree-one primes of K minus one, which is $a_K(p) - 1$; the five fixed rows give the uniform error term.

If F_p is a field, $n_p \geq 1$ would make its trace divisible by p , and is impossible. Otherwise the uniform projection above gives $\mathbb{P}_p(v_{\mathfrak{q}}(z) \geq k) = p^{-f_{\mathfrak{q}}k}$. Taking the union over at most three factors proves Equation (5.23). Finally, the trace map is onto, so the affine trace lattice has two integral coordinates. Projection to each unramified factor modulo p^{k+1} is determined by these coordinates modulo p^{k+1} . The valuations truncated at $k+1$, and hence the asserted threshold and finite-depth functions, have the stated congruence dependence. This remains true on the component-vanishing set with the convention above. $\square$

Proposition 5.7 (Multiplicities from local densities). *For $z \in F$ with $\text{Tr } z = m \neq 0$, a positive value of $M(z)$ requires z to be integral outside S . For such integral z ,*

$$M(z) \leq 32 \prod_{p \in S} \left(|D|_p^{-1/2} \frac{1 - p^{-1}}{\prod_{\mathfrak{p}|p, \mathfrak{p} \subset R} (1 - N(\mathfrak{p})^{-1})} f_p^*(z) \right) \prod_{p \notin S} b_p(z). \quad (5.25)$$

All products in this assertion are finite at every contributing z .

Proof. The field F and the condition $\text{Tr } z = m \neq 0$ imply $z \in F^\times$. At every prime the set of accepted cosets is finite by Lemma 5.4. At $p \notin S$, Lemma 5.6 proves the integrality requirement and bounds the local sum by $b_p(z)$. The set

$$E_z = S \cup \{p : z \notin R_{F,p}^\times\}$$

is finite, since a nonzero global z is a unit at all but finitely many primes. At $p \notin E_z$, $b_p = 1$, and each coset fraction is 0 or 1, so there is at most one accepted coset, of fraction 1. If some prime has no accepted coset, then $M(z) = 0$ and the claimed inequality is immediate. Otherwise the accepted coset is unique with fraction 1 outside E_z , and each prime inside the finite set E_z has finitely many accepted cosets. Thus there are only finitely many assignments of accepted cosets, and all local sums outside E_z equal 1.

Choose local references. Equation (5.12) identifies the weight of a global pair at each prime with its coset fraction α_p . By Lemma 5.5, at most 32 global pairs occupy any assignment of accepted cosets. Summing over the finite set of assignments, including ones not realized globally, bounds $M(z)$ by 32 times the product of the local sums of α_p . Use Equation (5.17) at $p \in S$ and Equation (5.21) outside S . This proves Equation (5.25) and finiteness. $\square$

The unramified mean in Equation (5.22) has the Euler factor needed for the residue κ . The remaining problem in Theorem 5.2 is to average the densities f_p^* for $p \in S$ before inserting these unramified factors. For a frequency $\alpha \in F_p$, testing the law $u\Psi(x)$ against the character $z \mapsto \psi_p(\text{Tr}_{F_p/\mathbb{Q}_p}(\alpha z))$ produces the quadratic phase $u \text{Tr}_{F_p/\mathbb{Q}_p}(\alpha \Psi(x))$. We express these forms through a single unimodular bilinear form and an action of F . The elements of F whose operators preserve L_V will then form the order that controls the local Fourier analysis.

5.4 The integral quadratic structure and its order

Let B_V be the polarization of Q_V without division by two:

$$B_V(x, y) = Q_V(x + y) - Q_V(x) - Q_V(y).$$

Over a splitting field, write V_b for the plane spanned by the two coordinate axes belonging to the partition b , the same coordinate planes used over $\mathbb{R}$ in Section 4. For an equivariant triple $\alpha = (\alpha_b) \in F$, let α_b scale both coordinate axes of V_b .

Lemma 5.8 (Self-adjoint quadratic structure). *The plane-scalar action descends to a rational action $\rho : F \rightarrow \text{End}_{\mathbb{Q}}(V)$ for which V has rank two over F . The form Q_V in an exterior $\mathbb{Z}$ -basis of L_V is the three split pairs in Equation (5.4); in particular B_V is unimodular on L_V . The action ρ is self-adjoint for B_V , and for $\alpha \in F$,*

$$Q_\alpha(x) := \text{Tr}_{F/\mathbb{Q}}(\alpha \Psi(x)) = \frac{1}{2} B_V(\rho(\alpha)x, x), \tag{5.26}$$

$$\begin{aligned} B_\alpha(x, y) &= B_V(\rho(\alpha)x, y), \\ \det_{\mathbb{Q}} \rho(\alpha) &= N_{F/\mathbb{Q}}(\alpha)^2, \quad \Psi(\rho(\alpha)x) = \alpha^2 \Psi(x). \end{aligned} \tag{5.27}$$

Here B_α is the polarization of Q_α . The same assertions hold after base change to every $\mathbb{Q}_p$.

Proof. Galois permutes the planes V_b , including possible alternating signs on their coordinate axes, and permutes the scalars α_b in the same way. The plane-scalar action therefore commutes with Galois and descends. After splitting there are three planes of dimension two, proving the rank assertion. Equation (5.4) shows that the matrix of B_V pairs the three opposite pairs by unit off-diagonal entries, with the displayed signs, so it is unimodular. On each plane, multiplying both coordinates by α_b is self-adjoint for its split pairing. The contribution of that plane to $\frac{1}{2}B_V(\rho(\alpha)x, x)$ is $\alpha_b\Psi_b(x)$, which proves Equation (5.26). Polarization gives the next identity. The determinant and the last quadratic identity follow by multiplying, respectively, the two equal eigenvalues and the two coordinate scalars on each plane. All these are rational identities, hence persist under base change. $\square$

Define the stabilizer order and its discriminant by

$$C_F = \{\alpha \in F : \rho(\alpha)L_V \subset L_V\}, \quad \Delta_F = \text{Disc}(C_F). \quad (5.28)$$

It is a ring containing 1 and a full $\mathbb{Z}$ -lattice: this follows from the rationality of ρ by clearing denominators on a basis of F . It is contained in R_F , since an α whose matrix on L_V is integral is a root of that integral monic characteristic polynomial. Thus it is an order. The stabilizer condition is a finite list of matrix-entry integrality conditions in a lattice basis, so it commutes with completion:

$$C_{F,p} = C_F \otimes_{\mathbb{Z}} \mathbb{Z}_p = \{\alpha \in F_p : \rho(\alpha)L_{V,p} \subset L_{V,p}\}. \quad (5.29)$$

The following comparison ensures that its discriminant tends to infinity with the discriminant of $\mathcal{O}$, even when the field K stays fixed.

Lemma 5.9 (Scale of the stabilizer order). *There is an absolute constant C such that*

$$\frac{1}{2}X^{1/2} \leq \Delta_F = D_F[R_F : C_F]^2 \leq CX^{28}. \quad (5.30)$$

More precisely, with $d = \text{Disc}(\mathcal{O}) = X^2$,

$$d^2R_F \subset C_F \subset R_F, \quad D_F \ll D^2.$$

In particular $C_{F,p} = R_{F,p}$ for $p \nmid Dq$.

Proof. Let $E = \bigwedge^2 A_{\mathcal{O}}$ be the exterior embedding matrix. Its entries are integral in every local splitting ring, and $\det E = \delta_{\mathcal{O}}^3$. Since $d = \delta_{\mathcal{O}}^2$,

$$d^2E^{-1} = \delta_{\mathcal{O}} \text{adj}(E)$$

also has integral entries. For $\alpha \in R_{F,p}$, its plane-scalar matrix over the splitting ring is integral. Consequently $d^2\rho(\alpha) = d^2E^{-1} \text{diag}(\alpha_b, \alpha_b)E$ is integral over the splitting ring. Its entries are in $\mathbb{Q}_p$, so they lie in $\mathbb{Z}_p$. This proves $d^2R_F \subset C_F$ and $[R_F : C_F] \leq d^6$.

If $p \nmid D$, the étale algebra R_p splits over an unramified extension of $\mathbb{Z}_p$; its splitting field and cubic resolvent are unramified. Hence every prime dividing D_F divides D . For $p > 3$, a cubic algebra is tamely ramified and its discriminant exponent is at most 2. At 2 and 3, the finitely many local extensions of degree at most three have bounded discriminant exponents [16, Proposition 7.64 and Remark 7.65]. It follows that $D_F \ll D^2$. The discriminant-index formula now gives

$$\Delta_F = D_F[R_F : C_F]^2 \ll D^2d^{12} \leq X^{28}.$$

The containment $d^2 R_F \subset C_F$ also proves equality of the local orders away from Dq .

For the reverse inequality, we first explain why a projection denominator controls $\text{Disc}(\mathcal{O})$. Write tr for matrix trace, put $\mathcal{E} = \text{End}_{\mathbb{Z}}(\mathcal{O})$, and let $\mathcal{W}_K \subset \text{End}_{\mathbb{Q}}(K)$ be the rational subspace of multiplication matrices. Its intersection $\mathcal{L} = \mathcal{W}_K \cap \mathcal{E}$ is saturated. It is multiplication by $\mathcal{O}$: if $\alpha\mathcal{O} \subset \mathcal{O}$, then $\alpha = \alpha \cdot 1 \in \mathcal{O}$, and the converse follows because $\mathcal{O}$ is a ring. The pairing $(T, S) \mapsto \text{tr}(TS)$ is unimodular on $\mathcal{E}$, since its matrix units are dual. Its restriction to $\mathcal{W}_K$ is the nondegenerate field trace pairing. Let $\mathcal{E}_K$ be the orthogonal projection onto $\mathcal{W}_K$.

For the restricted pairing one has

$$\mathcal{E}_K(\mathcal{E}) = \mathcal{L}^\vee. \quad (5.31)$$

One inclusion follows by pairing a projected integral matrix with $\mathcal{L}$. Conversely, an integer functional on the primitive sublattice $\mathcal{L}$ extends to one on $\mathcal{E}$. Unimodularity represents the extension by an element of $\mathcal{E}$; its orthogonal projection represents the original functional on $\mathcal{L}$. Nondegeneracy identifies that projection with the prescribed element of $\mathcal{L}^\vee$. Consequently, if a positive integer N satisfies $N\mathcal{E}_K(\mathcal{E}) \subset \mathcal{E}$, then

$$\mathcal{L}^\vee \subset N^{-1}\mathcal{L}, \quad X^2 = \text{Disc}(\mathcal{O}) = [\mathcal{L}^\vee : \mathcal{L}] \leq N^4.$$

It remains to produce the denominator $N = 2\Delta_F$.

Choose trace-dual bases a_i of C_F and b_i of $C_F^\vee$, where

$$C_F^\vee = \{\beta \in F : \text{Tr}_{F/\mathbb{Q}}(\beta C_F) \subset \mathbb{Z}\}.$$

The adjugate of the trace Gram matrix gives $\Delta_F b_i \in C_F$. Define on $\text{End}_{\mathbb{Q}}(V)$

$$\mathcal{E}_F(T) = \sum_{i=1}^3 \rho(a_i) T \rho(b_i). \quad (5.32)$$

For two embeddings σ, τ of F , the dual-basis identity is

$$\sum_i \sigma(a_i) \tau(b_i) = \begin{cases} 1, & \sigma = \tau, \\ 0, & \sigma \neq \tau. \end{cases}$$

Thus, after splitting, $\mathcal{E}_F$ extracts exactly the three diagonal blocks $\text{End}(V_b)$. It is the orthogonal projection for the matrix trace pairing, and

$$\Delta_F \mathcal{E}_F(\text{End}_{\mathbb{Z}}(L_V)) \subset \text{End}_{\mathbb{Z}}(L_V). \quad (5.33)$$

This denominator conclusion uses the trace-dual lattice only; it does not require that lattice to be an invertible C_F -ideal.

We transfer this projection to matrices on K explicitly. Let

$$j(T)(v \wedge w) = Tv \wedge w + v \wedge Tw \quad (T \in \text{End}_{\mathbb{Q}}(K)).$$

There is a natural contraction $\mathfrak{c} : \text{End}_{\mathbb{Q}}(\wedge^2 K) \rightarrow \text{End}_{\mathbb{Q}}(K)$. For clarity, in a basis $e_1, \dots, e_4$ write

$$S(e_j \wedge e_l) = \sum_{i < k} S_{jl}^{ik} e_i \wedge e_k \quad (j < l),$$

and extend these coefficients antisymmetrically in i, k and in j, l , with zero coefficients when either pair repeats. Then

$$\mathfrak{c}(S)_j^i = \sum_{k=1}^4 S_{jk}^{ik}.$$

This is the contraction of $\wedge^2 K \otimes \wedge^2 K^\vee$ in one covariant and one contravariant index, so it is independent of the basis. Both j and $\mathfrak{c}$ are integral in the $\mathcal{O}$ and exterior $\mathcal{O}$ -bases. Expanding their coefficients gives

$$\mathfrak{c}(j(T)) = 2T + \text{tr}(T) \text{id}_K. \quad (5.34)$$

Indeed an off-diagonal coefficient of T occurs for the two remaining indices k ; a diagonal coefficient gives $\sum_{k \neq i} (T_i^i + T_k^k) = 2T_i^i + \text{tr}(T)$.

In embedding coordinates, an off-diagonal entry of $j(T)$ changes one index of a two-element subset. Such subsets belong to distinct opposite-pair planes. It follows that $\mathcal{E}_F(j(T)) = j(\text{diag } T)$ over the splitting field. Equation (5.34) now proves the formula

$$\mathcal{E}_K(T) = \frac{1}{2}(\mathfrak{c}(\mathcal{E}_F(j(T)))) - \text{tr}(T) \text{id}_K \quad (5.35)$$

for the projection defined above: its right side is $\text{diag } T$ over the splitting field. A Galois-equivariant diagonal tuple is the tuple of embeddings of an element of K , so these diagonal matrices are precisely $\mathcal{W}_K$. Diagonal extraction is orthogonal for matrix trace. Finally, Equation (5.33) and integrality of j and $\mathfrak{c}$ give

$$2\Delta_F \mathcal{E}_K(\text{End}_{\mathbb{Z}}(\mathcal{O})) \subset \text{End}_{\mathbb{Z}}(\mathcal{O}).$$

Taking $N = 2\Delta_F$ in the preceding lattice bound gives

$$X^2 \leq (2\Delta_F)^4.$$

This is the lower inequality in Equation (5.30). $\square$

For $x \in L_{V,p}$ and $\alpha \in C_{F,p}$, Equation (5.26) gives two integral pairings:

$$\text{Tr } \Psi(x) = Q_V(x) \in \mathbb{Z}_p, \quad 2 \text{Tr}(\alpha \Psi(x)) = B_V(\rho(\alpha)x, x) \in \mathbb{Z}_p.$$

They remain integral after multiplying $\Psi(x)$ by a $\mathbb{Z}_p$ -unit. Thus the products pair integrally with the scalar 1 and with $2C_{F,p}$. Define the order and its trace dual

$$C_F^\sharp = \mathbb{Z} + 2C_F, \quad (C_F^\sharp)^\vee = \{z \in F : \text{Tr}_{F/\mathbb{Q}}(zC_F^\sharp) \subset \mathbb{Z}\}.$$

This trace dual gives the integral support for the additive count. The same support will bound the total depth of the unramified factors and prevent a nonzero real product coordinate from being too small.

Lemma 5.10 (Trace-dual support and product size). *One has*

$$[C_F : C_F^\sharp] = 4, \quad \Delta_F^\sharp := \text{Disc}(C_F^\sharp) = 16\Delta_F \ll X^{28}, \quad (5.36)$$

and $\text{Tr} : (C_F^\sharp)^\vee \rightarrow \mathbb{Z}$ is onto. For $p \notin S$, $C_{F,p}^\sharp = (C_{F,p}^\sharp)^\vee = R_{F,p}$. For every prime p , the density f_p^* on $F_p^\times$ is supported on $(C_{F,p}^\sharp)^\vee$. Consequently

$$M(z) > 0 \implies z \in (C_F^\sharp)^\vee.$$

The following stronger containment is used to average the majorant in Proposition 5.7: if $z \in F^\times$ is integral outside S and $\prod_{p \in S} f_p^*(z) > 0$, then $z \in (C_F^\sharp)^\vee$.

Fix $m \neq 0$ and a bounded $\Omega \subset \mathcal{H}_m$. Every $z \in (C_F^\sharp)^\vee \cap \Omega$ satisfies

$$\Delta_F^\sharp z \in R_F, \quad 1 \leq |\text{N}_{F/\mathbb{Q}}(\Delta_F^\sharp z)| \ll_\Omega X^{84}, \quad (5.37)$$

$$\prod_{p \notin S} p^{n_p(z)} \leq |\text{N}_{F/\mathbb{Q}}(\Delta_F^\sharp z)|, \quad |z_b| \gg_\Omega X^{-84} \quad (b \in \mathcal{B}). \quad (5.38)$$

The constants are uniform in all arithmetic data and in the point of Ω .

Proof. Since C_F is an order, $C_F \cap \mathbb{Q} = \mathbb{Z}$, so 1 is primitive in its lattice. A basis $1, a_2, a_3$ of C_F gives the basis $1, 2a_2, 2a_3$ of $C_F^\sharp$. This proves the index and discriminant assertions in Equation (5.36). The functional taking 1 to 1 and the other two basis vectors to 0 is represented in the trace dual. Hence trace from that dual is onto $\mathbb{Z}$. Lemma 5.9, and $2 \in S$, give $C_{F,p}^\sharp = R_{F,p}$ outside S .

The two integrality identities preceding the lemma show that the law in Equation (5.16) is supported on $(C_{F,p}^\sharp)^\vee$. For a contributing global pair, use its local representation $z = u\Psi(x)$ and Equation (5.14) to translate x into $L_{V,p}$, adjusting u by the inverse norm of that unit. This proves local trace-dual membership at every prime. Intersecting the local lattices proves the stated global support. For the second support assertion, the density factors give this membership at $p \in S$. At $p \notin S$, the order is $R_{F,p}$, whose trace pairing is unimodular because F_p is unramified; thus $R_{F,p}^\vee = R_{F,p}$, and the assumed integrality gives membership there as well.

The adjugate of the trace Gram matrix of $C_F^\sharp$ gives $\Delta_F^\sharp(C_F^\sharp)^\vee \subset C_F^\sharp \subset R_F$. If $z \in \Omega$, then $\text{Tr } z = m \neq 0$, so $z \neq 0$; since F is a field, all its embeddings are nonzero. The norm $N(\Delta_F^\sharp z)$ is consequently a nonzero integer. The boundedness of Ω and $\Delta_F^\sharp \ll X^{28}$ give

$$1 \leq |N(\Delta_F^\sharp z)| = (\Delta_F^\sharp)^3 \prod_{b \in \mathcal{B}} |z_b| \ll_\Omega X^{84}.$$

At $p \notin S$, z is integral and the local algebra is unramified. Hence

$$n_p(z) \leq \sum_{\mathfrak{q}|p} f_{\mathfrak{q}} v_{\mathfrak{q}}(z) \leq v_p(N(\Delta_F^\sharp z)).$$

The product of these inequalities gives the depth bound because $N(\Delta_F^\sharp z)$ is an integer. Finally, if the coordinates of Ω have absolute value at most B , with $B \geq 1$, then for each b

$$|z_b| = \frac{|N(\Delta_F^\sharp z)|}{(\Delta_F^\sharp)^3 \prod_{b' \neq b} |z_{b'}|} \geq \frac{1}{(\Delta_F^\sharp)^3 B^2} \gg_\Omega X^{-84}.$$

This proves Equation (5.38). $\square$

6 Local Fourier estimates

The exterior multiplicity bound in Section 5 uses an additive density at each prime in S . We now estimate its Fourier transform. Scalar frequencies detect only the trace and have an exact, order-independent value. For the remaining frequencies we record both decay and the support of the inverse transform on a fixed denominator shell. The support is needed because the denominator at a prime dividing the order index can be arbitrarily large.

6.1 Quadratic data and the trace marginal

Fix a rational prime p . We first specify the local hypotheses used in the argument. Let

$$E = \prod_{\nu=1}^r E_\nu, \quad R_E = \prod_{\nu=1}^r R_\nu$$

be a cubic étale $\mathbb{Q}_p$ -algebra and its maximal order. Here E_ν is a field, R_ν is its integer ring, and $r \leq 3$. Write $\mathfrak{q}_\nu = (\pi_\nu)$ for the maximal ideal of R_ν , $v_\nu(\pi_\nu) = 1$, $e_\nu = v_\nu(p)$, and $q_\nu = |R_\nu/\mathfrak{q}_\nu| = p^{f_\nu}$. Thus

$$\sum_\nu e_\nu f_\nu = 3, \quad |N_{E_\nu/\mathbb{Q}_p} a|_p = q_\nu^{-v_\nu(a)}.$$

Let $\mathfrak{d}_\nu = \mathfrak{q}_\nu^{d_\nu}$ be the different of $E_\nu/\mathbb{Q}_p$, and put $\mathfrak{d}_E = \prod_\nu \mathfrak{d}_\nu$.

Let W be a free E -module of rank two, with its unital action denoted by $\rho : E \longrightarrow \text{End}_{\mathbb{Q}_p}(W)$. Suppose that B is a nondegenerate symmetric $\mathbb{Q}_p$ -bilinear form satisfying

$$B(\rho(a)x, y) = B(x, \rho(a)y) \quad (a \in E, x, y \in W).$$

Let L be a full $\mathbb{Z}_p$ -lattice in W which is self-dual for B :

$$L = \{y \in W : B(y, L) \subset \mathbb{Z}_p\}.$$

Assume also that $Q_1(x) = B(x, x)/2$ is integral on L . For the scalar calculation we shall use the stronger property that in some integral coordinates on L ,

$$Q_1(x_1, y_1, x_2, y_2, x_3, y_3) = x_1y_1 - x_2y_2 + x_3y_3. \quad (6.1)$$

The division by 2 takes place in $\mathbb{Q}_p$. Equation (6.1) asserts the required integrality also when $p = 2$. The shell estimates below use only $Q_1(L) \subset \mathbb{Z}_p$; the exact split expression is used for the scalar marginal.

For $a \in E$, set $Q_a(x) = B(\rho(a)x, x)/2$. The nondegenerate trace pairing on E determines a unique quadratic map $\Psi_E : W \longrightarrow E$ by

$$\text{Tr}_{E/\mathbb{Q}_p}(a\Psi_E(x)) = Q_a(x) = \frac{1}{2}B(\rho(a)x, x) \quad (a \in E). \quad (6.2)$$

Define the stabilizer order and the auxiliary order

$$\mathcal{C}_L = \{a \in E : \rho(a)L \subset L\}, \quad \mathcal{C}_L^\# = \mathbb{Z}_p + 2\mathcal{C}_L.$$

Indeed $\mathcal{C}_L$ is an order in R_E . It is a subring containing 1, and a sufficiently large power of p times R_E preserves L . Conversely, an element preserving L satisfies the monic characteristic polynomial of its integral matrix on L . The action is faithful, so the element is integral over $\mathbb{Z}_p$ and belongs to R_E .

In the application, $E = F_p$, $W = V_p$, $L = L_{V,p}$, $B = B_V$, and $\mathcal{C}_L = C_{F,p}$. Lemma 5.8 supplies all the stated hypotheses and identifies Ψ_E with the base change of Ψ . We use the abstract notation temporarily to include every product algebra and every ramification type in one argument.

Choose an additive character ψ_p of $\mathbb{Q}_p$ with conductor $\mathbb{Z}_p$. We give W the self-dual Haar measure for $(x, y) \mapsto \psi_p(B(x, y))$; it gives L mass 1. On E use the self-dual measure for $(A, z) \mapsto \psi_p(\text{Tr}_{E/\mathbb{Q}_p}(Az))$, denoted by dz or dA according to the variable. If $\mathfrak{a}$ is a fractional R_E -ideal, its annihilator for this pairing is $\mathfrak{a}^\perp = \mathfrak{a}^{-1}\mathfrak{d}_E^{-1}$. In particular

$$\text{vol}(R_E) = \prod_\nu q_\nu^{-d_\nu/2} \leq 1. \quad (6.3)$$

To see the normalization, $R_E^\perp = \mathfrak{d}_E^{-1}$ and $\text{vol}(R_E)\text{vol}(R_E^\perp) = 1$, whereas $[\mathfrak{d}_E^{-1} : R_E] = \prod_\nu q_\nu^{d_\nu}$. For $E = F_p$, Equation (6.3) reads $\text{vol}(R_{F,p}) = |D_F|_p^{1/2}$.

Let μ_p be the law of

$$u\Psi_E(x), \quad x \in L \text{ with additive probability,} \quad u \in \mathbb{Z}_p^\times \text{ with Haar probability,} \quad (6.4)$$

with x and u independent. This is the law defining f_p^* in Lemma 5.4. The following regularity fact specifies the version of that density which we shall invert pointwise.

Lemma 6.1. *The measure μ_p has a density f_p^* with respect to dz . It has a finite locally constant version at every $z = (z_\nu) \in E$ with $z_\nu \neq 0$ for all ν . This version vanishes outside*

$$(\mathcal{C}_L^\sharp)^\vee = \{z \in E : \text{Tr}_{E/\mathbb{Q}_p}(z\mathcal{C}_L^\sharp) \subset \mathbb{Z}_p\}.$$

In particular $\text{Tr}_{E/\mathbb{Q}_p}(u\Psi_E(x)) \in \mathbb{Z}_p$ on the domain of Equation (6.4).

Proof. Define an E -valued bilinear form b_E by

$$\text{Tr}_{E/\mathbb{Q}_p}(ab_E(x, y)) = B(\rho(a)x, y) \quad (a \in E).$$

Self-adjointness of the action makes b_E symmetric and E -bilinear. It is nondegenerate because B is nondegenerate. Equation (6.2) then gives $\Psi_E(x) = b_E(x, x)/2$ and $d\Psi_{E,x}(h) = b_E(x, h)$. On the ν -th field factor, a nonzero component of $\Psi_E(x)$ implies that the component x_ν of x is nonzero. The nondegenerate form on the rank-two E_ν -space makes $h_\nu \mapsto b_E(x_\nu, h_\nu)$ a nonzero, hence onto, E_ν -linear functional. It follows that Ψ_E is a $\mathbb{Q}_p$ -submersion at every preimage of an output with nonzero components. Multiplication by a unit u is invertible on E , so the map $(x, u) \mapsto u\Psi_E(x)$ is a submersion at the same points.

For each ν the polynomial $\Psi_{E,\nu}$ is nonzero: a nondegenerate symmetric form in characteristic zero has a nonzero quadratic form. Choose a nonzero $\mathbb{Q}_p$ -coordinate polynomial of this component. Its zero set in the open lattice L has Haar measure zero, by the usual one-variable root bound followed by induction and Fubini. Off the union of these null sets, the local submersion theorem gives absolute continuity of the pushforward. For a fixed output with nonzero components, its preimage in the compact set $L \times \mathbb{Z}_p^\times$ is compact. Cover this preimage by finitely many compact-open submersion charts, refined to disjoint compact-open pieces within those charts. The image of the remaining compact set avoids a sufficiently small neighborhood of the output. After shrinking that neighborhood, integration in the fiber coordinates of the pieces gives a finite constant density on it. This proves the asserted pointwise version, including value zero when the neighborhood misses the image.

Finally, for $c \in \mathcal{C}_L$ and $r \in \mathbb{Z}_p$,

$$\text{Tr}_{E/\mathbb{Q}_p}((r + 2c)\Psi_E(x)) = rQ_1(x) + B(\rho(c)x, x) \in \mathbb{Z}_p \quad (x \in L).$$

Multiplication by $u \in \mathbb{Z}_p^\times$ preserves this inclusion. The image of the compact domain is therefore contained in the displayed trace dual, which proves the support and trace assertions. $\square$

We use the Fourier convention

$$\widehat{f}(A) = \int_E f(z)\psi_p(\text{Tr}_{E/\mathbb{Q}_p}(Az))dz, \quad \check{g}(z) = \int_E g(A)\psi_p(-\text{Tr}_{E/\mathbb{Q}_p}(Az))dA.$$

The transform of the measure in Equation (6.4) is

$$\gamma_p(A) = \widehat{f_p^*}(A) = \mathbb{E}_{u \in \mathbb{Z}_p^\times} \int_L \psi_p(uQ_A(x))dx. \quad (6.5)$$

Lemma 6.2 (The scalar trace density). *Suppose that the integral scalar form is exactly $Q_1 = x_1y_1 - x_2y_2 + x_3y_3$ as in Equation (6.1). Regard $a \in \mathbb{Q}_p$ as a scalar in E . In the quartic application this scalar form is $Q_1 = Q_V$. Then, for every prime p , including $p = 2$,*

$$\gamma_p(a) = \begin{cases} 1, & a \in \mathbb{Z}_p, \\ p^{-3k}, & v_p(a) = -k < 0. \end{cases} \quad (6.6)$$

The trace of the law in Equation (6.4) has density $d_p(t)\mathbf{1}_{\mathbb{Z}_p}(t)$ for additive measure giving $\mathbb{Z}_p$ mass 1. This density is continuous on $\mathbb{Z}_p$ and satisfies

$$|d_p(t) - 1| \leq \frac{1}{p(p+1)} \quad (t \in \mathbb{Z}_p). \quad (6.7)$$

In particular $5/6 \leq d_p(t) \leq 7/6$, and for every finite set of primes S and every choice $t_p \in \mathbb{Z}_p$,

$$\prod_{p \in S} d_p(t_p) \asymp 1$$

with absolute upper and lower constants.

Proof. For fixed $u \in \mathbb{Z}_p^\times$ and $v_p(a) = -k < 0$, integration in the second coordinate gives

$$\int_{\mathbb{Z}_p^2} \psi_p(uaxy) dx dy = \int_{\mathbb{Z}_p} \mathbf{1}_{\{uax \in \mathbb{Z}_p\}} dx = p^{-k}.$$

For $a \in \mathbb{Z}_p$ the integral is 1. The sign of a has no effect, so the three independent pairs in Equation (6.1) give Equation (6.6), even before averaging u . This calculation uses the integral split expression and does not divide by 2 in $\mathbb{Z}_p$.

The scalar coefficient is the Fourier coefficient of the trace law, since $\text{Tr}(az) = a \text{Tr}(z)$. That law is supported on $\mathbb{Z}_p$ by Lemma 6.1. The character group of $\mathbb{Z}_p$ is $\mathbb{Q}_p/\mathbb{Z}_p$, and it has $p^k - p^{k-1}$ characters of exact denominator p^k . Hence its nonconstant Fourier coefficients have total absolute sum

$$\sum_{k \geq 1} (p^k - p^{k-1})p^{-3k} = \frac{1 - p^{-1}}{p^2 - 1} = \frac{1}{p(p+1)}.$$

Fourier inversion on the compact group $\mathbb{Z}_p$ is therefore uniformly convergent and gives the continuous density and Equation (6.7). The lower bound is positive at every prime, and $\sum_p 1/(p(p+1)) < \infty$. Taking logarithms of the upper and lower bounds proves the last assertion. $\square$

6.2 Denominator shells and quadratic transformation

For $A = (A_\nu) \in E$, its denominator ideal is

$$\mathfrak{h}_p(A) = \prod_{\nu} \mathfrak{q}_{\nu}^{h_{\nu}(A)}, \quad h_{\nu}(A) = \max\{0, -v_{\nu}(A_{\nu})\},$$

where $v_{\nu}(0) = +\infty$. For an integral R_E -ideal $\mathfrak{h}_p = \prod_{\nu} \mathfrak{q}_{\nu}^{h_{\nu}}$, its shell is thus

$$\mathcal{S}_{\mathfrak{h}_p} = \prod_{h_{\nu}=0} R_{\nu} \times \prod_{h_{\nu}>0} \pi_{\nu}^{-h_{\nu}} R_{\nu}^{\times}.$$

The integral components of a shell may be zero. Define

$$H_p = N(\mathfrak{h}_p) = \prod_{\nu} q_{\nu}^{h_{\nu}}, \quad p^{g_p} = [\mathfrak{h}_p : \mathfrak{h}_p \cap \mathcal{C}_L]. \quad (6.8)$$

The shell is a compact open subset of $\mathfrak{h}_p^{-1}$. We shall shift its frequencies by integral scalars so that the quadratic form is nondegenerate, without changing the integral in Equation (6.5).

Lemma 6.3 (A scalar shift on a shell). *Put $\iota_p = 1$ for $p \leq 5$ and $\iota_p = 0$ for $p > 5$. For every $A \in \mathfrak{S}_{\mathfrak{h}_p}$ there is an $s \in \{0, 1, 2, 3\}$ such that $\alpha = A + s \in E^\times$ and*

$$v_\nu(\alpha_\nu) = -h_\nu \quad (h_\nu > 0), \quad 0 \leq v_\nu(\alpha_\nu) \leq e_\nu \iota_p \quad (h_\nu = 0).$$

Writing $\beta = \alpha^{-1}$, one has

$$\beta \in p^{-\iota_p} \mathfrak{h}_p, \quad p^{-3\iota_p} H_p \leq |N_{E/\mathbb{Q}_p} \alpha|_p \leq H_p. \quad (6.9)$$

In particular the norm is exactly H_p when $p > 5$. If

$$j_p = j_p(A, s) = \min\{j \geq 0 : p^j \beta \in \mathcal{C}_L\}, \quad (6.10)$$

then j_p is finite and $j_p \leq g_p + \iota_p$. Finally, for any generator $i \in E^\times$ of the ideal $\mathfrak{h}_p$, the element $x = i(A + s)$ is in R_E , is a unit of R_E if $p > 5$, and satisfies $x^{-1} \in p^{-\iota_p} R_E$.

Proof. An integral scalar shift leaves every pole valuation $-h_\nu < 0$ unchanged. For an integral component and $p > 5$, the condition that $A_\nu + s$ fail to be a unit forbids at most one residue $s \in \mathbf{F}_p$: it forbids none if the residue of $-A_\nu$ does not belong to the scalar subfield. There are at most three components, and $0, 1, 2, 3$ are distinct modulo p . Some choice therefore makes all integral components units.

For $p \leq 5$, normalize the extension of v_p by $v_p(p) = 1$, so that on E_ν it is v_ν/e_ν . Distinct elements s, s' of the list satisfy $v_p(s - s') \leq 1$. The ultrametric inequality shows that for a fixed integral component at most one candidate can satisfy $v_p(A_\nu + s) > 1$. Again there are at most three components, so a candidate remains for which all their valuations are at most 1. Thus $A_\nu + s$ is nonzero in every integral component.

The bounds on β follow componentwise. In the norm, the possible loss from the integral components is at most $p^{-\iota_p} \sum_\nu e_\nu f_\nu = p^{-3\iota_p}$, proving Equation (6.9). The group $\mathfrak{h}_p/(\mathfrak{h}_p \cap \mathcal{C}_L)$ has order p^{g_p} and is killed by p^{g_p} . Thus

$$p^{g_p} \mathfrak{h}_p \subset \mathcal{C}_L. \quad (6.11)$$

Multiplying the containment for β by $p^{g_p + \iota_p}$ proves the claim about j_p . Finally $v_\nu(i_\nu) = h_\nu$. The components of $i\alpha$ consequently have valuation zero on the pole factors and valuation between zero and $e_\nu \iota_p$ on the others. This proves the assertions about x and x^{-1} . $\square$

For the shift in Lemma 6.3, $Q_{A+s} = Q_A + sQ_1$ and $Q_1(L) \subset \mathbb{Z}_p$. Thus for every fixed unit u

$$\int_L \psi_p(uQ_{A+s}(x)) dx = \int_L \psi_p(uQ_A(x)) dx. \quad (6.12)$$

The following quadratic transformation turns the norm of $A + s$ into decay, and leaves an integral involving its inverse.

Lemma 6.4 (Quadratic Fourier transformation). *Let W_0 be a finite-dimensional $\mathbb{Q}_p$ -space with a nondegenerate symmetric form B_0 , and let $L_0 \subset W_0$ be a full $\mathbb{Z}_p$ -lattice self-dual for B_0 . Use the self-dual Haar measure, which gives L_0 mass 1. For an invertible B_0 -self-adjoint operator P , put $q_P(x) = B_0(Px, x)/2$. There is a phase $\omega(P)$ with $|\omega(P)| = 1$ such that*

$$\int_{L_0} \psi_p(q_P(x)) dx = \omega(P) |\det P|_p^{-1/2} \int_{L_0} \psi_p(-q_{P^{-1}}(x)) dx. \quad (6.13)$$

The normalized phase is invariant under linear isometry of the quadratic form: $\omega(T^*PT) = \omega(P)$ for every invertible T , where T^* is the B_0 -adjoint.

For the local data above, $\alpha \in E^\times$, $\beta = \alpha^{-1}$ and $u \in \mathbb{Z}_p^\times$, this gives

$$\int_L \psi_p(uQ_\alpha(x)) dx = \omega_p(u, \alpha) |N_{E/\mathbb{Q}_p} \alpha|_p^{-1} \int_L \psi_p(-u^{-1}Q_\beta(x)) dx, \quad |\omega_p(u, \alpha)| = 1. \quad (6.14)$$

For every $r \in E^\times$ one has $\omega_p(u, \alpha r^2) = \omega_p(u, \alpha)$. All assertions hold at $p = 2$.

Proof. We use Weil's Fourier formula for a quadratic character [27, Chapter I, §14, Theorem 2, p. 161]. We record its normalization for the transform

$$\mathcal{F}_{B_0} h(y) = \int h(x) \psi_p(B_0(x, y)) dx.$$

Completing the square gives

$$q_P(x) + B_0(x, y) = q_P(x + P^{-1}y) - q_{P^{-1}}(y).$$

Let $B_N = p^{-N}L_0$ and $c_N = \int_{B_N} \psi_p(q_P(x)) dx$. On any fixed compact set of y , translation by $P^{-1}y$ preserves B_N for all sufficiently large N . The transform of $\mathbf{1}_{B_N} \psi_p(q_P)$ on that set is therefore $c_N \psi_p(-q_{P^{-1}})$. If this identity is paired with a Schwartz–Bruhat function, its left side stabilizes with N , since the Fourier transform of the test function has compact support. Choosing a test function whose integral against $\psi_p(-q_{P^{-1}})$ is nonzero shows that c_N stabilizes as well. This proves the distributional shape of the formula.

Differencing the absolute square of c_N and integrating in one variable gives

$$|c_N|^2 = \text{vol}(B_N) \int_{B_N \cap P^{-1}B_N^\perp} \psi_p(q_P(h)) dh.$$

For sufficiently large N , the intersection is $P^{-1}p^N L_0$ and q_P is integral on it. The right side then equals $\text{vol}(p^{-N}L_0) \text{vol}(P^{-1}p^N L_0) = |\det P|_p^{-1}$. Hence the stabilized constant is $\omega(P) |\det P|_p^{-1/2}$ with unit phase. The transform of $\mathbf{1}_{L_0}$ is $\mathbf{1}_{L_0}$, so pairing the distribution identity with this function gives Equation (6.13).

Under $q_{T^*PT} = q_P \circ T$, changing variables multiplies the Gaussian constant by $|\det T|_p^{-1}$ and its normalizing determinant square root by $|\det T|_p$. Their product, the phase, is unchanged; this is also the isometry invariance in Weil [27, Chapter II, §25, p. 173]. For $P = u\rho(\alpha)$ its inverse is $u^{-1}\rho(\beta)$. Rank two over E gives $\det \rho(\alpha) = N_{E/\mathbb{Q}_p}(\alpha)^2$, and $|u|_p = 1$. This proves Equation (6.14). Taking $T = \rho(r)$ proves the last phase identity. The polar operator of q_P is P , because its polarization is $B_0(Px, y)$. In particular this proof never assumes that 2 is a unit of $\mathbb{Z}_p$. $\square$

6.3 The inverse transform on a shell

Extend the restriction of γ_p to a shell by zero, and denote its inverse transform by

$$\gamma_{p, \mathfrak{h}_p}(A) = \mathbf{1}_{\mathfrak{h}_p}(A) \gamma_p(A), \quad K_{p, \mathfrak{h}_p}(z) = \int_E \gamma_{p, \mathfrak{h}_p}(A) \psi_p(-\text{Tr}_{E/\mathbb{Q}_p}(Az)) dA. \quad (6.15)$$

The next lemma gives the estimates for these functions which will be used in Poisson summation. Its quantitative support assertions separate the case where the denominator ideal is contained in the stabilizer order.

Lemma 6.5 (Intrinsic shell estimates). *Let E be the cubic algebra above, let W have rank two over E with self-adjoint action for a nondegenerate symmetric form B , and let L be self-dual for B , with $Q_1(L) \subset \mathbb{Z}_p$. Use the stabilizer order $\mathcal{C}_L = \{a \in E : \rho(a)L \subset L\}$ and Ψ_E, γ_p from Equations (6.2) and (6.5). Fix an integral ideal $\mathfrak{h}_p = \prod_\nu \mathfrak{q}_\nu^{h_\nu}$, and use H_p, g_p from Equation (6.8). The zero-extended function $\gamma_{p, \mathfrak{h}_p}$ and its inverse $K_{p, \mathfrak{h}_p}$ are Schwartz–Bruhat functions. Moreover,*

$$\text{supp } K_{p, \mathfrak{h}_p} \subset (\mathcal{C}_L^\#)^\vee \subset \{z \in E : \text{Tr}_{E/\mathbb{Q}_p} z \in \mathbb{Z}_p\}.$$

For $A \in \mathcal{S}_{\mathfrak{h}_p}$ choose s and $j_p = j_p(A, s)$ as in Lemma 6.3. Then

$$|\gamma_p(A)| \leq p^{3\iota_p} H_p^{-1} p^{-j_p/2}, \quad \|K_{p, \mathfrak{h}_p}\|_\infty \leq p^{3\iota_p} \text{vol}(R_E) \leq p^{3\iota_p}. \quad (6.16)$$

Here $\iota_p = \mathbf{1}_{\{p \leq 5\}}$, so the first constant is 1 at every $p > 5$ and is at most 125 at the three remaining primes.

If $g_p > 0$ or $p \leq 5$, then

$$\text{supp } K_{p, \mathfrak{h}_p} \subset p^{-4(g_p+1)} \mathfrak{h}_p \mathfrak{d}_E^{-1}. \quad (6.17)$$

If $g_p = 0$ and $p > 5$, then support instead requires, for each ν ,

$$z_\nu \in \begin{cases} \mathfrak{d}_\nu^{-1}, & h_\nu = 0, \\ \mathfrak{q}_\nu^{h_\nu-1} \mathfrak{d}_\nu^{-1}, & h_\nu > 0. \end{cases} \quad (6.18)$$

For a z satisfying these support conditions, define its set of boundary primes by

$$\partial_{\mathfrak{h}_p}(z) = \{\mathfrak{q}_\nu : h_\nu > 0, z_\nu \notin \mathfrak{q}_\nu^{h_\nu} \mathfrak{d}_\nu^{-1}\}.$$

In this case

$$|K_{p, \mathfrak{h}_p}(z)| \leq 8 \text{vol}(R_E) \prod_{\mathfrak{q}_\nu \in \partial_{\mathfrak{h}_p}(z)} q_\nu^{-1/2}. \quad (6.19)$$

Every constant is uniform in the lattice, the order $\mathcal{C}_L$, the shell, and the field factors and their ramification.

Proof. For $c \in \mathcal{C}_L^\#$, Equation (6.2) shows that $Q_c(L) \subset \mathbb{Z}_p$. Consequently $\gamma_p(A+c) = \gamma_p(A)$. Each denominator shell is invariant under translation by R_E : adding an integral element preserves every negative component valuation and preserves the set of integral components. Since $\mathcal{C}_L^\# \subset R_E$, the zero-extended shell restriction retains this period. Its inverse is therefore supported in the annihilator $(\mathcal{C}_L^\#)^\vee$, whose elements have integral trace because $1 \in \mathcal{C}_L^\#$. The order $\mathcal{C}_L^\#$ is an open lattice, and the shell is compact open. Thus the zero extension in Equation (6.15) is Schwartz–Bruhat, as is its Fourier inverse.

For a fixed unit u put $I_\beta(u) = \int_L \psi_p(-u^{-1}Q_\beta(x)) dx$. Polarization and additive orthogonality on the self-dual lattice give

$$|I_\beta(u)|^2 = \int_{\substack{h \in L \\ \rho(\beta)h \in L}} \psi_p(-u^{-1}Q_\beta(h)) dh.$$

Indeed, in the difference $Q_\beta(y+h) - Q_\beta(y)$ the term linear in y is $B(\rho(\beta)h, y)$, whose integral is the indicator of $\rho(\beta)h \in L$. The absolute value of the last display is at most the relative mass of $L \cap \rho(\beta)^{-1}L$. If $p^{a_1}, \dots, p^{a_6}$ are the Smith elementary divisors of $\rho(\beta)$ on L , then

$$j_p = \max\{0, -\min_i a_i\}, \quad [L : L \cap \rho(\beta)^{-1}L] = p^{\sum_i \max(0, -a_i)} \geq p^{j_p}.$$

The equality for j_p is exactly the definition of $\mathcal{C}_L$. It follows that $|I_\beta(u)| \leq p^{-j_p/2}$. Apply Equations (6.12) and (6.14), then use Equation (6.9) and average over u . This proves the first bound in Equation (6.16). The shell is contained in $\mathfrak{h}_p^{-1}$, which has volume $H_p \text{vol}(R_E)$. Integration of the first bound, with $p^{-j_p/2} \leq 1$, proves the second.

We next prove Equation (6.17). Put $N = 4(g_p + 1)$ and take $t \in p^N \mathfrak{h}_p^{-1}$. For A in the shell choose s as in Lemma 6.3, and write $\alpha = A + s$, $\beta = \alpha^{-1}$. The component valuations of t are at least $-h_\nu + e_\nu N$. Translation by t therefore preserves the shell, and the same s still has the valuation properties of that lemma. Set $\zeta = t\beta$. Equation (6.9) gives

$$\zeta \in p^{N-\iota_p} R_E, \quad \alpha + t = \alpha(1 + \zeta), \quad (\alpha + t)^{-1} - \beta = -\frac{t\beta^2}{1 + \zeta} \in p^{N-2\iota_p} \mathfrak{h}_p.$$

Here $1 + \zeta$ is a unit. It is a square unit in each factor when p is odd, by Hensel's lemma on principal units. When $p = 2$, one has $\zeta \in 8R_E$. In the ν -th field the Hensel inequality for $Y^2 - (1 + \zeta_\nu)$ at $Y = 1$ is $v_\nu(\zeta_\nu) \geq 3e_\nu > 2e_\nu = 2v_\nu(2)$, so it is a square unit there as well, including for ramified dyadic factors.

Moreover $N - 2\iota_p \geq g_p + v_p(2)$. Equation (6.11) therefore places the inverse difference in $2\mathcal{C}_L$. Its Q -value on L is integral by Equation (6.2). In Equation (6.14), the inverse integral is unchanged by this difference. The norm is unchanged by the unit $1 + \zeta$, and the phase is unchanged because that unit is a square. The original integral, for each fixed u , is consequently unchanged under $A \mapsto A + t$. So is its unit average. Translation by $p^N \mathfrak{h}_p^{-1}$ preserves the shell as a set, hence preserves its zero-extended function. The inverse transform is supported on the annihilator of this group, namely $p^{-N} \mathfrak{h}_p \mathfrak{d}_E^{-1}$. This proves Equation (6.17) with the absolute exponent 4.

It remains to prove the sharper assertions for $g_p = 0$ and $p > 5$. The equality $g_p = 0$ says $\mathfrak{h}_p \subset \mathcal{C}_L$. For each factor with $h_\nu = 0$, its component idempotent 1_ν and its whole component ring $1_\nu R_\nu$ lie in $\mathfrak{h}_p$. The self-adjoint projections $\rho(1_\nu)$ preserve L and split off their lattices orthogonally. The complementary projection $1 - \sum_{h_\nu=0} \rho(1_\nu)$ also preserves L . Each resulting summand is self-dual, because the splitting is orthogonal and L is self-dual.

On an $h_\nu = 0$ summand, an integral A_ν preserves its lattice, so Q_{A_ν} is integral there since 2 is a unit. That factor of the integral in Equation (6.5) is 1. On the combined summand of all $h_\nu > 0$, the frequency A_+ is invertible. Its inverse, extended by zero on the other factors, lies in $\mathfrak{h}_p \subset \mathcal{C}_L$. Its inverse quadratic integral in Equation (6.13) is therefore also 1. On that summand the square root of the polar determinant has absolute value H_p , because each positive factor still has rank two. Applying the formula therefore shows that the remaining frequency function is H_p^{-1} times a unit-average of Gaussian phases. Its absolute value is at most H_p^{-1} . If there are no positive h_ν , this conclusion has $H_p = 1$ and the empty summand contributes 1.

Let χ_ν be the quadratic character of $(R_\nu/\mathfrak{q}_\nu)^\times$ for a factor with $h_\nu > 0$. At odd residue characteristic every principal unit is a square, so unit square classes are exactly the two residue square classes. Phase invariance in Lemma 6.4 shows that the bounded unit-average just obtained depends only on the tuple of these classes. This use of phase invariance concerns the ambient quadratic form; a unit need not preserve the lattice. Fourier expansion on the product of the two-element square-class groups expresses the shell function as a sum of at most $2^{|\{\nu: h_\nu > 0\}|} \leq 8$ terms of the form

$$c_\varepsilon H_p^{-1} \prod_{h_\nu=0} \mathbf{1}_{R_\nu}(A_\nu) \prod_{h_\nu>0} \mathbf{1}_{\pi_\nu^{-h_\nu} R_\nu^\times}(A_\nu) \chi_\nu(\overline{\pi_\nu^{h_\nu} A_\nu})^{\varepsilon_\nu}, \quad |c_\varepsilon| \leq 1,$$

where $\varepsilon_\nu \in \{0, 1\}$. The bound for the coefficients follows from the normalized finite Fourier expansion of a function of absolute value at most 1. It also covers any coupling of components caused by the common average over u .

We compute the inverse of each factor. Put $\psi_\nu = \psi_p \circ \text{Tr}_{E_\nu/\mathbb{Q}_p}$ and use the component self-dual measure. For $h_\nu = 0$, the integral over R_ν is $\text{vol}(R_\nu) \mathbf{1}_{\mathfrak{d}_\nu^{-1}}(z_\nu)$. For $h = h_\nu > 0$, the normalized inverse of the trivial unit character is

$$q_\nu^{-h} \int_{\pi_\nu^{-h} R_\nu^\times} \psi_\nu(-Az) dA = \text{vol}(R_\nu) \left(\mathbf{1}_{\mathfrak{q}_\nu^h \mathfrak{d}_\nu^{-1}}(z) - q_\nu^{-1} \mathbf{1}_{\mathfrak{q}_\nu^{h-1} \mathfrak{d}_\nu^{-1}}(z) \right).$$

This follows by subtracting the integral over $\pi_\nu^{-h+1} R_\nu$ from that over $\pi_\nu^{-h} R_\nu$ and using their annihilators. For the quadratic unit character, its function of $a = \pi_\nu^h A$ is invariant under addition by $\mathfrak{q}_\nu$. This forces the normalized inverse to vanish outside $\mathfrak{q}_\nu^{h-1} \mathfrak{d}_\nu^{-1}$. It also vanishes on $\mathfrak{q}_\nu^h \mathfrak{d}_\nu^{-1}$, where the additive character on R_ν is trivial and the sum of χ_ν is zero. At the intervening boundary the additive character descends to a nontrivial character η of the residue field k_ν , and the normalized integral is $\text{vol}(R_\nu) q_\nu^{-1}$ times its quadratic Gauss sum.

For completeness, if χ is the quadratic character of a finite field k of odd order q and η is a nontrivial additive character, then

$$\left| \sum_{a \in k^\times} \chi(a) \eta(a) \right|^2 = \sum_{t \in k^\times} \chi(t) \sum_{b \in k^\times} \eta((t-1)b) = (q-1) - \sum_{\substack{t \in k^\times \\ t \neq 1}} \chi(t) = q.$$

Thus the quadratic factor has magnitude $\text{vol}(R_\nu) q_\nu^{-1/2}$ at the boundary. The trivial factor there has the smaller magnitude $\text{vol}(R_\nu) q_\nu^{-1}$. Multiplying these component estimates, using $H_p = \prod_{h_\nu > 0} q_\nu^{h_\nu}$ and $\prod_\nu \text{vol}(R_\nu) = \text{vol}(R_E)$, and summing at most eight terms proves Equations (6.18) and (6.19). The annihilator computation retains the different in every ramified factor. $\square$

The density need not be locally constant on the component-vanishing locus. For the nonvanishing outputs that occur in the global field, the shell inverses nevertheless give a finite pointwise expansion.

Lemma 6.6 (Pointwise inverse expansion). *If $z = (z_\nu) \in E$ has $z_\nu \neq 0$ for every ν , then*

$$f_p^*(z) = \sum_{\mathfrak{h}_p} K_{p, \mathfrak{h}_p}(z), \quad (6.20)$$

where the sum is over integral R_E -ideals and only finitely many summands are nonzero.

Proof. Let $B_n = p^{-n} R_E$. Fubini and additive orthogonality give

$$\int_{B_n} \gamma_p(A) \psi_p(-\text{Tr}_{E/\mathbb{Q}_p}(Az)) dA = \text{vol}(B_n) \mu_p(z + p^n \mathfrak{d}_E^{-1}).$$

The self-dual normalizations give

$$B_n^\perp = p^n \mathfrak{d}_E^{-1}, \quad \text{vol}(B_n) \text{vol}(B_n^\perp) = 1.$$

Thus the right side is the average density on that neighborhood. By Lemma 6.1 it equals $f_p^*(z)$ for sufficiently large n . The ball B_n is the disjoint union of the finitely many shells with $h_\nu \leq ne_\nu$.

There can in fact be only finitely many nonzero shell inverses at this z . The injection $\mathfrak{h}_p/(\mathfrak{h}_p \cap \mathcal{C}_L) \hookrightarrow R_E/\mathcal{C}_L$ gives $g_p \leq v_p([R_E : \mathcal{C}_L])$. In Equation (6.17), a nonzero value requires

$$h_\nu \leq v_\nu(z_\nu) + d_\nu + 4e_\nu(g_p + 1).$$

In Equation (6.18), it requires $h_\nu \leq v_\nu(z_\nu) + d_\nu + 1$ whenever $h_\nu > 0$. All right sides are bounded for this z , because its component valuations are finite. The shell sum therefore stabilizes, and the preceding ball inversion proves Equation (6.20). $\square$

The exact scalar coefficients in Lemma 6.2 supply the trace marginal in the next section. The nonscalar estimate uses the pointwise decay in Equation (6.16), the inverse supports in Equations (6.17) and (6.18), and the boundary bound in Equation (6.19). In the case $g_p = 0$ and $p > 5$, the last upper bound contains a factor $q_\nu^{-1/2}$ for each actual boundary component where the support permits one fewer power of q_ν . Once the real boxes are specified, Section 7 combines the supports with the nonzero field norm to restrict the denominator profiles in Poisson summation.

7 Additive distribution of the conductor weights

The factors f_p^* introduced in Section 5 retain the local lattice condition of an order at the primes in S . We now average their product on the real trace plane. The estimate will also permit a residue class at a small modulus prime to S ; this is the form needed to average the factors at the remaining primes in Section 8.

Fix a nonzero integer m . In the three real embeddings of F , write $z = (z_1, z_2, z_3)$, so that $\text{Tr } z = z_1 + z_2 + z_3$. Choose two distinct indices i, j , a fixed nonnegative function $\Phi \in C_c^\infty(\mathbb{R}^2)$ which is at least one on $[-1, 1]^2$, and a fixed nonnegative $\chi \in C_c^\infty(\mathbb{R})$ with

$$\chi(m) = 1, \quad \text{supp } \chi \subset (m - \frac{1}{3}, m + \frac{1}{3}).$$

For $c = (c_i, c_j) \in \mathbb{R}^2$ and positive r_i, r_j , put

$$\phi(z) = \Phi\left(\frac{z_i - c_i}{r_i}, \frac{z_j - c_j}{r_j}\right) \chi(\text{Tr } z), \quad A_\phi = r_i r_j \int_{\mathbb{R}^2} \Phi(u, v) du dv. \quad (7.1)$$

Thus A_ϕ is the integral of ϕ on $\text{Tr } z = m$, using z_i, z_j as coordinates on that plane. Fix $r_* > 0$, keep $r_i, r_j \leq r_*$, and require the supports of their planar factors to lie in a fixed bounded region. The supports of all the resulting ϕ 's then lie in a fixed compact subset Ω_∞ of $\mathbb{R}^3$ on which $\text{Tr } z$ stays away from zero. All constants below may depend on m, Φ, χ , the upper bound for the scales, and this bounded region. They do not depend on the locations c or on the scales within their specified range.

For $p \notin S$, the algebra $F \otimes \mathbb{Q}_p$ is unramified and its integral trace map $R_{F,p} \rightarrow \mathbb{Z}_p$ is onto. The affine set

$$\mathcal{T}_{m,p} = \{z \in R_{F,p} : \text{Tr } z = m\}$$

is consequently a translate of a free rank-two direct summand of $R_{F,p}$. A residue class on this set modulo p^a means a translate of $p^a \ker(\text{Tr} : R_{F,p} \rightarrow \mathbb{Z}_p)$. Equivalently, if $z_p^0 \in \mathcal{T}_{m,p}$ represents the class, its members satisfy $z - z_p^0 \in p^a R_{F,p}$. There are p^{2a} such classes.

Proposition 7.1 (Additive distribution). *There are absolute constants $\tau, \theta > 0$ with the following property. Let $K, \mathcal{O}, F$ be as in Section 5, put $X = [\mathcal{O}_K : \mathcal{O}] \sqrt{|\text{Disc}(K)|}$, and take S as in that section for the fixed nonzero integer m . In Equation (7.1), choose any two real partition coordinates and scales*

$$X^{-\tau} \leq r_i, r_j \leq r_*$$

whose planar supports stay in the fixed bounded region above. Let $1 \leq l \leq X^\theta$ be prime to every prime in S . For each $p \mid l$ choose any $z_p^0 \in \mathcal{T}_{m,p}$, specifying a class modulo $p^{v_p(l)}$. Then

$$\sum_{\substack{z \in F, \text{Tr } z = m \\ z \in R_{F,p} \text{ (} p \notin S \text{)} \\ z - z_p^0 \in p^{v_p(l)} R_{F,p} \text{ (} p \mid l \text{)}}} \phi(z) \prod_{p \in S} f_p^*(z) = A_\phi l^{-2} \prod_{p \in S} d_p(m) (1 + O(X^{-\theta})). \quad (7.2)$$

Here d_p is the trace density of Lemma 6.2. The error is uniform over all the fields, orders, ordered real embeddings, permitted box locations and scales, moduli l , and chosen residue classes. The implied constant and the lower threshold for X depend only on the fixed real data listed above and on m .

7.1 Finite expansion and Poisson summation

We first assemble the local inverse transforms into a finite global expansion. This is the point at which the fact that the cubic algebra is a field is essential. Let $\mathfrak{d}_F$ be the different of F , and write $\mathfrak{d}_p = \mathfrak{d}_F R_{F,p}$ for its localization. For a choice of denominator shells at the primes of S , write

$$\mathfrak{h} = \prod_{p \in S} \mathfrak{h}_p, \quad H = \prod_{p \in S} H_p = N \mathfrak{h}, \quad G_0 = \prod_{p \in S} p^{g_p}.$$

Local ideals here and below are identified with global ideals supported at the corresponding primes. The functions $K_{p,\mathfrak{h}_p}$ and their boundary sets $\partial_{\mathfrak{h}_p}(z)$ are those of Lemma 6.5. Set

$$\mathcal{L}_{\mathfrak{h}} = \{p \in S : g_p > 0 \text{ or } p \leq 5\}.$$

Lemma 7.2 (Global bounds for denominator profiles). *For every profile one has*

$$G_0 \leq [R_F : C_F] \leq H G_0. \quad (7.3)$$

Suppose that $z \in F$ is integral outside S , its real image lies in Ω_∞ , and $\prod_{p \in S} K_{p,\mathfrak{h}_p}(z) \neq 0$. Define

$$T_\partial = T_\partial(\mathfrak{h}, z) = \prod_{\substack{p \in S \setminus \mathcal{L}_{\mathfrak{h}} \\ \mathfrak{q} \in \partial_{\mathfrak{h}_p}(z)}} N \mathfrak{q},$$

where an empty product is one. There is a constant depending only on Ω_∞ such that

$$H \ll D_F G_0^{24} T_\partial, \quad T_\partial \leq \left(\prod_{p \in S} p \right)^3, \quad (7.4)$$

$$\left| \prod_{p \in S} K_{p,\mathfrak{h}_p}(z) \right| \ll 8^{|S|} T_\partial^{-1/2}. \quad (7.5)$$

All such z lie in a fractional R_F -ideal $\mathfrak{a}_{\mathfrak{h}}$, independent of z , with

$$N(\mathfrak{a}_{\mathfrak{h}})^{-1} \leq 30^{12} D_F G_0^{24}. \quad (7.6)$$

Set $C_3 = 371$. For all sufficiently large X , depending only on the fixed real data and m , a profile contributing at any such z has $H \leq X^{C_3}$. For these X and every $z \in F$ integral outside S with real image in Ω_∞ ,

$$\prod_{p \in S} f_p^*(z) = \sum_{\substack{\mathfrak{h} = (\mathfrak{h}_p)_{p \in S} \\ H \leq X^{C_3}}} \prod_{p \in S} K_{p,\mathfrak{h}_p}(z). \quad (7.7)$$

The number of profiles in this sum is $O_\eta(X^\eta)$ for every fixed $\eta > 0$, uniformly in the arithmetic data.

Proof. Let $M_p = \mathfrak{h}_p \cap C_{F,p}$. The additive lattice identity

$$[R_{F,p} : C_{F,p}] = [R_{F,p} : C_{F,p} + \mathfrak{h}_p] [\mathfrak{h}_p : M_p]$$

has first factor between 1 and $[R_{F,p} : \mathfrak{h}_p] = H_p$. The index of C_F is supported on S by Lemma 5.9; multiplying gives Equation (7.3).

At $p \in \mathcal{L}_{\mathfrak{h}}$, the local support is contained in $p^{-4(g_p+1)}\mathfrak{h}_p\mathfrak{d}_p^{-1}$. At every other $p \in S$, the support loses at most one power of each boundary prime $\mathfrak{q}$, and loses no power in the other components. It follows that a supporting z belongs to

$$\mathfrak{b}_{\mathfrak{h},z} = \mathfrak{h}\mathfrak{d}_F^{-1} \prod_{p \in \mathcal{L}_{\mathfrak{h}}} (pR_F)^{-4(g_p+1)} \prod_{\substack{p \in S \setminus \mathcal{L}_{\mathfrak{h}} \\ \mathfrak{q} \in \partial_{\mathfrak{h}_p}(z)}} \mathfrak{q}^{-1}.$$

The real trace condition excludes $z = 0$. Since F is a field, its norm is nonzero, and containment of the principal ideal (z) in $\mathfrak{b}_{\mathfrak{h},z}$ gives

$$|\mathrm{N}_{F/\mathbb{Q}} z| = \mathrm{N}(\mathfrak{b}_{\mathfrak{h},z})[\mathfrak{b}_{\mathfrak{h},z} : (z)] \geq \frac{H}{D_F T_{\partial} \prod_{p \in \mathcal{L}_{\mathfrak{h}}} p^{12(g_p+1)}}.$$

The left side is bounded above on Ω_{∞} . For $g_p > 0$ we have $g_p + 1 \leq 2g_p$; the only additional factors, for $g_p = 0$, are at 2, 3, 5. Hence

$$\prod_{p \in \mathcal{L}_{\mathfrak{h}}} p^{12(g_p+1)} \leq 30^{12} G_0^{24}.$$

This proves the first inequality of Equation (7.4). Each boundary prime occurs once, and $\sum_{\mathfrak{q}|p} f_{\mathfrak{q}} \leq \sum_{\mathfrak{q}|p} e_{\mathfrak{q}} f_{\mathfrak{q}} = 3$; this proves the bound for T_{∂} . Equation (7.5) is the product of the boundary bounds of Lemma 6.5; its absolute implied constant accounts for the three small primes.

Dropping $\mathfrak{h}_p$ in the first support condition enlarges that support. In the second, the exponent $v_{\mathfrak{q}}(\mathfrak{h}_p) - 1$ at a boundary is nonnegative. Thus every supporting point belongs to the ideal

$$\mathfrak{a}_{\mathfrak{h}} = \mathfrak{d}_F^{-1} \prod_{p \in \mathcal{L}_{\mathfrak{h}}} (pR_F)^{-4(g_p+1)},$$

which gives Equation (7.6).

Lemma 5.9 gives $\Delta_F \ll X^{28}$, $D_F \leq \Delta_F$, and $[R_F : C_F] \leq \Delta_F^{1/2}$. The definition of S gives $\prod_{p \in S} p \ll_m X^2$. Equations (7.3)–(7.4) therefore give $H \ll X^{28+24 \cdot 14+6} = X^{370}$. The fixed implied constant is absorbed by X^{371} for sufficiently large X .

A nonzero field element has nonzero image in every component of $F \otimes \mathbb{Q}_p$. Lemma 6.6 therefore identifies the density $f_p^*(z)$ with its pointwise finite sum of inverse shells at all the present z . The norm bound just proved excludes every product with $H > X^{371}$, uniformly in z , and proves Equation (7.7). Finally, for any fixed $u > 0$, Rankin's inequality bounds the number of profiles with $H \leq X^{C_3}$ by

$$X^{C_3 u} \prod_{p \in S} (1 - p^{-u})^{-3}.$$

There are at most three prime-ideal components over each p . The radical bound on S implies $|S| = O(\log X / \log \log X)$, since the product of k distinct primes is at least $k!$. The displayed Euler product is then $X^{o(1)}$ for fixed u . Taking u arbitrarily small proves the asserted bound for every $\eta > 0$. $\square$

We apply this expansion before imposing Poisson summation. For $p \notin S$ let

$$\eta_{p,l}(z) = \begin{cases} \mathbf{1}_{z_p^0 + p^{v_p(l)} R_{F,p}}(z), & p \mid l, \\ \mathbf{1}_{R_{F,p}}(z), & p \nmid l. \end{cases}$$

Let $\eta_{\mathfrak{h},l}$ be the finite-adelic test function whose local factor is $K_{p,\mathfrak{h}_p}$ at $p \in S$ and $\eta_{p,l}$ elsewhere. By Lemma 6.5, every $K_{p,\mathfrak{h}_p}$ is supported on elements of integral trace. The factors outside S are supported in $R_{F,p}$ and also have integral trace. Thus any $z \in F$ with $\eta_{\mathfrak{h},l}(z) \neq 0$ has rational trace integral at every finite prime, hence an integer. Only m meets $\text{supp } \chi$. The finite profile expansion in Lemma 7.2 consequently shows that the left side of Equation (7.2), denoted by Σ , equals

$$\Sigma = \sum_{\substack{\mathfrak{h} \\ H \leq X^{C_3}}} S_{\mathfrak{h}}, \quad S_{\mathfrak{h}} = \sum_{z \in F} \phi(z) \eta_{\mathfrak{h},l}(z). \quad (7.8)$$

Here $C_3 = 371$ is fixed. The preceding lemma makes the profile interchange finite. Each inner sum also has finitely many terms, since its finite conditions place z in a fractional ideal and its real support is compact.

In the local results of Section 6, choose the finite components of the standard character of $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$; they all have conductor $\mathbb{Z}_p$. Compose this global character with the field trace. At a finite place the self-dual measure satisfies $\text{vol}(R_{F,p}) = |D_F|_p^{1/2}$, while at infinity it is Lebesgue measure in the three embeddings. The real covolume of R_F is $\sqrt{D_F}$ and $\prod_p \text{vol}(R_{F,p}) = D_F^{-1/2}$. Thus the diagonal copy of F has covolume one for these measures. Write

$$\widehat{\phi}(A) = \int_{\mathbb{R}^3} \phi(z) \psi_{\infty}(\text{Tr}(Az)) dz$$

for the real transform. Field Poisson summation in this normalization [26, Section 4.2, Lemma 4.2.4] gives

$$S_{\mathfrak{h}} = \sum_{A \in F} \widehat{\phi}(A) \widehat{\eta}_{\mathfrak{h},l}(A), \quad \widehat{\eta}_{\mathfrak{h},l}(A) = \prod_{p \in S} \gamma_{p,\mathfrak{h}_p}(A) \prod_{p \notin S} \widehat{\eta}_{p,l}(A). \quad (7.9)$$

Here $\gamma_{p,\mathfrak{h}_p}$ is the shell restriction of γ_p , extended by zero, and hats use the chosen character. Each finite test is Schwartz–Bruhat, so the formula is also ordinary lattice Poisson summation with congruence classes. At $p \notin S$, trace duality is unimodular and

$$\widehat{\eta}_{p,l}(A) = p^{-3v_p(l)} \psi_p(\text{Tr}(Az_p^0)) \mathbf{1}_{p^{-v_p(l)} R_{F,p}}(A) \quad (p \mid l);$$

for $p \nmid l$ it is $\mathbf{1}_{R_{F,p}}$. In particular every such transform is bounded in absolute value by one. The finite Fourier factors in Equation (7.9) are supported on the fractional ideal

$$\Gamma_{l,\mathfrak{h}} = l^{-1} \mathfrak{h}^{-1}, \quad P = l^3 H, \quad N(\Gamma_{l,\mathfrak{h}}) = P^{-1}. \quad (7.10)$$

The following elementary geometry of this ideal will be used both to count frequencies and to express the local congruences in global integer coordinates.

Lemma 7.3 (Lattice bounds for a cubic ideal). *Let $\mathfrak{a}$ be a fractional ideal of a totally real cubic field of discriminant D_F , and write $N(\mathfrak{a}) = P^{-1}$ with $P > 0$. Let $B_Y = \{x \in \mathbb{R}^3 : \|x\| \leq Y\}$. In embedding coordinates it has covolume $\sqrt{D_F}/P$, first Euclidean minimum $\gg P^{-1/3}$, and, for $Y > 0$,*

$$\#\{A \in \mathfrak{a} : \|A\| \leq Y\} \ll 1 + YP^{1/3} + Y^2 P^{2/3} + \frac{Y^3 P}{\sqrt{D_F}}. \quad (7.11)$$

There is a $\mathbb{Z}$ -basis b_1, b_2, b_3 of $\mathfrak{a}$ in which every $A = \sum n_i b_i$ with $\|A\| \leq Y$ satisfies

$$|n_i| \ll Y P^{1/3} \quad (1 \leq i \leq 3).$$

All constants here are absolute.

Proof. The covolume formula follows from the discriminant. For $0 \neq A \in \mathfrak{a}$, containment $(A) \subset \mathfrak{a}$ implies $|\mathrm{N}_{F/\mathbb{Q}} A| \geq \mathrm{N} \mathfrak{a} = P^{-1}$. The product inequality in three real coordinates gives $|\mathrm{N} A| \leq (\|A\|/\sqrt{3})^3$, proving the first-minimum bound.

Let $\lambda_1 \leq \lambda_2 \leq \lambda_3$ be the Euclidean successive minima. A lattice in fixed dimension has a basis of bounded orthogonality defect, by Mahler's basis theorem [15, Section 2, Theorem 1]. For such a basis, order its lengths increasingly and write its defect as $\omega = \prod_i \|b_i\| / \mathrm{covol}(\mathfrak{a}) \ll 1$. Cramer's rule and Hadamard's inequality bound the norm of the i -th coordinate functional by $\omega / \|b_i\|$. Since $\|b_i\| \geq \lambda_i$, counting its possible integer coordinates gives

$$\#(\mathfrak{a} \cap B_Y) \ll \prod_{i=1}^3 (1 + Y/\lambda_i).$$

Minkowski's second theorem for the Euclidean ball, in the formulation of [28, Theorem 4.3], gives $\lambda_1 \lambda_2 \lambda_3 \asymp \mathrm{covol}(\mathfrak{a})$. Expanding the last product, using $\lambda_2 \geq \lambda_1$, and inserting the first-minimum and covolume bounds proves Equation (7.11). The same coordinate functional bound with $\|b_i\| \geq \lambda_1$ proves the final assertion. $\square$

7.2 The nonscalar contribution

Scalar frequencies $A \in \mathbb{Q} \subset F$ in Equation (7.9) record only the trace. The main work is to show that the remaining frequencies contribute an absolute power saving. We prove that estimate before evaluating the scalar terms.

Lemma 7.4 (Nonscalar contribution). *There are absolute $\tau, \theta, \epsilon_F > 0$, with $\epsilon_F > 2\tau + 3\theta$, such that for $C_3 = 371$ and the data of Proposition 7.1 with these τ, θ ,*

$$\sum_{\substack{\mathfrak{h} \\ H \leq X^{C_3}}} \left| \sum_{A \in F \setminus \mathbb{Q}} \widehat{\phi}(A) \widehat{\eta}_{\mathfrak{h}, l}(A) \right| \ll X^{-\epsilon_F}. \quad (7.12)$$

The constants have the same uniformity as in that proposition.

Proof. We use $\lambda = 1/3$. Lemma 5.9, which gives $\Delta_F \geq X^{1/2}/2$, implies

$$\Delta_F \geq X^\lambda \quad (7.13)$$

for all sufficiently large X .

After bounding the real Fourier tail, we split according to the field discriminant. If D_F is large, comparing it with the discriminant of the order generated by a retained nonscalar frequency after clearing its denominators forces H large. Otherwise the lower bound for Δ_F makes the index $[R_F : C_F]$ large. For profiles with small G_0 , this forces H large; inverse support and boundary decay then bound the possibly signed original sum $S_{\mathfrak{h}}$. Poisson expresses its full nonscalar Fourier sum as $S_{\mathfrak{h}}$ minus the separately estimated scalar series. For profiles with large G_0 , the inverse depths attached to each frequency and its chosen shifts provide either sufficient pointwise

decay or congruences; in the latter range a further discriminant comparison supplies the required denominator size.

The positive exponents $d, e, e_1, e_2, w, \theta, \tau$ below will be fixed at the end of the proof; put $Y = X^w$ and $s = w + \theta$.

First the frequencies outside $\|A\| \leq Y$ can be discarded. For every fixed integer $r \geq 0$, integration by parts in the fixed real coordinates gives, uniformly over the box family,

$$|\widehat{\phi}(A)| \ll_r X^{r\tau} (1 + \|A\|)^{-r}. \quad (7.14)$$

Indeed each derivative of order r costs at most a constant times $X^{r\tau}$, and all supports are in one compact set. Lemma 6.5 and the bounds for $p \notin S$ give $|\widehat{\eta}_{\mathfrak{h},l}(A)| \ll H^{-1}$. The first-minimum bound also gives the rough count $\#(\Gamma_{l,\mathfrak{h}} \cap B_R) \ll (1 + RP^{1/3})^3$. For $R \geq Y \geq 1$ this is $O(R^3P)$, since $P = l^3H \geq 1$. Dyadic summation of Equation (7.14), for $r > 3$, yields

$$\sum_{\substack{A \in F \\ \|A\| > Y}} |\widehat{\phi}(A) \widehat{\eta}_{\mathfrak{h},l}(A)| \ll_r X^{r\tau} l^3 Y^{3-r} \leq X^{r\tau+3\theta+w(3-r)} O_r(1). \quad (7.15)$$

This also proves absolute convergence in Equation (7.9). The profile count costs only $X^{o(1)}$. When bounding Fourier terms directly, we may therefore restrict to the ball and use the uniform bound $|\widehat{\phi}| \ll 1$ there.

Suppose first that $D_F \geq X^d$. A nonscalar A generates F , since the field has degree three. Moreover $HR_F \subset \mathfrak{h}$: the integer $H = |R_F/\mathfrak{h}|$ annihilates this finite group. Thus $lHA \in R_F$ for $A \in \Gamma_{l,\mathfrak{h}}$. The discriminant of the order $\mathbb{Z}[lHA]$ is a positive integer multiple of D_F , whereas the product of the squared differences of its three real conjugates is $O((lHY)^6)$. Existence of a nonscalar frequency in the ball therefore implies

$$D_F \ll (lHY)^6, \quad H \gg X^{d/6-s}.$$

Apply Equation (7.11) to $\Gamma_{l,\mathfrak{h}}$ and multiply it by H^{-1} . The four resulting terms are bounded by constants times

$$X^{-d/6+s} + X^{-d/9+5s/3} + X^{-d/18+7s/3} + X^{-d/2+3s}. \quad (7.16)$$

All have a fixed power saving once $s < d/100$. If the ball contains no nonscalar frequency there is nothing to count for that profile.

We now assume $D_F < X^d$. When $d \leq \lambda/3$, Equation (7.13) gives

$$[R_F : C_F] = (\Delta_F/D_F)^{1/2} \geq X^{\lambda/3}.$$

Profiles with $G_0 \leq X^e$ are best estimated in their original form $S_{\mathfrak{h}}$. Equation (7.3) implies $H \geq X^{\lambda/3-e}$. All supporting points lie in $\mathfrak{a}_{\mathfrak{h}}$, whose reciprocal norm is $O(D_F G_0^{24})$ by Equation (7.6). The first-minimum bound of Lemma 7.3, applied in the fixed real region, gives $O(D_F G_0^{24})$ possible points. At each such point, Equations (7.4)–(7.5) give

$$\left| \prod_{p \in S} K_{p,\mathfrak{h}_p}(z) \right| \ll X^{o(1)} \left(\frac{D_F G_0^{24}}{H} \right)^{1/2}.$$

Here and below $X^{o(1)}$ denotes $O_{\eta}(X^{\eta})$ for every fixed $\eta > 0$, uniformly in the arithmetic data; it includes factors $8^{|S|}$. It follows that

$$|S_{\mathfrak{h}}| \ll X^{-\lambda/6+(3/2)d+(73/2)e+o(1)}. \quad (7.17)$$

The use of the original sum has gained the boundary factor from the inverse transform. It remains to check that its scalar portion is small as well, so that this bounds the nonscalar portion.

If a rational scalar has denominator p^k at p , its denominator ideal in $R_{F,p}$ is $p^k R_{F,p}$; its component exponents are $e_q k$, so its norm is p^{3k} , including at ramified primes. Thus a scalar in this profile has its S -part denominator $t = H^{1/3}$, or the profile contains no scalar. Its denominator away from S divides l , so there are $O(1 + YlH^{1/3})$ such rationals in the ball. Their absolute contribution is at most

$$O(H^{-1} + YlH^{-2/3}). \quad (7.18)$$

By $H \geq X^{\lambda/3-e}$ this also saves a power for the choices below. Equation (7.9) expresses the full dual sum as $S_{\mathfrak{h}}$. Subtracting its scalar frequencies in the ball and using Equation (7.15) for the scalar tail therefore bounds the full nonscalar sum for these profiles by Equations (7.17)–(7.18) and the tail. No positivity of an individual inverse shell is needed.

It remains to treat $D_F < X^d$ and $G_0 > X^e$. For each frequency in the denominator shell of the profile choose at each $p \in S$ one of the shifts s_p of Lemma 6.3, and write

$$\alpha_p = A + s_p, \quad \beta_p = \alpha_p^{-1}, \quad J = \prod_{p \in S} p^{j_p},$$

where j_p is the least nonnegative integer with $p^{j_p} \beta_p \in C_{F,p}$. All sums involving J below are over frequencies in this profile. The pointwise local estimate now reads

$$|\hat{\eta}_{\mathfrak{h},l}(A)| \ll H^{-1} J^{-1/2}.$$

For frequencies with $J \geq X^{e_2}$, the rough lattice count immediately gives

$$\sum_{\substack{\|A\| \leq Y \\ J \geq X^{e_2}}} |\hat{\phi}(A) \hat{\eta}_{\mathfrak{h},l}(A)| \ll \frac{(1 + YP^{1/3})^3}{H} X^{-e_2/2} \ll X^{3s-e_2/2}. \quad (7.19)$$

We used $H, l, Y \geq 1$. This leaves $J < X^{e_2}$, where the extra factor in the Fourier bound may be too small. The order discriminant then forces a large denominator ideal, and the definition of j_p supplies congruences among its possible frequencies.

For a nonscalar A let $\text{disc}(A) = \text{Disc}(1, A, A^2) \in \mathbb{Q}^\times$. The equality $F = \mathbb{Q}[A]$ remains $F \otimes \mathbb{Q}_p = \mathbb{Q}_p[A]$ after tensoring. Shifting by a scalar and inverting an element with nonzero components preserve the generated algebra, so $p^{j_p} \beta_p$ generates $F \otimes \mathbb{Q}_p$. The full-rank order $\mathbb{Z}_p[p^{j_p} \beta_p]$ is contained in $C_{F,p}$. The identities

$$\text{disc}(\beta_p) = \frac{\text{disc}(A)}{N_{F_p/\mathbb{Q}_p}(\alpha_p)^4}, \quad \text{disc}(p^{j_p} \beta_p) = p^{6j_p} \text{disc}(\beta_p)$$

follow by taking the three differences of conjugates and by scaling. Discriminant divisibility therefore gives

$$v_p(\text{disc}(A)) - 4v_p(N\alpha_p) + 6j_p \geq v_p(\Delta_F).$$

The shift lemma says $v_p(N\alpha_p) = -\log_p H_p$ for $p > 5$, and differs from this by at most three at 2, 3, 5. For $p \notin S$, lA is integral and $C_{F,p} = R_{F,p}$ is unramified, so $v_p(\text{disc}(A)) \geq -6v_p(l)$. Combining these valuation bounds for the nonzero rational $\text{disc}(A)$, with only an absolute factor from 2, 3, 5, gives

$$Y^6 \gg |\text{disc}(A)| \gg \Delta_F J^{-6} H^{-4} l^{-6}. \quad (7.20)$$

The first inequality uses $\|A\| \leq Y$. Thus for $J < X^{e_2}$, Equations (7.13) and (7.20) imply

$$H \gg X^{\lambda/4 - (3/2)(s+e_2)}. \quad (7.21)$$

In particular, once $s + e_2 < \lambda/12$ with a margin, every nonscalar frequency under consideration has $H \geq X^{\lambda/8}$ for large X .

For the remaining count, group frequencies according to their j_p 's and the chosen s_p 's. The number of exponent lists with $J < X^{e_2}$ is $X^{o(1)}$, by the same Rankin argument as for profiles; the bounded lists of shifts cost $4^{|S|} = X^{o(1)}$. Groups containing no nonscalar frequency contribute zero and are omitted. Fix one of the other groups. Apply the basis part of Lemma 7.3 to $\Gamma_{l,\mathfrak{h}}$. All frequencies in the ball have integer coordinates in a cube $[-L, L]^3$, where we choose L comparable to the upper bound $Y^{1/3}$. This choice, together with Equation (7.21), gives

$$L \asymp Y^{1/3}, \quad L \gg X^{\lambda/24}, \quad L^3/H \ll X^{3s}. \quad (7.22)$$

At $p \in S$ choose a generator i_p of the maximal-order ideal $\mathfrak{h}_p$, and set $x = i_p(A + s_p)$. Localizing the global basis of $\Gamma_{l,\mathfrak{h}}$ at p , then multiplying it by i_p , gives a $\mathbb{Z}_p$ -basis of $R_{F,p}$, because $p \nmid l$. Accordingly the map from the three integer coordinates to x is an affine $\mathrm{GL}_3(\mathbb{Z}_p)$ change of coordinates. This statement holds also for a product of ramified local fields.

Smith normal form for $\mathfrak{h}_p \cap C_{F,p}$ inside $\mathfrak{h}_p$ gives a primitive linear form $\ell_p : R_{F,p} \rightarrow \mathbb{Z}_p$, meaning that its image is $\mathbb{Z}_p$, and an integer $g'_p \geq g_p/3$ such that

$$\ell_p(i_p^{-1}(\mathfrak{h}_p \cap C_{F,p})) \subset p^{g'_p}\mathbb{Z}_p.$$

Indeed the three elementary-divisor exponents sum to g_p , and we use the coordinate with largest exponent. If $p > 5$, the shift lemma makes x a unit and $\beta_p = i_p x^{-1} \in \mathfrak{h}_p$. The condition $p^{j_p}\beta_p \in C_{F,p}$ implies

$$\ell_p(x^{-1}) \equiv 0 \pmod{p^k} \quad \text{for every } 1 \leq k \leq (g'_p - j_p)_+. \quad (7.23)$$

Inversion permutes the units in the finite algebra $R_{F,p}/p^k R_{F,p}$. A primitive linear form has p^{2k} zeros among the p^{3k} additive residues. Thus the unit condition and Equation (7.23) retain at most the fraction p^{-k} of all residues modulo p^k . Nilpotents in the finite algebra do not affect this argument.

For $p = 2, 3, 5$, the shift lemma gives $x \in R_{F,p}$ and $x^{-1} \in p^{-1}R_{F,p}$. Put $y = px^{-1}$. Then $p^{j_p}i_p y = p(p^{j_p}\beta_p)$ still lies in $\mathfrak{h}_p \cap C_{F,p}$, and the congruence in Equation (7.23) holds for y in place of x^{-1} . On the permitted locus the inverse identities

$$y(x) - y(x') = px^{-1}(x' - x)(x')^{-1}, \quad x - x' = p^{-1}x(y(x') - y(x))x'$$

show respectively that $x \bmod p^{k+1}$ determines $y \bmod p^k$, and that a fixed residue of $y \bmod p^k$ confines x to one residue modulo p^{k+1} . The permitted locus has $v_q(x) = 0$ on the pole components and $0 \leq v_q(x) \leq e_q$ on the other components. These conditions are determined modulo p^2 , since a change in $p^2 R_{F,p}$ has component valuation at least $2e_q$. At modulus p^{k+1} , each of the at most p^{2k} residues in the linear kernel therefore allows at most p^6 residues of x . The retained fraction is at most p^{3-k} , hence $O(p^{-k})$ with an absolute constant at these three primes. This proves the required congruence fraction with only one extra power of a small prime in its modulus.

Write $a_p = (g'_p - j_p)_+$. The available congruence powers satisfy

$$\prod_{p \in S} p^{a_p} \geq G_0^{1/3}/J > X^{e/3 - e_2}. \quad (7.24)$$

We will choose $e/3 - e_2 > 2e_1$. If every prime with $a_p > 0$ is at most X^{e_1} , multiply available factors p , one at a time, until their product M_1 first reaches $X^{e_1/2}$. Equation (7.24) ensures this is possible, and

$$X^{e_1/2} \leq M_1 \leq X^{3e_1/2} < X^{2e_1}.$$

Impose just the corresponding congruences. The Chinese remainder theorem combines their retained fractions into $O(M_1^{-1})$. Their joint modulus is M_1 times at most $2 \cdot 3 \cdot 5$, because only the small-prime tests need one extra power. The affine integral coordinate changes above preserve these fractions in the global coordinates. If $2e_1 < \lambda/24$, this joint modulus is $o(L)$ by Equation (7.22). Counting each admissible residue class in the cube then gives

$$O(L^3/M_1) = O(L^3 X^{-e_1/2}) \quad (7.25)$$

possible frequencies.

Otherwise some available prime satisfies $p > X^{e_1}$; for large X it exceeds 5. Use its congruence modulo p . Set $B = R_{F,p}/pR_{F,p}$, a three-dimensional commutative $\mathbf{F}_p$ -algebra, possibly with nilpotents, and reduce ℓ_p modulo p . Choose an $\mathbf{F}_p$ -basis of B and let x be the universal element of $B \otimes_{\mathbf{F}_p} \mathbf{F}_p[X_1, X_2, X_3]$ in this basis. Let M_x be its multiplication matrix and put $\mathcal{N}_B(x) = \det M_x$. This is a nonzero polynomial of degree three, since $\mathcal{N}_B(1) = 1$. On the unit open set $\mathcal{N}_B \neq 0$,

$$\ell_p(x^{-1}) = \frac{P_\ell(x)}{\mathcal{N}_B(x)}, \quad P_\ell(x) = \ell_p(\text{adj}(M_x)1_B),$$

where P_ℓ has degree at most two. This numerator is nonzero. Indeed the affine space parametrizing elements of the vector space B has coordinate ring $\mathbf{F}_p[X_1, X_2, X_3]$, which is a domain even when multiplication in B has nilpotents. The unit open set is dense, and inversion is an automorphism of that open set. The nonzero linear polynomial ℓ_p stays nonzero on the open set and after composition with inversion. The displayed formula, or injectivity of localization at $\mathcal{N}_B$, then forces $P_\ell \neq 0$. The affine $\text{GL}_3(\mathbf{F}_p)$ change from global lattice coordinates preserves nonvanishing.

Let $N_L = 2\lfloor L \rfloor + 1 \asymp L$ be the number of possible integers in one coordinate of the cube. A nonzero polynomial of bounded degree has $O(N_L^2)$ zeros on a product of three residue intervals of length $N_L < p$; this follows by induction on the number of variables, viewing it as a polynomial in the last variable and bounding the slices on which all its coefficients vanish. If $N_L \geq p$, partition the cube into boxes of side at most p and use the same bound there. The number of global integer coordinates satisfying $P_\ell = 0 \pmod p$ is consequently

$$O(L^3/p + L^2) = O(L^3(X^{-e_1} + X^{-\lambda/24})). \quad (7.26)$$

This bounds the subset for which x is a unit as well. Multiplying Equations (7.25)–(7.26) by H^{-1} , and using Equation (7.22), gives respectively the bounds

$$O(X^{3s-e_1/2}), \quad O(X^{3s-e_1} + X^{3s-\lambda/24})$$

for the fixed group of j_p 's and shifts. Their $X^{o(1)}$ number is harmless once these exponents are negative with a margin. This completes the count of nonscalar frequencies in the ball.

We finish by making the parameter choices and all margins explicit. Put $\lambda_0 = \min(\lambda, 1)$ and take

$$\begin{aligned} d = e &= \frac{\lambda_0}{1000}, & e_1 &= \frac{e}{100}, & e_2 &= \frac{e_1}{100}, \\ w &= \frac{e_2}{100}, & \theta = \tau &= \frac{w}{100}, & \epsilon_F &= \frac{e_2}{4}. \end{aligned} \quad (7.27)$$

Then $s = 101e_2/10000$,

$$s < d/100, \quad d < \lambda/3, \quad e/3 - e_2 > 2e_1, \quad 2e_1 < \lambda/24, \quad s + e_2 < \lambda/12.$$

The saving in Equation (7.17) is at least

$$\lambda/6 - (3/2)d - (73/2)e \geq (1/6 - 0.038)\lambda_0,$$

and Equation (7.18) has a still larger margin than ϵ_F , since $e \leq \lambda/1000$. All four exponents in Equation (7.16) also have margins at least $29d/900 > \epsilon_F$, by $s < d/100$. Among the remaining ball estimates the smallest margin is in Equation (7.19), namely

$$e_2/2 - 3s = \frac{4697}{10000}e_2 > \epsilon_F;$$

the congruence margins $e_1/2 - 3s$, $e_1 - 3s$, and $\lambda/24 - 3s$ are larger. With $r = 100$, the exponent in Equation (7.15) is

$$r\tau + 3\theta + w(3 - r) = -\frac{9597}{10000}e_2.$$

All $X^{o(1)}$ costs from profiles, shifts, and exponent lists can thus be absorbed while retaining $\epsilon_F = e_2/4$. This proves Equation (7.12). Finally $2\tau + 3\theta = e_2/2000 < \epsilon_F$, as asserted. $\square$

7.3 The scalar term and completion

Proof of Proposition 7.1. Use the constants in Equation (7.27). Equations (7.8) and (7.9), together with Lemma 7.4, leave the scalar frequencies with absolute error $O(X^{-\epsilon_F})$. We first remove the profile cutoff from those frequencies.

For $a \in \mathbb{Q}$, let $t = \prod_{p \in S} p^{\max(0, -v_p(a))}$ be its S -part denominator, with $t = 1$ for $a = 0$. As noted above, its unique profile has $H = t^3$. Lemma 6.2 gives the exact product $\prod_{p \in S} \gamma_p(a) = t^{-3}$, including at 2, 3, 5. The other denominator divides l , so all such scalars with a given t lie in $(tl)^{-1}\mathbb{Z}$. The change of real variables $(z_i, z_j, z_k) \mapsto (z_i, z_j, \text{Tr } z)$ has absolute Jacobian one. Writing $\mathbf{1} = (1, 1, 1)$, it gives the exact identity

$$\widehat{\phi}(a\mathbf{1}) = A_\phi \widehat{\chi}(a).$$

For any fixed $r > 1$, rapid decay of the fixed function χ implies $\sum_{n \in \mathbb{Z}} |\widehat{\chi}(n/(tl))| \ll_r tl$. The scalars omitted by $H \leq X^{C_3}$ therefore contribute at most

$$\sum_{t > X^{C_3/3}} t^{-3} \sum_{n \in \mathbb{Z}} |\widehat{\phi}((n/(tl))\mathbf{1})| \ll A_\phi l X^{-C_3/3}. \quad (7.28)$$

We have enlarged the set of t 's to all positive integers, which is an upper bound. With $C_3 = 371$, this is $O(X^{-\epsilon_F})$, since $A_\phi \ll 1$, $l \leq X^\theta$, and $371/3 - \theta > \epsilon_F$. The same calculation without the cutoff proves absolute convergence of the unrestricted scalar series.

That series is one-dimensional Poisson summation for the trace marginals. At infinity the marginal is $A_\phi \chi(t)$. At $p \in S$ it is $d_p(t) \mathbf{1}_{\mathbb{Z}_p}(t)$. At $p \nmid l$, the onto integral trace map and two kernel coordinates show that the indicator $z_p^0 + p^{v_p(l)} R_{F,p}$ has trace density

$$p^{-2v_p(l)} \mathbf{1}_{m+p^{v_p(l)}\mathbb{Z}_p}(t)$$

with respect to Haar measure of mass one on $\mathbb{Z}_p$. At every other finite prime the marginal is $\mathbf{1}_{\mathbb{Z}_p}$. Their Fourier transforms at a rational a are exactly the finite factors in Equation (7.9) evaluated at the scalar a , because $\text{Tr}(az) = a \text{Tr } z$.

For completeness, this one-dimensional application of Poisson does not require a smoothness assumption on d_p . Its Fourier coefficients on $\mathbb{Q}_p/\mathbb{Z}_p$ of denominator p^k are p^{-3k} , and there are $p^k - p^{k-1}$ such characters. Their absolute sum is finite. Truncating each expansion gives a compactly supported locally constant density on $\mathbb{Z}_p$, to which Poisson applies. The densities converge uniformly by Lemma 6.2, and the preceding absolute scalar bound permits passage to the limit on the Fourier side. On the original side, the finite marginals force the rational trace to be an integer. Only m meets the real cutoff. Hence the unrestricted scalar contribution is precisely

$$A_\phi l^{-2} \prod_{p \in S} d_p(m).$$

Lemma 6.2 bounds the product of $d_p(m)$ above and below by positive absolute constants. Also $A_\phi \gg X^{-2\tau}$ by Equation (7.1), and $l^{-2} \geq X^{-2\theta}$. Dividing the combined absolute error $O(X^{-\epsilon_F})$ by the main term therefore costs at most $O(X^{2\tau+2\theta})$. Since $\epsilon_F > 2\tau + 3\theta$, the relative error is $O(X^{-\theta})$. Every estimate used fixed real derivative and support bounds and constants independent of the fields, orders, residues, and box locations and scales. This proves Equation (7.2) with its stated uniformity. $\square$

8 Averaging the unramified factors

Proposition 7.1 distributes the nonnegative mass $\phi(z) \prod_{p \in S} f_p^*(z)$ among residue classes at primes outside S . To prove Theorem 5.2, we must insert the remaining factors $b_p(z)$ while retaining their local means. Some of those means are less than one, so the argument must also keep track of the probability that a prime does not divide any component of z . We first obtain a weighted sieve estimate with this feature and then combine its local means with the Euler factors in Proposition 5.7.

8.1 Residue functions and the local bounds

Fix $m \in \mathbb{Z} \setminus \{0\}$ and a bounded region of the real trace- m plane. Throughout the sieve argument, ϕ is one of the nonnegative smooth box functions in Proposition 7.1, with its planar support in this region. Its two scales lie between $X^{-\tau}$ and fixed constants. Define a measure on F by

$$\nu_\phi = \sum_{\substack{z \in F, \text{Tr } z = m \\ z \text{ integral outside } S}} \phi(z) \prod_{p \in S} f_p^*(z) \delta_z, \quad V_\phi = A_\phi \prod_{p \in S} d_p(m). \quad (8.1)$$

Here δ_z is unit point mass at z . All its coefficients are nonnegative, and Proposition 7.1 with modulus one gives a finite total mass. Lemma 6.2 gives

$$V_\phi \asymp A_\phi \quad (8.2)$$

uniformly in S : for large primes the logarithms of $d_p(m) = 1 + O(p^{-2})$ have an absolutely convergent sum, and the finitely many small-prime densities have uniform positive upper and lower bounds.

For a positive integer l prime to every prime of S , the integral trace- m coset has l^2 residue classes modulo l in its two integral coordinates. Write $\mathbb{E}_l W$ for the uniform mean of a function W on these classes, and also write $W(z)$ for its pullback to points integral outside S .

Lemma 8.1 (Nonnegative residue functions). *With ϕ and V_ϕ as in Equation (8.1), let $1 \leq l \leq X^\theta$ be an integer prime to every prime of S . For every nonnegative function W on the residue classes of the integral trace- m coset modulo l ,*

$$\int W d\nu_\phi = V_\phi \mathbb{E}_l W + O(V_\phi X^{-\theta} \mathbb{E}_l W). \quad (8.3)$$

The implied constant is uniform in the fields, orders, embedding orderings, permitted box locations and scales, l , and W . It may depend on m , the fixed bounded region, and the fixed smooth box function.

Proof. For each residue class a , Equation (7.2) gives

$$\nu_\phi(a) = V_\phi l^{-2}(1 + \epsilon_a), \quad |\epsilon_a| \leq C_{\text{add}} X^{-\theta},$$

with the same C_{add} for every a and l in the stated range. Multiply by $W(a)$ and sum. Since $W(a) \geq 0$, the absolute value of the summed error is at most $C_{\text{add}} V_\phi X^{-\theta} l^{-2} \sum_a W(a) = C_{\text{add}} V_\phi X^{-\theta} \mathbb{E}_l W$. This also covers $\mathbb{E}_l W = 0$. In particular the error retains the local mean even when that mean is very small. $\square$

We recall the information about the functions to be inserted. For $p \notin S$, let $\mathbb{E}_p$ and $\mathbb{P}_p$ denote expectation and probability for normalized additive Haar measure on $\{z \in R_{F,p} : \text{Tr } z = m\}$. On $R_{F,p} \cap F_p^\times$, recall that $n_p(z) = \max_{\mathfrak{q}|p} v_{\mathfrak{q}}(z)$ is the largest component valuation, with $v_{\mathfrak{q}}(p) = 1$; see Equation (5.20). On the null set where a component vanishes, keep the convention $n_p = \infty$, $b_p = 0$. Lemma 5.6 permits one fixed integer $C \geq 1$ such that, on the nonvanishing locus,

$$b_p \geq 0, \quad b_p = 1 \text{ if } n_p = 0, \quad b_p \leq C(1 + n_p)^C, \quad (8.4)$$

$$\mathbb{P}_p(n_p \geq j) \leq Cp^{-j} \quad (j \geq 1), \quad \vartheta_p := \mathbb{P}_p(n_p \geq 1) \leq C/p, \quad \vartheta_p \leq \frac{1}{2}. \quad (8.5)$$

The event $\{n_p \geq j\}$ and the function $b_p \mathbf{1}_{\{n_p=j\}}$ are determined by the two integral coordinates modulo $p^{C(j+1)}$ for every integer $j \geq 1$. They have extensions defined by congruence classes on the whole local coset, agreeing with their values at every global point. A prime with $n_p(z) \geq 1$ will be called active at z . In particular,

$$\mathbb{E}_p b_p \geq 1 - \vartheta_p \geq \frac{1}{2}. \quad (8.6)$$

Lemma 5.10 places the support of ν_ϕ in the trace- m coset of $(C_F^\sharp)^\vee$. On this support all $n_p(z)$, $p \notin S$, are finite. The lemma's norm and depth conclusions bound the product of the depth factors by $O(X^{84})$ in our fixed bounded real region. Thus, for a fixed $B > 84$ and all sufficiently large X ,

$$\prod_{p \notin S} p^{n_p(z)} \leq X^B \quad (\nu_\phi(\{z\}) > 0). \quad (8.7)$$

Two consequences will be used repeatedly. There is a fixed $A_0 > 1$ such that $b_p(z) \leq A_0^{n_p(z)}$. This follows from Equation (8.4) for $n_p \geq 1$, since $\sup_{j \geq 1} (C(1+j)^C)^{1/j} < \infty$, and from $b_p = 1$ at depth zero. Also, with

$$\mathcal{W}(z) = \prod_{p \notin S} b_p(z),$$

the product is finite on the support of ν_ϕ and satisfies

$$\mathcal{W}(z) \ll_\epsilon X^\epsilon \quad (\epsilon > 0). \quad (8.8)$$

Indeed, for any $t > 0$, the inequality $C(1+j)^C \leq p^{tj}$ holds for all $j \geq 1$ once p is sufficiently large. For each of the remaining finitely many primes it holds with a constant factor independent of j . Multiplying these inequalities and using Equation (8.7), with $t = \epsilon/B$, proves Equation (8.8). All constants here are uniform in the arithmetic data and in the permitted boxes.

Proposition 8.2 (Weighted averaging). *For the measure ν_ϕ in Equation (8.1), there is a fixed $\delta > 0$ such that*

$$\int \mathcal{W}(z) d\nu_\phi(z) \ll V_\phi \prod_{\substack{p \leq X^\delta \\ p \notin S}} \mathbb{E}_p b_p. \quad (8.9)$$

The choice of δ and the implied constant are independent of the field, order, embedding ordering, and the permitted box location and scales. They may depend on m , the fixed bounded real region, and the fixed smooth box function.

The proof first counts bounded products of prescribed positive depths, while requiring inactivity at any other chosen small primes. The inactivity condition preserves the factors in Equation (8.6). We then bound the contribution of two exceptional sets of points and truncate the remaining depth products within the modulus range of Lemma 8.1.

8.2 Prescribed depths and inactive primes

Write $\mathcal{P}_y = \{p \leq y : p \notin S\}$. Chebyshev's estimate $\sum_{p \leq y} (\log p)/p \ll \log y$ and Equation (8.5) give a fixed $C_* \geq 1$ such that

$$\sum_{p \in \mathcal{P}_y} \vartheta_p \log p \leq C_* \log y \quad (y \geq 2).$$

Choose fixed positive parameters satisfying

$$20(C+1)\eta < \theta, \quad 0 < \delta < \min\{\eta/4, \eta/(4C_*)\}, \quad T_0 = X^\eta, \quad Y = X^\delta. \quad (8.10)$$

Here T_0 truncates the squarefree Selberg divisors and is the stopping threshold for depth products, whose crossing prefixes may reach T_0^2 ; Y bounds the small primes outside S handled by the sieve. For every $\mathcal{I} \subset \mathcal{P}_Y$, Mertens' estimate and $-\log(1 - \vartheta_p) \leq 2\vartheta_p \leq 2C/p$ give, for a fixed C_1 ,

$$\prod_{p \in \mathcal{I}} (1 - \vartheta_p) \gg (\log X)^{-C_1}. \quad (8.11)$$

The bound is uniform in the subset $\mathcal{I}$ and in S .

Lemma 8.3 (Prescribed depths and inactive primes). *Let $\mathcal{A}, \mathcal{I}$ be disjoint subsets of $\mathcal{P}_Y$, and assign an integer $j_p \geq 1$ to each $p \in \mathcal{A}$. Suppose that $\prod_{p \in \mathcal{A}} p^{j_p} \leq T_0^2$. Then*

$$\int \prod_{p \in \mathcal{A}} b_p \mathbf{1}_{\{n_p = j_p\}} \prod_{p \in \mathcal{I}} \mathbf{1}_{\{n_p = 0\}} d\nu_\phi \ll V_\phi \prod_{p \in \mathcal{A}} \mathbb{E}_p (b_p \mathbf{1}_{\{n_p = j_p\}}) \prod_{p \in \mathcal{I}} (1 - \vartheta_p). \quad (8.12)$$

The constant is uniform in the two subsets and all their assigned depths.

Proof. We use the Selberg upper-bound square [22]; see also [11, Chapter 21]. Its calculation is included because the relative error in Lemma 8.1 must preserve the assigned means. If $\vartheta_p = 0$, the

activity event at p is a union of residue classes of Haar measure zero, hence is empty on global points. Such a prime can be removed from $\mathcal{I}$. For the remaining primes put

$$\vartheta(d) = \prod_{p|d} \vartheta_p, \quad h(d) = \prod_{p|d} \frac{\vartheta_p}{1 - \vartheta_p}$$

for squarefree d supported on $\mathcal{I}$. Both functions take value one at $d = 1$. Let $\mathcal{D}$ be the set of those integers with $d \leq T_0$, and put $\mathcal{H} = \sum_{d \in \mathcal{D}} h(d)$.

For coefficients λ_d supported on $\mathcal{D}$ with $\lambda_1 = 1$, the square

$$\left(\sum_{d \in \mathcal{D}} \lambda_d \prod_{p|d} \mathbf{1}_{\{n_p \geq 1\}} \right)^2$$

majorizes the indicator that every prime of $\mathcal{I}$ is inactive. Its expectation in the independent product of the local probability spaces is

$$\sum_{d, e \in \mathcal{D}} \lambda_d \lambda_e \vartheta([d, e]) = \sum_{l \in \mathcal{D}} \frac{1}{h(l)} \left(\sum_{\substack{d \in \mathcal{D} \\ l|d}} \lambda_d \vartheta(d) \right)^2. \quad (8.13)$$

Here $[d, e]$ denotes the least common multiple. The identity follows by expanding the right side and using $\vartheta([d, e]) = \vartheta(d)\vartheta(e) \sum_{l|(d, e)} h(l)^{-1}$.

The set $\mathcal{D}$ is closed under divisors. Finite Möbius inversion therefore permits the choice

$$\sum_{\substack{d \in \mathcal{D} \\ l|d}} \lambda_d \vartheta(d) = \frac{\mu_{\text{Mob}}(l) h(l)}{\mathcal{H}}, \quad l \in \mathcal{D},$$

where μ_{Mob} is the Möbius function. Explicitly,

$$\lambda_d = \frac{\mu_{\text{Mob}}(d) h(d)}{\vartheta(d) \mathcal{H}} \sum_{\substack{r \leq T_0/d \\ (r, d)=1 \\ dr \in \mathcal{D}}} h(r).$$

Consequently $\lambda_1 = 1$, the value of Equation (8.13) is $\mathcal{H}^{-1}$, and

$$|\lambda_d| \leq \frac{h(d)}{\vartheta(d)} = \prod_{p|d} (1 - \vartheta_p)^{-1} \leq 2^{\omega(d)} \leq T_0. \quad (8.14)$$

To estimate $\mathcal{H}$, give every squarefree integer supported on $\mathcal{I}$ probability proportional to $h(d)$, without the bound $d \leq T_0$. In this finite product distribution, prime p is selected with probability ϑ_p . Thus

$$\mathbb{E} \log d = \sum_{p \in \mathcal{I}} \vartheta_p \log p \leq C_* \log Y < \frac{1}{4} \log T_0.$$

Markov's inequality shows that at least half the mass lies at $d \leq T_0$. It follows that

$$\mathcal{H} \geq \frac{1}{2} \prod_{p \in \mathcal{I}} (1 + h(p)), \quad \mathcal{H}^{-1} \leq 2 \prod_{p \in \mathcal{I}} (1 - \vartheta_p). \quad (8.15)$$

It remains to average this square against ν_ϕ together with the assigned weights. Write

$$A_{\mathcal{A}} = \prod_{p \in \mathcal{A}} \mathbb{E}_p(b_p \mathbf{1}_{\{n_p=j_p\}}).$$

The assigned weight is defined modulo

$$\prod_{p \in \mathcal{A}} p^{C(j_p+1)} \leq \left(\prod_{p \in \mathcal{A}} p^{j_p} \right)^{2C} \leq T_0^{4C}.$$

For a pair $d, e \in \mathcal{D}$, its activity indicators require only the additional modulus $[d, e]^{2C} \leq T_0^{4C}$. The two prime sets are disjoint, so the total modulus is at most

$$T_0^{8C} \leq X^\theta \quad (8.16)$$

and is prime to every prime of S . The function before multiplication by $\lambda_d \lambda_e$ is nonnegative, and its residue mean is $A_{\mathcal{A}} \vartheta([d, e])$ by the Chinese remainder theorem. Lemma 8.1 therefore gives the corresponding main term $V_\phi A_{\mathcal{A}} \vartheta([d, e])$ and an error of absolute value at most $CV_\phi X^{-\theta} A_{\mathcal{A}} \vartheta([d, e])$. There are at most T_0^2 pairs and, by Equation (8.14), each coefficient product has absolute value at most T_0^2 . Summing the errors absolutely costs at most

$$CV_\phi A_{\mathcal{A}} X^{-\theta} T_0^4. \quad (8.17)$$

This estimate keeps $A_{\mathcal{A}}$ even if it is very small. The main term is $V_\phi A_{\mathcal{A}} \mathcal{H}^{-1}$ by Equation (8.13). Finally, $X^{-\theta} T_0^4 = o((\log X)^{-C_1})$ by Equation (8.10). Equations (8.11) and (8.15) absorb Equation (8.17) and prove the lemma. $\square$

8.3 Truncating and summing the depths

Proof of Proposition 8.2. Put $P_Y = \prod_{p \in \mathcal{P}_Y} \mathbb{E}_p b_p$. Equations (8.6) and (8.11) imply

$$P_Y \gg (\log X)^{-C_1}. \quad (8.18)$$

First observe that primes exceeding Y cost only a fixed factor. Equations (8.7) and (8.8), together with $b_p \leq A_0^{n_p}$, give

$$\prod_{\substack{p > Y \\ p \notin S}} b_p(z) \leq A_0^{\sum_{p > Y, p \notin S} n_p(z)} \leq \exp(B \log A_0 / \delta). \quad (8.19)$$

Consider the points for which $\prod_{p \in \mathcal{P}_Y} p^{n_p(z)} \leq T_0$. Assign all their active primes in $\mathcal{P}_Y$ and require inactivity at the other primes in $\mathcal{P}_Y$. These assignments satisfy Lemma 8.3. After applying it, we may enlarge the sum of its nonnegative upper bounds by allowing every subset and every positive depth. The resulting product is

$$V_\phi \prod_{p \in \mathcal{P}_Y} \left(1 - \vartheta_p + \sum_{j \geq 1} \mathbb{E}_p(b_p \mathbf{1}_{\{n_p=j\}}) \right) = V_\phi P_Y.$$

Together with Equation (8.19), this bounds the contribution from these points.

For a remaining point, the product through $\mathcal{P}_Y$ exceeds T_0 . Order its active primes in $\mathcal{P}_Y$ increasingly and stop at the first prefix whose product exceeds T_0 . If every individual factor $p^{n_p(z)}$

for $p \in \mathcal{P}_Y$ is at most T_0 , the prefix has product in $(T_0, T_0^2]$, since its preceding product and its last factor are both at most T_0 . If also $\prod_{p \in \mathcal{P}_{y_0}} p^{n_p(z)} \leq T_0$, for a fixed cutoff $y_0 > 4$ to be chosen below independently of X , its last prime exceeds y_0 . The next two estimates allow these restrictions at negligible weighted cost.

First, if $p^{n_p(z)} > T_0$ for $p \in \mathcal{P}_Y$, then $n_p(z) \geq J_p$, where $J_p = \lfloor \log T_0 / \log p \rfloor + 1$. The threshold is determined modulo

$$p^{C(J_p+1)} \leq T_0^C Y^{2C} \leq X^\theta.$$

Lemma 8.1 and Equation (8.5) bound its ν_ϕ -measure by $O(V_\phi p^{-J_p}) = O(V_\phi/T_0)$. Summing over $p \in \mathcal{P}_Y$ gives $O(V_\phi Y/T_0)$. By Equation (8.8), the contribution after multiplication by $\mathcal{W}$ is $O_\epsilon(V_\phi X^{\epsilon+\delta-\eta})$. Choose $\epsilon < \eta - \delta$. This is smaller than $V_\phi P_Y$ by a power of X , in view of Equation (8.18).

We also need the analogous removal for a fixed nonempty finite set $\mathcal{Q}$ of primes, even though some of them may belong to the varying set S . Put $N = |\mathcal{Q}|$. If $\prod_{p \in \mathcal{Q} \setminus S} p^{n_p(z)} > T_0$, some prime in $\mathcal{Q} \setminus S$ has $p^{n_p(z)} > T_0^{1/N}$. Put $J_{p,\mathcal{Q}} = \lfloor \log T_0 / (N \log p) \rfloor + 1$. The event $n_p \geq J_{p,\mathcal{Q}}$ is defined modulo at most $T_0^{C/N} p^{2C} \leq X^\theta$ for large X , and its local probability is at most $Cp^{-J_{p,\mathcal{Q}}} \leq CT_0^{-1/N}$. Lemma 8.1 therefore bounds the measure of the union by $O_{\mathcal{Q}}(V_\phi T_0^{-1/N})$. After multiplication by Equation (8.8) with $\epsilon < \eta/N$, its weighted contribution again saves a power of X relative to V_ϕ . If $\mathcal{Q} \setminus S$ is empty the event is empty. Thus this removal is uniform in the possible intersections of $\mathcal{Q}$ with S .

Choose a fixed $M_0 > 0$ and then a fixed $y_0 > 4$ such that

$$\eta M_0 > 2B \log A_0 + 1, \quad M_0 / \log y_0 < \frac{1}{4}. \quad (8.20)$$

Apply the finite-set removal with $\mathcal{Q} = \{p : p \leq y_0\}$. These choices precede the choice of the small ϵ used in that removal, and all are independent of X .

The retained first-crossing prefixes therefore have product in $(T_0, T_0^2]$ and last prime greater than y_0 . Group these assignments by the interval containing their last prime:

$$\sqrt{y} < p \leq y, \quad y = Y^{1/2^i}, \quad i = 0, 1, 2, \dots \quad (8.21)$$

Only groups with $y > y_0$ occur. Every unassigned active prime is greater than the last assigned prime and hence greater than $\sqrt{y}$. The total weight of those primes, including primes beyond Y , is therefore at most

$$\exp\left(\frac{2B \log A_0}{\log y} \log X\right). \quad (8.22)$$

For each assignment in the group, require inactivity at its unassigned primes in $\mathcal{P}_{\sqrt{y}}$ and apply Lemma 8.3. We use Rankin's power majorant [20]: since the assigned product exceeds T_0 , its upper bound can be multiplied by

$$T_0^{-s} \prod_{\text{assigned } p} p^{sj_p} \geq 1, \quad s = M_0 / \log y.$$

Now enlarge the sum of these nonnegative upper bounds to all subsets and all positive depths through y . This enlarges only a numerical sum after the lemma has been applied to the genuine prefixes; it invokes no residue estimate for the newly admitted products above T_0^2 .

For $p \in \mathcal{P}_{\sqrt{y}}$, the option of no assignment has weight $1 - \vartheta_p$ and the assigned options have total weight $\sum_{j \geq 1} \mathbb{E}_p(b_p \mathbf{1}_{\{n_p=j\}}) p^{sj}$. Their sum is $\mathbb{E}_p(b_p p^{sn_p})$. For $\sqrt{y} < p \leq y$ outside S , inactivity was

not imposed, so the option of no assignment has weight one. The total there is $\mathbb{E}_p(b_p p^{sn_p}) + \vartheta_p$. Thus the group's contribution is at most

$$CV_\phi T_0^{-s} \exp\left(\frac{2B \log A_0}{\log y} \log X\right) \prod_{p \in \mathcal{P}_{\sqrt{y}}} \mathbb{E}_p(b_p p^{sn_p}) \prod_{\substack{\sqrt{y} < p \leq y \\ p \notin S}} (\mathbb{E}_p(b_p p^{sn_p}) + \vartheta_p). \quad (8.23)$$

We estimate these local moments uniformly for $y > y_0$. By Equation (8.20), $s < 1/4$. The total contribution of depths $j \geq 2$ to $\mathbb{E}_p(b_p p^{sn_p})$ is at most

$$C^2 \sum_{j \geq 2} (1+j)^C p^{-(1-s)j} \ll_C p^{-3/2}.$$

At depth one, the increase over $\mathbb{E}_p b_p$ is bounded by

$$\frac{C'}{p} (p^s - 1) \ll_{M_0} \frac{\log p}{p \log y} \quad (p \leq y),$$

because $0 \leq s \log p \leq M_0$. Chebyshev's estimate shows that these increases have bounded sum over $p \leq y$. Mertens' estimate also gives $\sum_{\substack{\sqrt{y} < p \leq y \\ p \notin S}} \vartheta_p \ll 1$. Since $\mathbb{E}_p b_p \geq 1/2$, comparison of the products in Equation (8.23) with their base means gives

$$\prod_{p \in \mathcal{P}_{\sqrt{y}}} \mathbb{E}_p(b_p p^{sn_p}) \prod_{\substack{\sqrt{y} < p \leq y \\ p \notin S}} (\mathbb{E}_p(b_p p^{sn_p}) + \vartheta_p) \ll_{M_0} \prod_{p \in \mathcal{P}_y} \mathbb{E}_p b_p.$$

For a fixed C_2 , a final use of Equation (8.6) and Mertens' estimate yields

$$\prod_{p \in \mathcal{P}_y} \mathbb{E}_p b_p \leq P_Y \prod_{\substack{y < p \leq Y \\ p \notin S}} (1 - \vartheta_p)^{-1} \ll P_Y \left(\frac{\log Y}{\log y}\right)^{C_2}.$$

By Equation (8.20), the bound for the group in Equation (8.23) is consequently at most

$$C_{M_0} V_\phi P_Y \exp\left(-\frac{\log X}{\log y}\right) \left(\frac{\log Y}{\log y}\right)^{C_2}.$$

For the values of y in Equation (8.21), its sum is bounded by

$$C_{M_0} V_\phi P_Y \sum_{i \geq 0} \exp(-2^i/\delta) 2^{iC_2} \ll V_\phi P_Y.$$

Together with the small-product part and the two discarded sets, this proves Equation (8.9). $\square$

8.4 The planar exterior count

Proof of Theorem 5.2. Fix the nonzero trace and bounded region in that theorem. Let $\mathcal{R}$ be any of its rectangles, in the specified two partition coordinates. Choose an allowed smooth function ϕ that majorizes $\mathbf{1}_{\mathcal{R}}$, with its two scales equal to the side lengths of $\mathcal{R}$. Its support lies in a fixed enlargement of the bounded region and

$$A_\phi \asymp \text{area}(\mathcal{R}).$$

The constants are uniform over the positions, side lengths, and aspect ratios of the rectangles. Proposition 5.7 and Proposition 8.2 give

$$\sum_{\substack{\text{Tr } z=m \\ z \in \mathcal{R}}} M(z) \ll \sqrt{D} A_\phi \prod_{p \in S} (1 - p^{-1}) Z_p \prod_{p \in \mathcal{P}_Y} \mathbb{E}_p b_p, \quad Z_p = \prod_{\substack{\mathfrak{p}|p \\ \mathfrak{p} \subset R}} (1 - N(\mathfrak{p})^{-1})^{-1}. \quad (8.24)$$

Here $\prod_{p \in S} |D|_p^{-1/2} = \sqrt{D}$, because S contains every prime divisor of D , and Equation (8.2) absorbed the trace densities. The restriction to integral points outside S in the multiplicity proposition agrees with the support of ν_ϕ .

The exact unramified mean in Lemma 5.6 is

$$\mathbb{E}_p b_p = (1 - p^{-1})(1 - p^{-3}) Z_p \quad (p \notin S).$$

Since $1 - p^{-3} \leq 1$, the two Euler products on the right of Equation (8.24) are at most the product of $(1 - p^{-1}) Z_p$ over $p \in S \cup \mathcal{P}_Y$. Ideal factorization in the degree-four maximal order gives, uniformly also at ramified primes,

$$\log Z_p = \frac{a_K(p)}{p} + O(p^{-2}).$$

Thus

$$\prod_{p \in S} (1 - p^{-1}) Z_p \prod_{p \in \mathcal{P}_Y} \mathbb{E}_p b_p \ll \exp \left(\sum_{p \in S \text{ or } p \leq Y} \frac{a_K(p) - 1}{p} \right). \quad (8.25)$$

For fixed m , the radical of S is polynomial in X . Choose a fixed $c \geq 1/2$ large enough that, for all sufficiently large X , every prime in S and every prime at most Y is at most $Z = X^c$. Since $a_K(p) \geq 0$, the exponent in Equation (8.25) is at most

$$\sum_{p \leq Z} \frac{a_K(p)}{p} - \sum_{p \leq Y} \frac{1}{p}.$$

We have $Z \geq D^{1/4}$ because $X = q\sqrt{D} \geq \sqrt{D}$. Equation (3.14) and Mertens' estimate now give

$$\exp \left(\sum_{p \in S \text{ or } p \leq Y} \frac{a_K(p) - 1}{p} \right) \ll \frac{\kappa \log Z}{\log Y} = \frac{c}{\delta} \kappa.$$

This also applies when D remains bounded and q tends to infinity, by the stated uniform range of Lemma 3.3. Substitution in Equation (8.24) proves

$$\sum_{\substack{\text{Tr } z=m \\ z \in \mathcal{R}}} M(z) \ll \kappa \sqrt{D} \text{ area}(\mathcal{R}).$$

The only box-dependent quantity in the proof is A_ϕ , which is comparable to the area with a fixed constant. All other estimates are uniform over the box translations and the two scales between $X^{-\tau}$ and the fixed upper bounds. The local multiplicity majorant has the same constant for every chosen ideal representative system, and its right side is independent of that choice. Thus the bound is uniform in the representative system as well. This proves the precise uniformity asserted in Theorem 5.2. $\square$

9 Exclusion of the obstruction

We apply Theorem 5.2 to the exterior planes in Proposition 4.6. Four small exterior coordinates give two small signed products, whose range in the trace plane has area of order r^4 . The integral along a diagonal orbit contributes logarithmic factors; we will show that they are summable at this scale.

9.1 The exterior vector integral

Fix a partition $b_1 = ij|kl$ of $\{1, 2, 3, 4\}$, and denote the other two partitions by b_2, b_3 . For $m \in \mathbb{Z} \setminus \{0\}$, $L > 1$, and $0 < r < 1$, define $N_{m,L,r}(\Lambda)$ to be the number of $v \in \Lambda^2 \Lambda$ satisfying

$$Q(v) = m, \quad |v_{ij}|, |v_{kl}| < L, \quad |v_{ab}| < r \quad (ab \neq ij, kl). \quad (9.1)$$

This count is finite because $\Lambda^2 \Lambda$ is a lattice in $\Lambda^2 \mathbb{R}^4$ and the specified box is bounded. To see that it is Borel, choose a local continuous determinant-one basis around a lattice with k witnesses and keep their distinct integer exterior labels fixed. They remain distinct, and their Q -values remain m because $Q((\Lambda^2 g)w) = \det(g)Q(w)$. Their coordinates vary continuously, so the strict inequalities persist in a neighborhood. Thus $\{N_{m,L,r} \geq k\}$ is open for every integer $k \geq 1$. Recall that the three real embeddings of F are indexed by $\mathcal{B}$. For a nonzero $z \in F$, set

$$\mathcal{L}_{L,r}(z) = \log_+ \frac{L^2}{|z_{b_1}|} \log_+ \frac{r^2}{|z_{b_2}|} \log_+ \frac{r^2}{|z_{b_3}|}. \quad (9.2)$$

All three conjugates of a nonzero element of the field F are nonzero, so the expression is defined.

In Equation (5.8), $M(z)$ sums over the fixed maximal-order ideal representatives I and the U^+ -orbits of nonzero raw $w \in \Lambda_{\mathbb{Z}}^2 I$ with $\Psi(w)/NI = z$. Each orbit has weight $\prod_p P_p(I, w)$, the average membership of w in $\Lambda_{\mathbb{Z}}^2 L'$ over the local labels $L' \in \mathcal{L}_{\mathcal{O}}(I)$.

Lemma 9.1 (Exterior unfolding). *For every primitive totally real quartic field K , order $\mathcal{O}$, and ordering of its real embeddings, let $d_U = [U : U^+]$. With the preceding choices of b_1, m, L, r , use the same fixed system of maximal-order ideal representatives in the packet model and in the definition of M . Then*

$$\int_{X_4} N_{m,L,r}(\Lambda) d\mu_{\mathcal{O},\sigma}(\Lambda) = \frac{4}{d_U \sqrt{D} \kappa} \sum_{t \in \{m, -m\}} \sum_{\substack{z \in F \\ \text{Tr } z = t}} M(z) \mathcal{L}_{L,r}(z). \quad (9.3)$$

The index satisfies $2 \leq d_U \leq 16$.

Proof. We use the packet integral in Equation (2.9). For its maximal-order ideal representative I and local label $L' \in \mathcal{L}_{\mathcal{O}}(I)$, the normalized lattice at the sign vector $s = (s_1, \dots, s_4) \in \{\pm 1\}^4$ and the logarithmic parameter $(u_1, \dots, u_4) \in \mathbb{R}^4$ with $u_1 + \dots + u_4 = 0$ is

$$(XNI)^{-1/4} \sigma(L') \text{diag}(s_1 e^{u_1}, \dots, s_4 e^{u_4}).$$

For a raw vector $w \in \Lambda^2 I$, its normalized exterior coordinates are

$$y_{ab} = (XNI)^{-1/2} s_a s_b e^{u_a + u_b} w_{ab}.$$

If $z = \Psi(w)/NI$ and $\epsilon(s) = \prod_a s_a$, then Equation (5.6) gives

$$(y_{12}y_{34}, -y_{13}y_{24}, y_{14}y_{23}) = \epsilon(s)z, \quad Q(y) = \epsilon(s) \text{Tr } z. \quad (9.4)$$

There are eight sign vectors for each value of $\epsilon(s)$.

The kernel of the signature map on U is U^+ , so $d_U = |\operatorname{sgn} U| \leq 16$; the signature of -1 gives $d_U \geq 2$. After averaging over L' , the vector-count integrand in Equation (2.9) is invariant under $v \mapsto \sigma(e)v$ for $e \in U$, since $L' \mapsto eL'$ permutes the labels and transports their exterior lattices. Replacing the fundamental domain for U in T_∞ by one for U^+ therefore multiplies its integral by d_U , which we compensate by division by d_U . The latter domain consists of all sign components, with a fundamental domain for the positive unit logarithms in each component.

For a given raw w , its average membership over the local labels is

$$\frac{1}{N_{\mathcal{O}}} \sum_{L' \in \mathcal{L}_{\mathcal{O}}(I)} \mathbf{1}_{\{w \in \Lambda^2 L'\}} = \prod_p P_p(I, w),$$

the weight in Equation (5.8). This weight is constant on its U^+ -orbit. That action is free: every embedding coordinate of the nonzero raw vector $w \in \bigwedge_{\mathbb{Z}}^2 I \subset \bigwedge_{\mathbb{Q}}^2 K$ is nonzero by Lemma 5.1. A stabilizing positive unit e would therefore satisfy $\sigma_a(e)\sigma_b(e) = 1$ for every pair ab , forcing $e = 1$. Summing the translates of each raw vector thus unfolds the positive-unit fundamental domain to all of the three-dimensional space $u_4 = -u_1 - u_2 - u_3$, with multiplicity one. The packet normalization remaining after the uniform label average is $h_K \operatorname{vol}(\mathcal{F}_U) = \sqrt{D} \kappa$ by Equation (2.6). Hence this unfolding has coefficient $1/(d_U \sqrt{D} \kappa)$ before the sign sum and the integral in u . All the sums and integrals are nonnegative, so Tonelli's theorem applies.

Choose one coordinate from each opposite pair. The logarithms of the absolute values of these coordinates are affine functions of (u_1, u_2, u_3) . For the choices 12, 13, 14, their linear parts are

$$u_1 + u_2, \quad u_1 + u_3, \quad -u_2 - u_3.$$

Their matrix has determinant of absolute value two. Choosing the opposite member of a pair negates its row, and ordering the partitions only permutes rows, so the same absolute determinant holds for every choice. If the two entries in an opposite pair have product of absolute value $|z_b|$ and both must be less than $h > 0$ in absolute value, the logarithm of the chosen entry lies in an interval of length $\log_+(h^2/|z_b|)$. Therefore, for a raw vector with the required trace after the sign change, its integral over (u_1, u_2, u_3) is $\frac{1}{2} \mathcal{L}_{L,r}(z)$.

The U^+ -orbit sum of the weights, including the sum over I , is exactly $M(z)$ by Equation (5.8). Equation (9.4) allows raw trace m for the eight positive determinant signs and raw trace $-m$ for the other eight. Multiplying eight by the Jacobian factor $1/2$ gives the factor 4 in Equation (9.3). The unit -1 , although it acts trivially on exterior vectors, has already been accounted for in the index d_U . $\square$

Proposition 9.2 (Small exterior boxes). *For every fixed $m \in \mathbb{Z} \setminus \{0\}$ and $L > 1$, there are constants $r_0 > 0$ and $C_{m,L} > 0$ with the following property. For every sequence of the fields, orders, and embedding orderings in Theorem 1.1 with $X_i \rightarrow \infty$, and every fixed $0 < r < r_0$,*

$$\limsup_{i \rightarrow \infty} \int_{X_4} N_{m,L,r}(\Lambda) d\mu_{\mathcal{O}_i, \sigma_i}(\Lambda) \leq C_{m,L} r^4. \quad (9.5)$$

The constants are independent of the sequence, of r , and of the choice of partition b_1 .

Proof. Take $r_0 < 1$ so small that $2r_0^2 \leq |m|/2$. For each $t \in \{m, -m\}$, fix the coordinate rectangle

$$\Omega_t = \{z \in \mathcal{H}_t : |z_{b_2}|, |z_{b_3}| \leq 1\}.$$

Now let $0 < r < r_0$. In the trace- t sum in Equation (9.3), only points with $|z_{b_2}|, |z_{b_3}| < r^2$ contribute, and these points lie in Ω_t . Since their three coordinates have sum t ,

$$|z_{b_1}| \geq |m| - 2r^2 \geq |m|/2.$$

Thus the first logarithm in Equation (9.2) is bounded in terms of m, L on the contributing points, independently of r . For every $z \in \Omega_t$ with $M(z) > 0$, Lemma 5.10 gives $|z_b| \gg_{\Omega_t} X^{-84}$ for each b . Hence the last two logarithms in Equation (9.2) are $O(\log X)$, uniformly for $0 < r < 1$.

We first remove the points in the two strips

$$|z_{b_2}| \leq 4X^{-\tau} \quad \text{or} \quad |z_{b_3}| \leq 4X^{-\tau}. \quad (9.6)$$

For large X , each strip within Ω_t is covered by a rectangle of width $8X^{-\tau}$ and length 2 in the coordinates z_{b_2}, z_{b_3} , and these rectangles lie in Ω_t . Both sides meet the minimum length in Theorem 5.2 for large X . That theorem bounds the sum of $M(z)$ over their union, divided by $\sqrt{D}\kappa$, by $O(X^{-\tau})$. The logarithmic bounds just proved show that their contribution to Equation (9.3) is

$$O_{m,L}(X^{-\tau}(\log X)^2) = o(1).$$

The use of the rectangle estimate here controls the weighted mass directly; it does not require a pointwise bound for $M(z)$.

For the remaining points, separate the signs of z_{b_2}, z_{b_3} and use the dyadic intervals

$$r^2 2^{-a-1} < |z_{b_2}| \leq r^2 2^{-a}, \quad r^2 2^{-b-1} < |z_{b_3}| \leq r^2 2^{-b}, \quad a, b \geq 0.$$

Each signed dyadic rectangle lies in Ω_t . Each interval has width half its upper endpoint. If it meets $|z_{b_j}| > 4X^{-\tau}$, that width exceeds $2X^{-\tau}$. Hence every dyadic rectangle meeting the remaining region satisfies the minimum side lengths of Theorem 5.2; any boundary rectangle may be included in full. Its area is $O(r^4 2^{-a-b})$, while the last two factors of Equation (9.2) are at most $(a+1)\log 2$ and $(b+1)\log 2$. Theorem 5.2 gives

$$\frac{1}{\sqrt{D}\kappa} \sum_{z \text{ in that rectangle}} M(z) \ll_{m,L} r^4 2^{-a-b}.$$

The sum over the four choices of signs and over all a, b is therefore at most

$$C_{m,L} r^4 \sum_{a,b \geq 0} (a+1)(b+1) 2^{-a-b} \ll_{m,L} r^4.$$

All constants are independent of r : the rectangles Ω_t and the uniform rectangle constant were fixed before choosing r . The two strips tend to zero for each fixed r , and $4/d_U \leq 2$ in Equation (9.3). These observations prove Equation (9.5) for both traces and all three partitions. $\square$

9.2 Identification of every probability limit

Proof of Theorem 1.1. For the sequence in that theorem, $X_i = \sqrt{|\text{Disc}(\mathcal{O}_i)|}$ tends to infinity. By Proposition 3.6, the packet probabilities are tight, and every subsequence has a further weakly convergent subsequence with an A_4 -invariant probability limit μ . Proposition 3.8 gives the local ball bound required by Corollary 4.3. Hence almost every component in the A_4 -ergodic decomposition of μ has positive entropy for a diagonal flow. Proposition 4.6 reduces the identification of these components to the exclusion of its three exterior-plane events.

Fix a partition b_1 , a nonzero integer m , and $L > 1$. The event

$$E_{m,L,r} = \{\Lambda \in X_4 : N_{m,L,r}(\Lambda) \geq 1\}$$

is open by the observation following Equation (9.1).

The open-set inequality for weak convergence and Proposition 9.2 now give, along the convergent subsequence,

$$\mu(E_{m,L,r}) \leq \liminf_i \mu_{\mathcal{O}_i, \sigma_i}(E_{m,L,r}) \leq \limsup_i \int N_{m,L,r} d\mu_{\mathcal{O}_i, \sigma_i} \leq C_{m,L} r^4$$

for every sufficiently small fixed $r > 0$. In particular,

$$\mu \left(\bigcap_{n \geq 2} E_{m,L,1/n} \right) \leq \inf_{\substack{n \geq 2 \\ 1/n < r_0}} C_{m,L} n^{-4} = 0.$$

Every vector in one of the obstruction planes with $Q \neq 0$ has a nonzero integer Q -value by Equation (4.7), and its two entries are less than some integer $L \geq 2$ in absolute value. An exact witness belongs to $E_{m,L,1/n}$ for every $n \geq 2$. Thus the Borel event in Equation (4.9) satisfies

$$\mathcal{E}_{b_1} \subset \bigcup_{m \in \mathbb{Z} \setminus \{0\}} \bigcup_{\substack{L \in \mathbb{Z} \\ L \geq 2}} \bigcap_{n \geq 2} E_{m,L,1/n}.$$

The right side is a countable union of null sets, so $\mu(\mathcal{E}_{b_1}) = 0$. The same argument applies to all three partitions. Integrating these null events over the ergodic decomposition shows that the proper alternative in Proposition 4.6 has zero total component weight. Almost every component is therefore m_4 , and hence $\mu = m_4$.

Every subsequential probability limit is now identified as m_4 . Tightness implies that the entire sequence converges weakly to m_4 , which gives the asserted convergence against every $f \in C_c(X_4)$. The same tightness statement supplies, for each $\epsilon > 0$, one compact set carrying at least $1 - \epsilon$ of every sufficiently late packet probability. This is the required no-escape assertion. $\square$

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